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Supplementary appendix 2

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

Supplement to: Ren Z, Xu J, Bai Y, et al. Sintilimab plus a bevacizumab biosimilar (IBI305) versus sorafenib in unresectable hepatocellular carcinoma (ORIENT-32): a randomised, open-label, phase 2–3 study. *Lancet Oncol* 2021; published online June 15. http://dx.doi.org/10.1016/S1470-2045(21)00252-7.

Supplementary Methods

Proportional hazard assumption in the Cox Model

The following three methods were used for checking the proportional hazard assumption in the Cox model on the co-primary endpoints of overall survival (OS) and progression-free survival (PFS) assessed by the independent radiological review committee (IRRC): visual inspection, including time dependent covariates in Cox Model and Schoenfeld residuals:

(1) Per the graph of log(-log(survival)) versus log of survival time, the 2 lines are approximately in parallel;

(2) Time dependent covariates (trtpt) are not significant (p-value<0.05);

(3) The lines on plot of Schoenfeld residuals is approximately horizontal.

Sensitive analysis

Sensitive analyses of OS were performed based on two analysis sets, full analysis set (FAS, defined as patients with at least one baseline measurable lesion, and received at least one study drug) and per-protocol set (PPS, defined as patients not using prohibited combination drug in protocol, completing predefined minimum exposure dose of study drug, no serious protocol deviation that influenced the efficacy assessment), and based on unstratified Cox model; sensitive analysis of IRRC-assessed PFS by RECIST version 1.1 were conducted based on different censored rules (ignoring new tumour therapies, ignoring missing tumour radiographic assessment, or taking loss to follow-up as progression), two other analysis sets (FAS, PP) and unstratified Cox model.

Assessment of immunogenicity

For patients in sintilimab–bevacizumab biosimilar group, immunogenicity of sintilimab was assessed through the serum antidrug antibody (ADA) assay, which was measured in a specified central laboratory. ADAs in patient serum samples were detected using a validated and semiquantitative electrochemiluminescence (ECL) immunoassay. The assay included two phases.

(1) *Pretreatment of serum samples*: after acid dissociation of the serum matrix, the free ADAs were incubated with biotin-simtilimab, and then captured by streptavidin on magnetic bead. The captured ADA was performed with acid dissociation again, generating purified free ADA.

(2) *Detection of samples*: incubated the purified free ADA and biotin- and ruthenium-labeled simtilimab to generate the immunocomplex, which was then combined with streptavidin-coated MSD plate. Putting the plate into MSD reader to trigger ECL reaction and generate the luminescence signal.

***Table S1*: The ORIENT-32 study group**

	First name	Last name	Affiliation	City, Country	Degree
41	Ying	Liu	Gastroenterology Department, Henan Cancer Hospital	Zhengzhou, China	Prof, MD
42	Tianqiang	Song	Hepatobiliary Oncology, Tianjin Medical University Cancer Institute and Hospital	Tianjing, China	Prof, MD
43	Xianhai	Mao	Liver Surgery Department II, Hunan Provincial People's Hospital	Changsha, China	Prof, MD
44	Chunyi	Hao	Hepatobiliary and Pancreatic Surgery II, Beijing Cancer Hospital	Beijing, China	Prof, MD
45	Hongming	Pan	Medical Oncology, Sir Run Run Shaw Hospital, Zhejiang University School of Medicine	Hangzhou, China	Prof, MD
46	Yunpeng	Liu	Oncology Department, The First Hospital of China Medical University	Shenyang, China	Prof, MD
47	Xiaoli	Liao	Department of Digestive Oncology, Guangxi Medical University Affiliated Tumour Hospital	Nanning, China	Prof, MD
48	Xiwen	Fan	Department of Interventional Medicine, Affiliated Tumour Hospital of Xinjiang Medical University	Urumchi, China	Prof, MD
49	Xiaoli	Zhu	Department of Interventional Medicine, The First Affiliated Hospital of Soochow University	Suzhou, China	Prof, MD
50	Qi	Li	Oncology Department, Shanghai General Hospital	Shanghai, China	Prof, MD
51	Liangfang	Shen	Oncology Department, Xiangya Hospital of Central South University	Changsha, China	Prof, MD
52	Zhengxiang	Han	Oncology Department, The Affiliated Hospital of Xuzhou Medical University	Xuzhou, China	Prof, MD
53	Guanghai	Dai	The Second Section of Oncology Department, Chinese PLA General Hospital	Beijing, China	Prof, MD
54	Bangwei	Cao	Oncology Department, Beijing Friendship Hospital, Capital Medical University	Beijing, China	Prof, MD

***Table S2*: Enrollment of phase 3 in study sites**

	Investigator name	Affiliation	Enrolled patient number
1	Zhenggang Ren*	Department of Hepatic Oncology, Liver Cancer Institute, Zhongshan Hospital, Fudan University, and Key Laboratory of Carcinogenesis and Cancer Invasion, Ministry of Education	19#
2	Jianming Xu*	Digestive Oncology Department, The Fifth Medical Centre of PLA General Hospital	16
3	Yuxian Bai	Department of Internal Medicine, Harbin Medical University Cancer Hospital	36
4	Aibing Xu	Oncological internal medicine, Nantong Tumour Hospital	26
5	Shundong Cang	Internal Medicine-Oncology, Henan Provincial People's Hospital	25

6	Chengyou Du	Hepatobiliary Surgery, The First Affiliated Hospital of Chongqing Medical University	20
7	Qiu Li	Abdominal Tumour Department, West China Hospital, Sichuan University	19
8	Yinying Lu	Treatment and Research Centre for Liver Cancer Department 2, The Fifth Medical Centre of PLA General Hospital	18
9	Yajin Chen	Hepatobiliary Surgery, Sun Yat-Sen Memorial Hospital Sun Yat-Sen University	18
10	Yabing Guo	Tumours of Liver, Nan Fang Hospital	17
11	Zhendong Chen	Oncology Department, The Second Hospital of Anhui Medical University	17
12	Baorui Liu	Department of Oncology, Nanjing Drum Tower Hospital of Nanjing University Medical School	16
13	Weidong Jia	Genera Surgery Department, Anhui Provincial Hospital	16
14	Jian Wu	Hepatobiliary and Pancreatic Surgery, The First Affiliated Hospital, Zhejiang University	16
15	Junye Wang	Department of Oncology, Affiliated Hospital of Jining Medical University	16
16	Guoliang Shao	Department of Interventional Therapy, Zhejiang Cancer Hospital	15
17	Bixiang Zhang	Hepatobiliary Surgery, Tongji Medcial College Huazhong University of Science & Technology	15
18	Yunfeng Shan	Hepatobiliary Surgery, The First Affiliated Hospital of Wenzhou Medical University	14
19	Zhiqiang Meng	Department of Traditional Chinese Medicine/Integrative Oncology, Fudan University Shanghai Cancer Centre	13
20	Jianbing Wu	Oncology Department, The Second Affiliated Hospital of Nanchang University	13
21	Shanzhi Gu	Radioactive Interventional Department, Hunan Cancer Hospital	12
22	Wei Yang	Department of Hepatobiliary Surgery, The First Affiliated Hospital of Xi'an Jiao Tong University	11
23	Chao Liu	Pancreaticobiliary surgery, Sun Yat-sen Memorial Hospital Sun Yat-sen University	11
24	Xuetao Shi	Department of Hepatobiliary Surgery, Shandong Cancer Hospital	10
25	Zhenyuan Gao	Medical Oncology, The First Affiliated Hospital of Bengbu Medical College	10
26	Tao Yin	Hepatobiliary Surgery, Hubei Cancer Hospital	10
27	Jiuwei Cui	Oncology Department, Bethune First Hospital of Jilin University	9
28	Ming Huang	Department of Minimally Invasive Interventional Medicine, Yunnan Cancer Hospital	9
29	Baocai Xing	Hepatobiliary Pancreatic Surgery 1, Beijing Cancer Hospital	8
30	Yilei Mao	Live Surgery Ward, Peking Union Medical College Hospital	8
31	Gaojun Teng	Radiology Department, Zhongda Hospital Southeast University	8

32	Yanru Qin	Internal Medicine Oncology, The First Affiliated Hospital of Zhengzhou University	8
33	Jinhai Wang	Gastroenterology, The Second Affiliated Hospital of Xi'an Jiao Tong University	8
34	Feng Xia	Department of Hepatobiliary Surgery, The First Hospital Affiliated to AMU (Southwest Hospital)	8
35	Guowen Yin	Intervention Department, Jiangsu Cancer Hospital	7
36	Yong Yang	Medical Science and Strategy Oncology, Innovent Biologics, Inc·	Funder author
37	Mingxia Chen	Department of Biostatistics and Information, Innovent Biologics, Inc·	Funder author
38	Yan Wang	Medical Science and Strategy Oncology, Innovent Biologics, Inc·	Funder author
39	Hui Zhou	Medical Science and Strategy Oncology, Innovent Biologics, Inc·	Funderauthor
40	Jia Fan* (Corresponding author)	Department of Liver Surgery & Transplantation, Liver Cancer Institute, Zhongshan Hospital, Fudan University, and Key Laboratory of Carcinogenesis and Cancer Invasion, Ministry of Education	19#
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42	Tianqiang Song	Hepatobiliary Oncology, Tianjin Medical University Cancer Institute and Hospital	7
43	Xianhai Mao	Liver Surgery Department II, Hunan Provincial People's Hospital	7
44	Chunyi Hao	Hepatobiliary and Pancreatic Surgery II, Beijing Cancer Hospital	6
45	Hongming Pan	Medical Oncology, Sir Run Run Shaw Hospital, Zhejiang University School of Medicine	6
46	Yunpeng Liu	Oncology Department, The First Hospital of China Medical University	6
47	Xiaoli Liao	Department of Digestive Oncology, Guangxi Medical University Affiliated Tumour Hospital	6
48	Xiwen Fan	Department of Interventional Medicine, Affiliated Tumour Hospital of Xinjiang Medical University	6
49	Xiaoli Zhu	Department of Interventional Medicine, The First Affiliated Hospital of Soochow University	5
50	Qi Li	Oncology Department, Shanghai General Hospital	4
51	Liangfang Shen	Oncology Department, Xiangya Hospital of Central South University	3
52	Zhengxiang Han	Oncology Department, The Affiliated Hospital of Xuzhou Medical University	3
53	Guanghai Dai	The Second Section of Oncology Department, Chinese PLA General Hospital	2
54	Bangwei Cao	Oncology Department, Beijing Friendship Hospital, Capital Medical University	1

*, Zhenggang Ren, Jianming Xu and Jia Fan are Safety Committee Members; #, Zhenggang Ren and Jia Fan co-enrolled 19 patients.

***Table S3*: Patients had failure of screening of phase 3**

	Screen Failure
Patients did not meet at least one inclusion criteria or met at least one exclusion criteria	305
Inclusion criteria	181 (59%)
I8: Sufficient organ and bone marrow functions	88 (29%)
I6: At least one measurable lesion according to RECIST V1·1, or measurable lesions showing definite progression following the local treatment based on RECIST V1·1	33 (11%)
I11: Patients who have signed ICF and are able to perform follow-up visits and relevant procedures required in the protocol	29 (10%)
I5: Absence of systemic anti-tumour treatment for HCC before the first dose	11 (4%)
I7: Child-Pugh score of ≤ 7	8 (3%)
I9: Expected survival of ≥ 12 weeks	7 (2%)
I4: Barcelona Clinic Liver Cancer (BCLC) stage of C· Stage B for those unsuitable for radical surgery and/or local treatment	6 (2%)
I1: Histologically/cytologically confirmed HCC or cirrhosis meeting the clinical diagnostic criteria of HCC by American Association for the Study of Liver Diseases (AASLD)	4 (1%)
Exclusion criteria	138 (45%)
E4: Acute or chronic active hepatitis B or C infection	42 (14%)
E9: Involvement of both the main portal vein and the left and right branches by portal vein tumour thrombus, or of both the main trunk and the superior mesenteric vein concurrently.· Presence of tumour thrombus of inferior vena cava	31 (10%)
E6: Presence of bleeding events from esophageal or gastric varices caused by portal hypertension within the past 6 months· Presence of known severe (G3) varicose veins in endoscopy within 3 months before the first dose· Evidence of portal hypertension (including the finding of splenomegaly in imaging studies) with a high risk of bleeding assessed by the investigator	15 (5%)
E33: Acute or chronic diseases, psychiatric disorders, or laboratory abnormalities that may lead to the following consequences: increased study participation or drug-related risks, or interference with interpreting trial results, and considered ineligible for participating in the trial by the investigator	7 (2%)
E1: Histologically/cytologically confirmed fibrolamellar hepatocellular carcinoma, sarcomatoid hepatocellular carcinoma, and cholangiocarcinoma	6 (2%)
E28: Other malignancies diagnosed within 5 years before the first dose, excluding radically treated basal cell carcinoma of skin, squamous cell carcinoma of skin, and/or	6 (2%)

radically resected carcinoma *in situ*·	
E27: Uncontrolled/uncorrectable metabolic disorders, other non-malignant organ diseases, systemic diseases, or cancer-related secondary diseases with the potential to cause a relatively high medical risk and/or survival evaluation uncertainties unsuitable for subject enrollment as judged by the investigator; other circumstances unsuitable for subject enrollment as judged by the investigator	4 (1%)
E32: Pregnant or breastfeeding female patients	4 (1%)
E11: Uncontrolled hypertension (systolic greater than 140 mmHg or diastolic greater than 90 mmHg) after the optimal medical treatment, history of hypertensive crisis or hypertensive encephalopathy	3 (1%)
E17: History or current experience of pulmonary fibrosis and such lung diseases as interstitial pneumonia, pneumoconiosis, drug-related pneumonia, and severely impaired lung function·	3 (1%)
E18: Active tuberculosis (TB), currently receiving anti-tuberculosis treatment or received such treatment within 1 year before the first dose	3 (1%)
E29: Previous receipt of any antibody treatment involving anti-PD-1, anti-PD-L1/L2, or anti-CTLA4 or other immunotherapies· Previous receipt of anti-VEGF and/or VEGFR, RAF, MEK, PDGFR, and FGFR targeted therapy	2 (<1%)
E7: Presence of any life-threatening bleeding events within the past 3 months, including the need for transfusion, surgery or local treatment, and continuous medication therapy	3 (1%)
E8: Any arterial/venous thromboembolic events within 6 months, including myocardial infarction, unstable angina, cerebrovascular accident or transient cerebral ischemic attack, pulmonary embolism, deep vein thrombosis, or any other history of serious thromboembolism	3 (1%)
E14: Serious hemorrhagic tendency or coagulopathy, or currently receiving thrombolytic therapy·	2 (<1%)
E20: Active or poorly clinically controlled serious infections· Severe infections within 4 weeks before the first dose, including but not limited to hospitalization caused by infection, bacteremia, or severe pneumonia complication	2 (<1%)
E25: Receipt of local treatment for liver cancer within 4 weeks before the first dose·	2 (<1%)
E13: Symptomatic congestive cardiac failure (NYHA Class II–IV)· Symptomatic or poorly controlled arrhythmia· History of congenital long QT syndrome or corrected Qtc > 500 ms (calculated using Fridericia formula) during screening	1 (<1%)
E16: Receipt of radiotherapy within 3 weeks before the first dose·	1 (<1%)
E21: Presence of active autoimmune diseases requiring systemic treatment (e·g·, use of disease-modifying drugs, corticosteroids, or immunosuppressants) within 2 years before the first dose	1 (<1%)

E26: Receipt of systemic treatment with traditional Chinese medicines with cancer indications or immunomodulators (including thymosin, interferon, and interleukin, barring local use for controlling pleural fluid or ascites) within 2 weeks before the first dose·	1 (<1%)
E3: Symptomatic pleural effusion, ascites, and pericardial effusion that require drainage	1 (<1%)

***Table S4*: Progression-free survival, time to response and time to progression in phase 3**

Endpoints	Treatment group	Events	Stratified HR (95% CI)	P-value	Median, months (95% CI)
Progression-free survival*					
Investigator per RECIST version 1.1	Sintilimab–bevacizumab biosimilar (n=380)	238 (63%)	0·61 (0·49–0·75)	<0·0001	5·5 (4·2–6·3)
	Sorafenib (n=191)	141 (74%)			3·0 (2·8–4·1)
IRRC per mRECIST	Sintilimab–bevacizumab biosimilar (n=380)	248 (65%)	0·56 (0·45–0·69)	<0·0001	4·5 (4·2–5·7)
	Sorafenib (n=191)	142 (74%)			2·8 (2·7–3·1)
Time to response#					
IRRC per RECIST version 1.1	Sintilimab–bevacizumab biosimilar (n=365)	75 (21%)	—	—	2·8 (2·4–3·0)
	Sorafenib (n=172)	7 (4%)			1·4 (1·2–4·2)
Investigator per RECIST version 1.1	Sintilimab–bevacizumab biosimilar (n=367)	72 (20%)	—	—	2·8 (2·7–3·0)
	Sorafenib (n=173)	5 (3%)			2·8 (1·2–5·1)
IRRC per mRECIST	Sintilimab–bevacizumab biosimilar (n=366)	89 (24%)	—	—	2·4 (1·5–2·8)
	Sorafenib (n=172)	13 (8%)			1·5 (1·4–4·2)
Time to progression*					
IRRC per RECIST version 1.1	Sintilimab–bevacizumab biosimilar (n=380)	240 (63%)	0·57 (0·45–0·70)	<0·0001	5·2 (4·2–5·8)
	Sorafenib (n=191)	140 (73%)			2·8 (2·7–3·2)
Investigator per RECIST version 1.1	Sintilimab–bevacizumab biosimilar (n=380)	231 (61%)	0·61 (0·49–0·76)	<0·0001	5·5 (4·2–6·5)
	Sorafenib (n=191)	137 (72%)			3·1 (2·8–4·1)
IRRC per mRECIST	Sintilimab–bevacizumab biosimilar (n=380)	244 (64%)	0·56 (0·45–0·70)	<0·0001	4·6 (4·2–5·7)
	Sorafenib (n=191)	140 (73%)			2·8 (2·7–3·1)

Data are n (%) or median (95% CI) unless stated otherwise. IRRC=independent radiological review committee; RECIST=Response Evaluation Criteria in Solid Tumours,

version 1·1; mRECIST=hepatocellular carcinoma-specific modified RECIST; *, intention-to-treat population; #, evaluble patients who had measurable disease according to Response Evaluation Criteria in Solid Tumours, version 1.1 (RECIST version 1.1) by independent radiological review committee (IRRC) and investigator, and according to hepatocellular carcinoma-specific modified RECIST (mRECIST) by IRRC.

***Table S5*: Summary of events in the safety population in phase 3**

	Sintilimab–bevacizumab biosimilar (n=380)	**Sorafenib (n=185)**
TEAEs (all Grades)	376 (99%)	181 (98%)
Treatment-related AEs (all Grades)	337 (89%)	173 (94%)
Grade 3-4 TEAE	207 (54%)	87 (47%)
Treatment-related Grade 3-4 AE	128 (34%)	66 (36%)
Serious AE	123 (32%)	36 (19%)
Treatment-related serious AE	65 (17%)	19 (10%)
TEAE leading to death	10 (3%)	6 (3%)
Treatment-related AEs leading to death	6 (2%)	2 (1%)
TEAE leading to interruption of any study treatment	188 (49%)	75 (41%)
TEAE leading to dose modification of sorafenib	NA	66 (36%)
TEAE leading to discontinuation of any study treatment	52 (14%)	11 (6%)

Data are n (%). AE=adverse event; TEAE=treatment-emergent adverse event.

***Table S6*: Treatment-emergent adverse events leading to discontinuation of any study treatment in phase 3**

MedDRA Preferred Term	**Sintilimab–bevacizumab biosimilar (n=380)**	**Sorafenib (n=185)**
Any treatment-emergent adverse events leading to discontinuation	52 (14%)	11 (6%)
Gastrointestinal disease	19 (5%)	6 (3%)
Upper gastrointestinal haemorrhage	7 (2%)	0

Abdominal pain	2 (<1%)	0
Esophageal variceal	2 (<1%)	0
Esophageal variceal bleeding	2 (<1%)	0
Vomiting	1 (<1%)	1 (<1%)
Nausea	1 (<1%)	1 (<1%)
Chronic gastritis	1 (<1%)	0
Ulcerative colitis	1 (<1%)	0
Colonitis	1 (<1%)	0
Volvulus	1 (<1%)	0
Enteritis	1 (<1%)	0
Gastrointestinal haemorrhage	1 (<1%)	2 (1%)
Abdominal wall variceal	1 (<1%)	0
Haematemesis	0	1 (<1%)
Gastrorrhagia	0	1 (<1%)
Ascites	0	1 (<1%)
Laboratory examination	15 (4%)	0
Decreased platelet count	6 (2%)	0
Elevated bilirubin	5 (1%)	0
Aspartate aminotransferase increased	2 (<1%)	0
Amylase increased	2 (<1%)	0
Blood alkaline phosphatase increased	2 (<1%)	0
C-reactive protein increased	1 (<1%)	0
Gamma-glutamyltransferase increased	1 (<1%)	0

Weight decreased	1 (<1%)	0
Urinary urobilinogen increased	1 (<1%)	0
Urine bilirubin increased	1 (<1%)	0
Detection of urinary protein	1 (<1%)	0
Presence of urine ket	1 (<1%)	0
Conjugated bilirubin increased	1 (<1%)	0
Serum lactate dehydrogenase increased	1 (<1%)	0
Serum creatin phosphokinase increased	1 (<1%)	0
Blood cholinesterase decreased	1 (<1%)	0
Disease of hepatobiliary system	10 (3%)	3 (2%)
Abnormal liver function	5 (1%)	2 (1%)
Hepatic failure	2 (<1%)	0
Immune mediated hepatitis	1 (<1%)	0
Liver damage	1 (<1%)	0
Hepatorenal syndrome	1 (<1%)	0
Jaundice	0	1 (<1%)
Metabolic and nutritional diseases	3 (<1%)	0
Hypochloridemia	1 (<1%)	0
Hypoalbuminemia	1 (<1%)	0
Hypocalcemia	1 (<1%)	0
Hyponatremia	1 (<1%)	0
Hyperuricemia	1 (<1%)	0
Hyperglycemia	1 (<1%)	0

Systemic disease and reactions at drug administration site	3 (<1%)	1 (<1%)
Death	2 (<1%)	0
Pyrexia	1 (<1%)	0
Fatigue	0	1 (<1%)
Disease of respiratory system, chest and mediastinum	3 (<1%)	1 (<1%)
Hemoptysis	1 (<1%)	0
Pulmonary embolism	1 (<1%)	0
Interstitial pulmonary disease	1 (<1%)	0
Epistaxis	0	1 (<1%)
Disease of the kidney and urinary system	3 (<1%)	0
Proteinuria	2 (<1%)	0
Nephrotic syndrome	1 (<1%)	0
Diseases of the nervous system	2 (<1%)	1 (<1%)
Cerebral infarction	1 (<1%)	0
Hepatic encephalopathy	1 (<1%)	0
Cerebral hemorrhage	0	1 (<1%)
Infection and infectious diseases	2 (<1%)	0
Infectious pneumonia	1 (<1%)	0
Genital fungal infection	1 (<1%)	0
Disease of the Hemic and Lymphatic System	2 (<1%)	0
Anaemia	2 (<1%)	0
Endocrine system disease	1 (<1%)	0
Hypothyroidism	1 (<1%)	0

Cardiac disease	1 (<1%)	0
Second-degree atrioventricular block	1 (<1%)	0
Immune mediated myocarditis	1 (<1%)	0
Disease of the genital system and mammary gland	1 (<1%)	0
Balanoposthitis	1 (<1%)	0
Cutaneous and subcutaneous disease	1 (<1%)	0
Cutaneous reaction	1 (<1%)	0
Benign, malignant and unknown tumours (including cystic and polypoid tumour)	1 (<1%)	0
Tumour rupture	1 (<1%)	0
Disease of musculoskeletal and connective tissue	0	1 (<1%)
Backache	0	1 (<1%)
Psychosis disease	0	1 (<1%)
Anorexia nervosa	0	1 (<1%)

Data are n (%).

***Table S7*: Treatment-related adverse events in any treatment group in phase 3**

MedDRA Preferred Term	**Sintilimab–bevacizumab biosimilar (n=380)**				**Sorafenib (n=185)**			
	Grade 1–2	**Grade 3**	**Grade 4**	**Grade 5**	**Grade 1–2**	**Grade 3**	**Grade 4**	**Grade 5**
Any treatment-related adverse event	206 (54%)	117 (31%)	8 (2%)	6 (2%)	105 (57%)	62 (34%)	4 (2%)	2 (1%)
Proteinuria	136 (36%)	18 (5%)	0	0	25 (14%)	2 (1%)	0	0
Aspartate aminotransferase increased	77 (20%)	3 (<1%)	1 (<1%)	0	37 (20%)	5 (3%)	0	0
Platelet count decreased	77 (20%)	19 (5%)	3 (<1%)	0	38 (21%)	1 (<1%)	1 (<1%)	0
Hypertension	60 (16%)	48 (13%)	0	0	18 (10%)	9 (5%)	0	0

Alanine aminotransferase increased	59 (16%)	3 (<1%)	1 (<1%)	0	33 (18%)	2 (1%)	0	0
Blood bilirubin increased	52 (14%)	5 (1%)	2 (<1%)	0	40 (22%)	5 (3%)	0	0
Hypothyroidism	47 (12%)	0	0	0	11 (6%)	0	0	0
White blood cell count decreased	43 (11%)	4 (1%)	0	0	20 (11%)	2 (1%)	1 (<1%)	0
Blood thyroid-stimulating hormone increased	38 (10%)	0	0	0	15 (8%)	0	0	0
Pyrexia	32 (8%)	0	0	0	6 (3)	0	0	0
Asthenia	28 (7%)	1 (<1%)	0	0	20 (11%)	0	0	0
Diarrhoea	22 (6%)	2 (<1%)	0	0	60 (32%)	3 (2%)	0	0
Rash	22 (6%)	1 (<1%)	0	0	20 (11%)	2 (1%)	0	0
Palmar-plantar erythrodysaesthesia syndrome	1 (<1%)	0	0	0	80 (43%)	22 (12%)	0	0
Alopecia	0	0	0	0	38 (21%)	0	0	0
Detection of proteinuria	28 (7%)	2 (<1%)	0	0	13 (7%)	2 (1%)	0	0
Elevation of blood pressure	23 (6%)	8 (2%)	0	0	6 (3%)	3 (2%)	0	0
Gamma-glutamyltransferase	22 (6%)	8 (2%)	0	0	16 (9%)	1 (<1%)	0	0
Neutrophil count decreased	18 (5%)	4 (1%)	0	0	11 (6%)	2 (1%)	1 (<1%)	0
Upper gastrointestinal haemorrhage	2 (<1%)	6 (2%)	1 (<1%)	1 (<1%)	2 (1%)	1 (<1%)	1 (<1%)	0
Hypokalemia	11 (3%)	1 (<1%)	0	0	4 (2%)	2 (1%)	0	0
Abnormal liver function	8 (2%)	4 (1%)	1 (<1%)	1 (<1%)	3 (2%)	3 (2%)	1 (<1%)	0
Jaundice	0	1 (<1%)	0	0	0	2 (1%)	0	0

Data are n (%). Grade 1-2 treatment-related adverse events occurring in ≥10% of patients in either group, and grade 3, 4, 5 treatment-related adverse events occurring in ≥1% of patients in either group are listed.

***Table S8*: Serious adverse events in any treatment group in phase 3**

MedDRA Preferred Term	Sintilimab–bevacizumab (n=380)	Sorafenib (n=185)
Serious adverse events of any grade	123 (32%)	36 (19%)
Death	10 (3%)	6 (3%)
Life-threaten	8 (2%)	2 (1%)
Hospitalization and prolongation of hospitalization	115 (30%)	32 (17%)
Birth defect	1 (<1%)	0
Significant adverse event	3 (<1%)	0

Data are n (%).

***Table S9*: Adverse events of special interest in any treatment group in phase 3**

MedDRA Preferred Term	Sintilimab–bevacizumab biosimilar (n=380)				Sorafenib (n=185)			
	Grade 1-2	Grade 3	Grade 4	Grade 5	Grade 1-2	Grade 3	Grade 4	Grade 5
Proteinuria	140 (37%)	20 (5%)	0	0	31 (17%)	3 (2%)	0	0
Platelet count decreased	124 (33%)	26 (7%)	5 (1%)	0	50 (27%)	4 (2%)	1 (<1%)	0
Hypertension	66 (17%)	55 (14%)	0	0	19 (10%)	11 (6%)	0	0
Hypothyroidism	53 (14%)	0	0	0	14 (8%)	0	0	0
Protein urine present	34 (9%)	2 (<1%)	0	0	15 (8%)	2 (1%)	0	0
Blood urine present	29 (8%)	1 (<1%)	0	0	12 (6%)	0	0	0
Blood pressure increased	25 (7%)	10 (3%)	0	0	6 (3%)	3 (2%)	0	0
Hyperthyroidism	25 (7%)	0	0	0	3 (2%)	0	0	0
Amylase increased	24 (6%)	3 (<1%)	0	0	5 (3%)	0	0	0
Gingival bleeding	22 (6%)	0	0	0	4 (2%)	0	0	0
Epistaxis	13 (3%)	0	0	0	2 (1%)	1 (<1%)	0	0

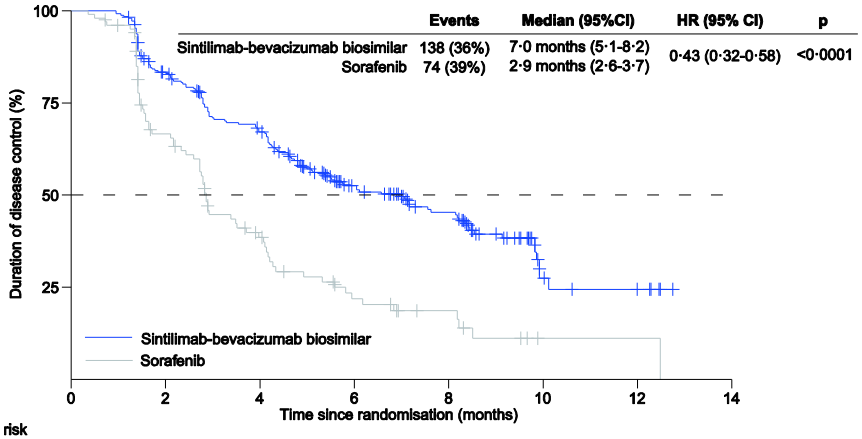
Haematuria	13 (3%)	0	0	0	8 (4%)	0	0	0
Hepatitis B DNA increased	13 (3%)	0	0	0	1 (1%)	0	0	0
Haemoptysis	7 (2%)	0	0	0	1 (1%)	1 (1%)	0	0
Dermatitis	6 (2%)	0	0	0	0	0	0	0
Hepatitis B reactivation	4 (1%)	1 (<1%)	0	0	2 (1%)	0	0	0
Skin exfoliation	3 (1%)	0	0	0	8 (4%)	0	0	0
Upper gastrointestinal haemorrhage	3 (1%)	7 (2%)	1 (<1%)	1 (<1%)	2 (1%)	1 (<1%)	1 (<1%)	0
Gastrointestinal haemorrhage	2 (1%)	3 (<1%)	0	1 (<1%)	2 (1%)	2 (1%)	0	1 (<1%)

Data are n (%). Grade 1-2 adverse events of special interest occurring in ≥1% of patients in either group are listed.

***Table S10*: Sintilimab ADA results in the study in phase 3**

Visit	**Baseline**		
	Negative	**Positive**	**Total**
Baseline, n (%)	325 (86)	53 (14)	378 (100)
Post-baseline			
n [a]	313	50	363
Treatment-emergent ADA positive, n (%) [b]	12 (4)	0	12 (3)

[a], number of ADA-evaluable patients with baseline and at least one post-baseline ADA measurements was used as denominator of treatment-emergent ADA positive. [b], included the patients with ADA-negative at baseline and tested as ADA-positive in at least one visit after baseline; or ADA-positive at baseline, and the ADA titer was 4 folds or more than that at baseline. ADA=antidrug antibody.



Number at risk (number censored)								
Sintilimab-bevacizumab biosimilar	256 (0)	205 (9)	159 (16)	92 (51)	61 (71)	10 (109)	6 (112)	0 (118)
Sorafenib	102 (0)	59 (11)	30 (16)	14 (20)	8 (24)	1 (28)	1 (28)	0 (28)

Figure S1 Kaplan–Meier curve of the duration of disease control, including patients who achieved a complete response, partial response and stable disease, as assessed by IRRC per RECIST version 1.1 in the intention-to-treat population. Vertical bars indicate censored events (death or progression). CI=Confidence interval; HR=hazard ratio; IRRC=independent radiological review committee; RECIST=Response Evaluation Criteria in Solid Tumours.

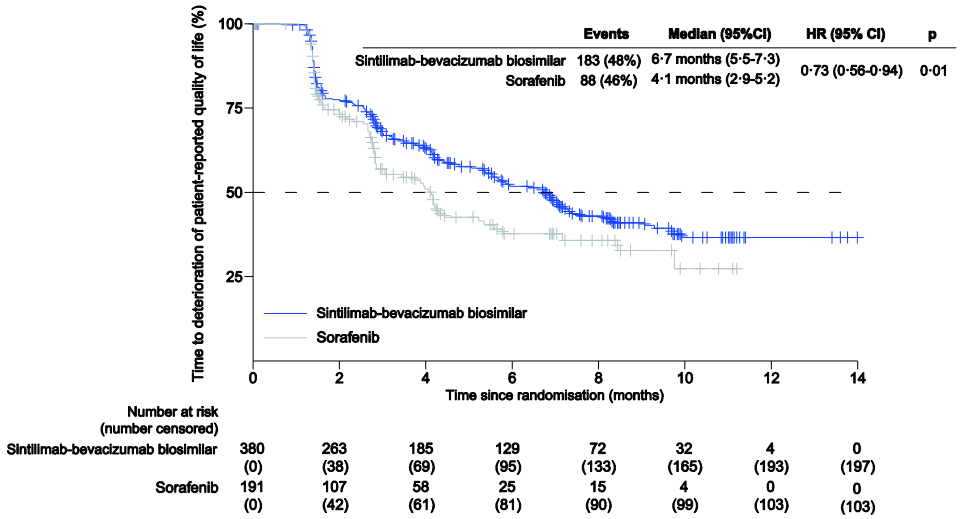


Figure S2 Kaplan–Meier curve of time to deterioration of global health status (QLQ-C30) in the intention-to-treat population. Vertical bars indicate censored events (deterioration). CI=Confidence interval; HR=Hazard ratio.

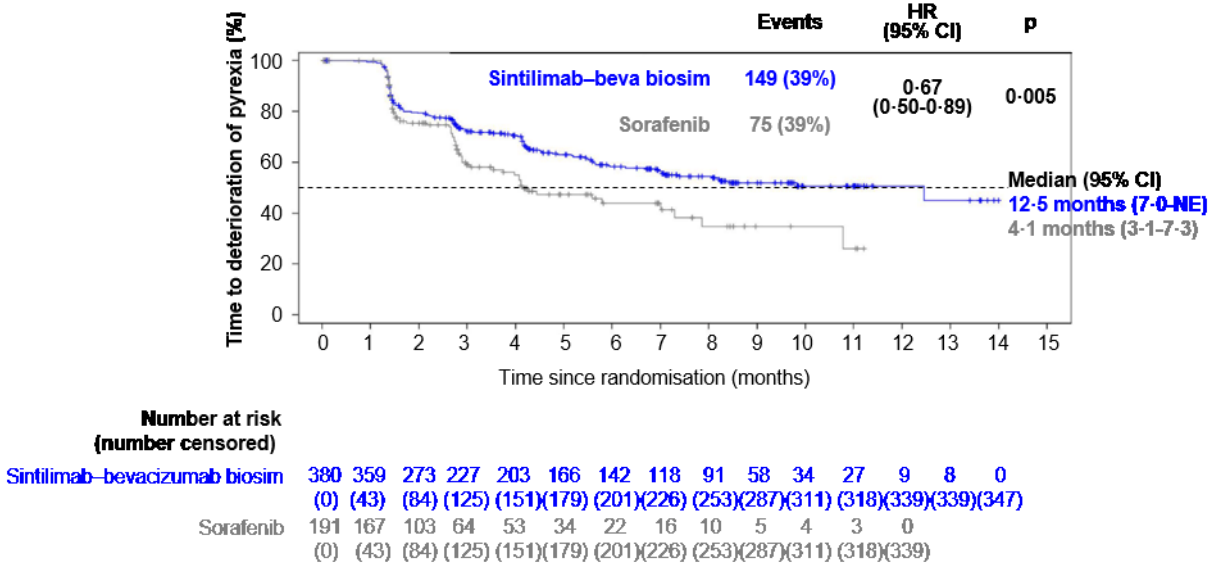


Figure S3 Kaplan–Meier curve of time to deterioration of pyrexia (QLQ-HCC18) in the intention-to-treat population. Vertical bars indicate censored events (deterioration). CI=Confidence interval; HR=Hazard ratio.

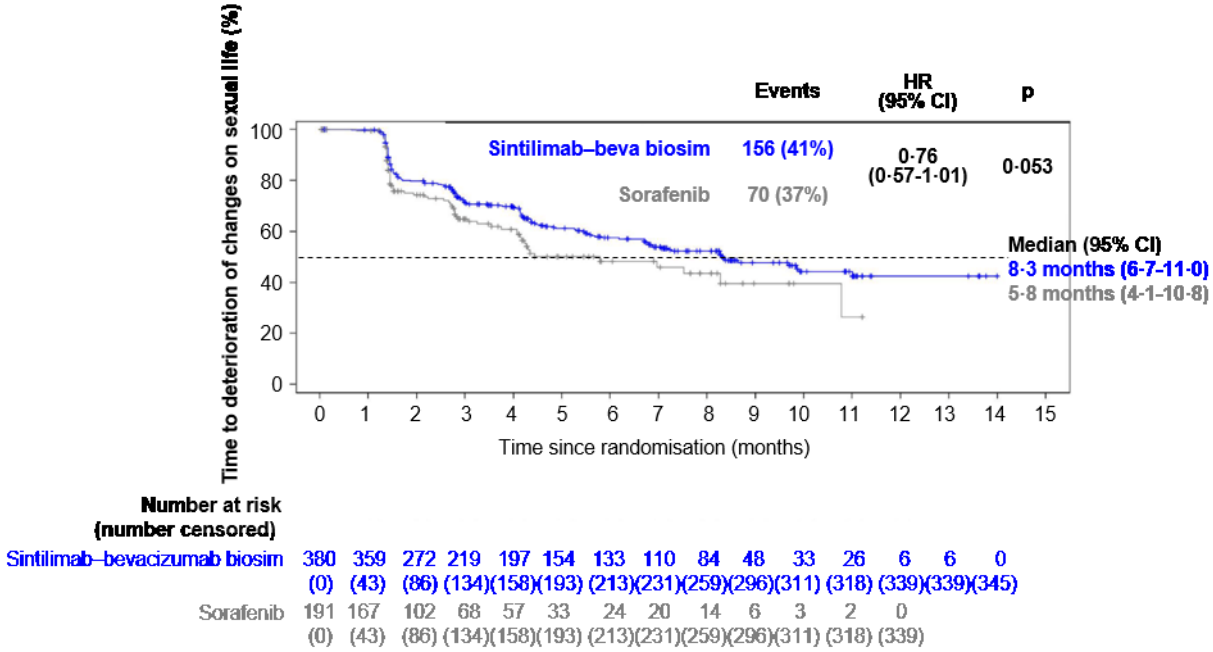


Figure S4 Kaplan–Meier curve of time to deterioration of changes on sexual life (QLQ-HCC18) in the intention-to-treat population. Vertical bars indicate censored events (deterioration)· CI=Confidence interval; HR=Hazard ratio.

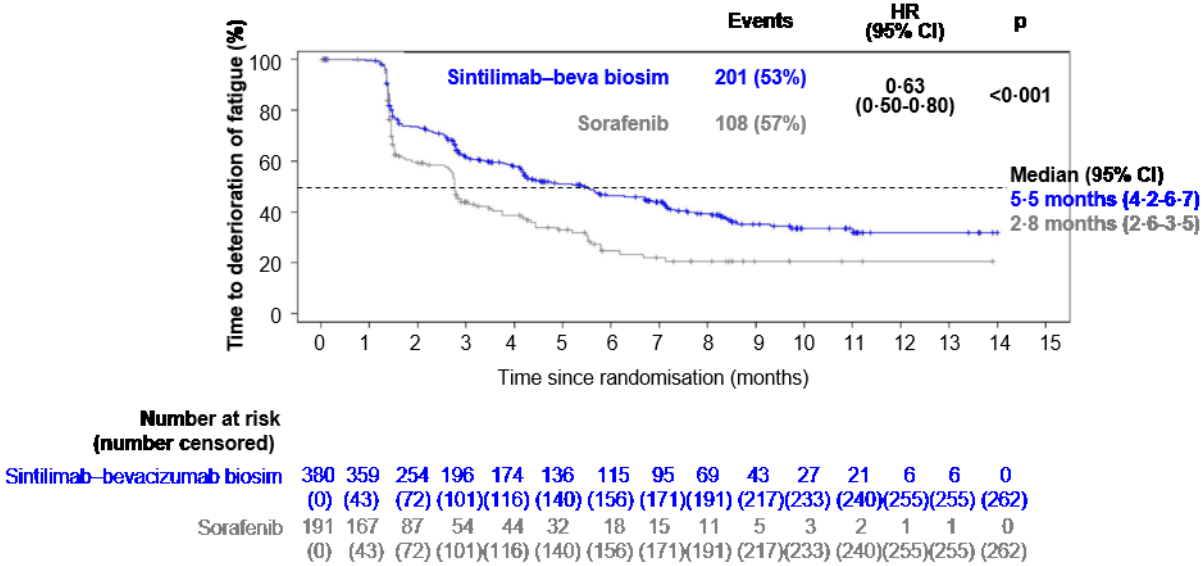
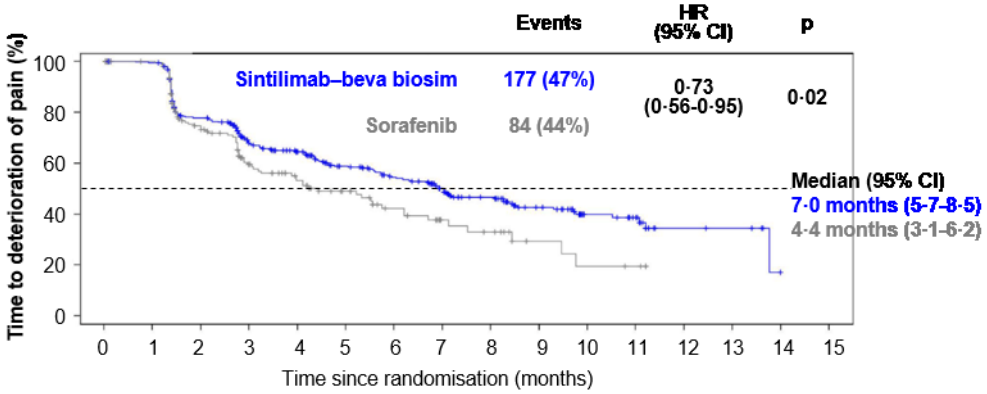


Figure S5 Kaplan–Meier curve of time to deterioration of fatigue (QLQ-HCC18) in the intention-to-treat population. Vertical bars indicate censored events (deterioration)· CI=Confidence interval; HR=Hazard ratio.



Number at risk (number censored)															
Sintilimab–bevacizumab biosim	380	359	265	210	185	152	131	103	81	55	29	22	6	5	0
	(0)	(43)	(82)	(122)	(144)	(171)	(189)	(216)	(232)	(258)	(281)	(288)	(305)	(306)	(310)
Sorafenib	191	167	106	69	54	41	28	16	13	6	4	3	0		
	(0)	(43)	(82)	(122)	(144)	(171)	(189)	(216)	(232)	(258)	(281)	(288)	(305)		

Figure S6 Kaplan–Meier curve of time to deterioration of pain (QLQ-HCC18) in the intention-to-treat population. Vertical bars indicate censored events (deterioration)· CI=Confidence interval; HR=Hazard ratio.

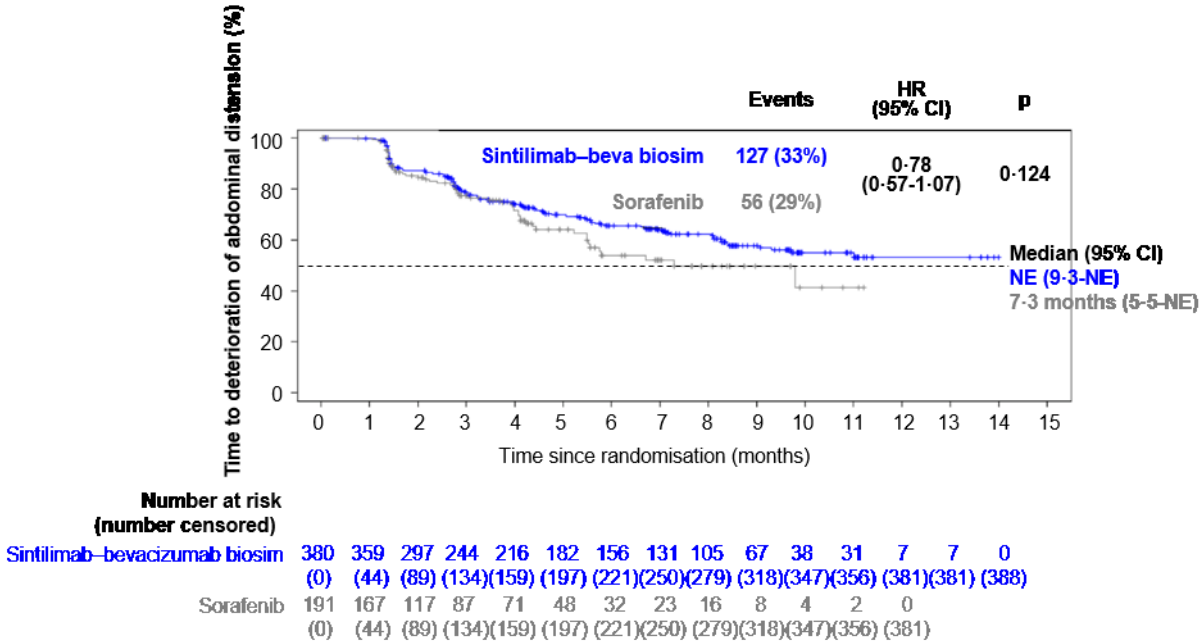


Figure S7 Kaplan–Meier curve of time to deterioration of abdominal distension (QLQ-HCC18) in the intention-to-treat population. Vertical bars indicate censored events (deterioration). CI=Confidence interval; HR=Hazard ratio.·

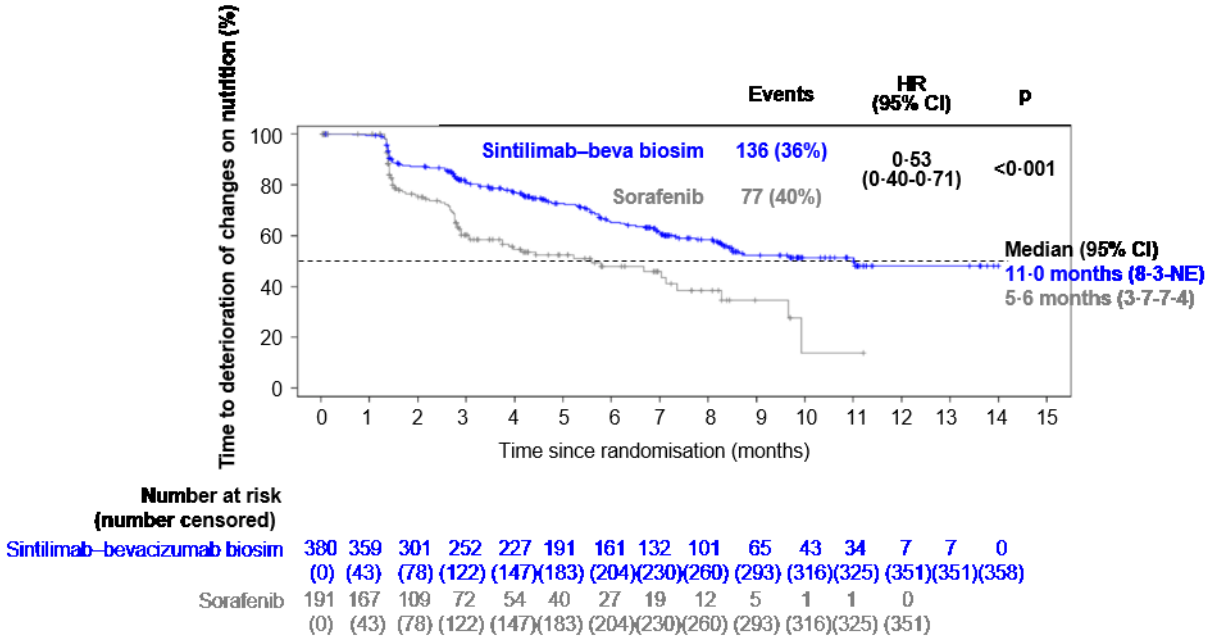


Figure S8 Kaplan–Meier curve of time to deterioration of changes on nutrition (QLQ-HCC18) in the intention-to-treat population. Vertical bars indicate censored events (deterioration)· CI=Confidence interval; HR=Hazard ratio.

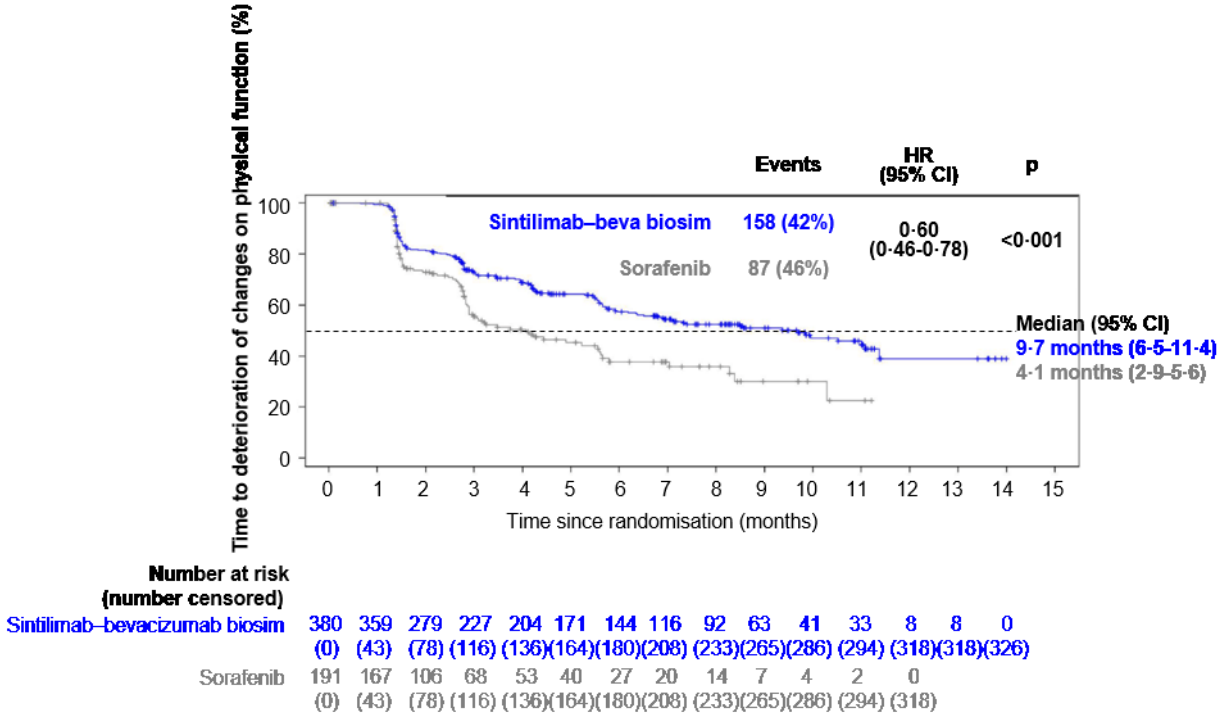


Figure S9 Kaplan–Meier curve of time to deterioration of changes on physical function (QLQ-HCC18) in the intention-to-treat population. Vertical bars indicate censored events (deterioration). CI=Confidence interval; HR=Hazard ratio.

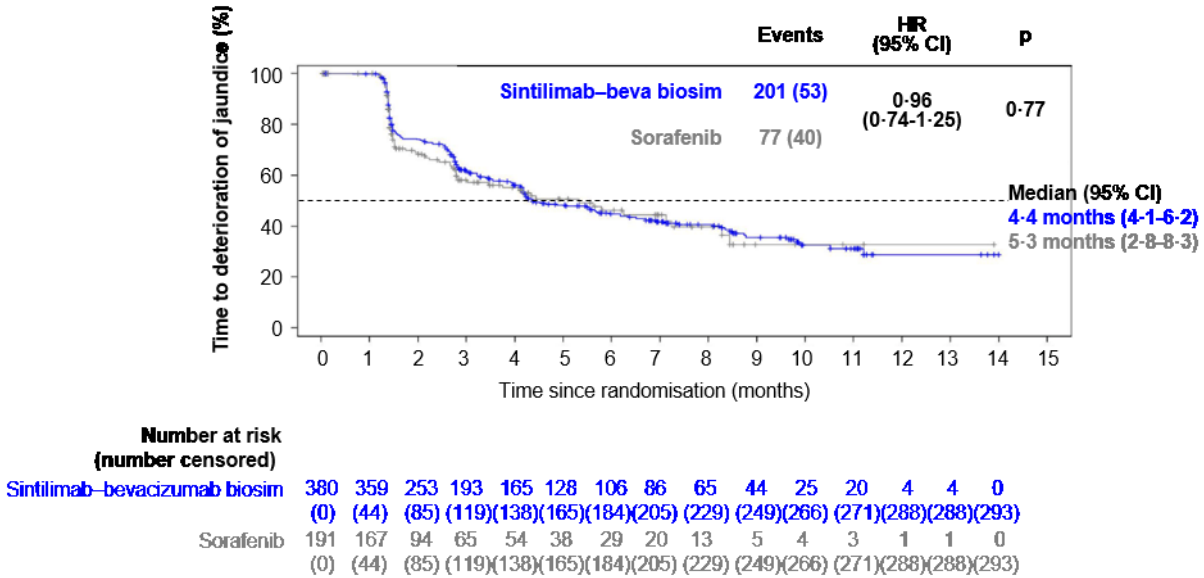


Figure S10 Kaplan–Meier curve of time to deterioration of jaundice (QLQ-HCC18)in the intention-to-treat population. Vertical bars indicate censored events (deterioration). CI=Confidence interval; HR=Hazard ratio.

Table S11: Treatment-emergent adverse events in any treatment group in phase 3

MedDRA Preferred Term	Sintilimab–bevacizumab biosimilar (n=380)			Sorafenib (n=185)		
	Any Grade	Grade 3–4	Grade 5	Any Grade	Grade 3–4	Grade 5
Proteinuria	160 (42%)	20 (5%)	0	34 (18%)	3 (2%)	0
Platelet count decreased	155 (41%)	31 (8%)	0	55 (30%)	5 (3%)	0
Increased aspartate aminotransferase	135 (36%)	7 (2%)	0	75 (41%)	10 (5%)	0
Hypertension	121 (32%)	55 (15%)	0	30 (16%)	11 (6%)	0
Increased blood bilirubin	112 (30%)	19 (5%)	0	66 (36%)	6 (3%)	0
Increased alanine aminotransferase	99 (26%)	5 (1%)	0	52 (28%)	5 (3%)	0
Decreased white blood cell count	77 (20%)	6 (2%)	0	30 (16%)	4 (2%)	0
Pyrexia	66 (17%)	1 (<1%)	0	20 (11%)	1 (1%)	0
Increased Gamma-glutamyltransferase	63 (17%)	19 (5%)	0	29 (16%)	3 (2%)	0
Hypoalbuminaemia	63 (17%)	0	0	18 (10%)	1 (1%)	0
Asthenia	61 (16%)	3 (<1%)	0	36 (20%)	2 (1%)	0
Abdominal pain	57 (15%)	4 (1%)	0	25 (14%)	0	0
Decreased weight	57 (15%)	2 (<1%)	0	31 (17%)	0	0
Diarrhoea	55 (15%)	6 (2%)	0	77 (42%)	5 (3%)	0
Hypothyroidism	53 (14%)	0	0	14 (8%)	0	0
Anaemia	51 (13%)	10 (3%)	0	15 (8%)	2 (1%)	0
Increased blood alkaline phosphatase	47 (12%)	5 (1%)	0	14 (8%)	1 (1%)	0
Abdominal distension	44 (12%)	2 (<1%)	0	26 (14%)	1 (1%)	0
Decreased appetite	43 (11%)	2 (<1%)	0	22 (12%)	0	0
Increased blood thyroid-stimulating hormone	43 (11%)	0	0	17 (9%)	0	0
Increased bilirubin conjugated	40 (11%)	10 (3%)	0	20 (11%)	2 (1%)	0
Decreased neutrophil count	39 (10%)	8 (2%)	0	18 (10%)	3 (2%)	0

Pruritus	39 (10%)	0	0	10 (5%)	0	0
Elevation of blood pressure	35 (9%)	10 (3%)	0	9 (5%)	3 (2%)	0
Hyponatremia	31 (8%)	7 (2%)	0	10 (5%)	3 (2%)	0
Hypokalemia	28 (7%)	7 (2%)	0	15 (8%)	2 (1%)	0
Nausea	28 (7%)	2 (<1%)	0	19 (10%)	0	0
Upper respiratory infection	27 (7%)	5 (1%)	0	10 (5%)	3 (2%)	0
Lymphocyte count decreased	26 (7%)	7 (2%)	0	14 (8%)	2 (1%)	0
Rash	26 (7%)	1 (<1%)	0	23 (12%)	2 (1%)	0
Vomiting	26 (7%)	1 (<1%)	0	19 (10%)	0	0
Abnormal liver function	24 (6%)	11 (3%)	2 (<1%)	10 (5%)	5 (3%)	0
Backache	22 (6%)	4 (1%)	0	7 (4%)	1 (<1%)	0
Increased blood lactate dehydrogenase	20 (5%)	0	0	25 (14%)	0	0
Ascites	15 (4%)	7 (2%)	0	5 (3%)	1 (<1%)	0
Infectious pneumonia	15 (4%)	5 (1%)	0	3 (2%)	0	0
Upper gastrointestinal haemorrhage	12 (3%)	8 (2%)	1 (<1%)	4 (2%)	2 (1%)	0
Hyperglycemia	12 (3%)	4 (1%)	0	1 (<1%)	0	0
Hepatic failure	6 (2%)	3 (<1%)	2 (<1%)	3 (2%)	0	2 (1%)
Gastrointestinal haemorrhage	6 (2%)	3 (<1%)	1 (<1%)	5 (3%)	2 (1%)	1 (<1%)
Jaundice	3 (<1%)	1 (<1%)	0	2 (1%)	2 (1%)	0
Unknown death	2 (<1%)	0	2 (<1%)	2 (1%)	0	2 (1%)
Palmar-plantar erythrodysaesthesia syndrome	1 (<1%)	0	0	102 (55%)	22 (12%)	0
Alopecia	0	0	0	38 (21%)	0	0
Hepatic abscess	0	0	0	2 (1%)	2 (1%)	0

Data are n (%). Any grade treatment-emergent adverse events occurring in ≥10% of patients in either group; and grade 3, 4, 5 treatment-emergent adverse events occurring in ≥1% of patients in either group are listed, the safety population included all patients who received at least one dose of study treatment. The severity of adverse events was graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 5.0.

Table S12: Response and duration of response in the full analysis set*

Patiets with at least 1 target lesion at the baseline of corresponding assessment*	IRRC per RECIST version 1.1			Investigator per RECIST version 1.1			mRECIST		
	Sintilimab–bevacizumab biosimilar (n=376)	Sorafenib (n=183)	Difference *P* value [a]	Sintilimab–bevacizumab biosimilar (n=380)	Sorafenib (n=185)	Difference *P* value [a]	Sintilimab–bevacizumab biosimilar (n=377)	Sorafenib (n=183)	Difference *P* value [a]
Confirmed objective response[†]	72 (19%, 15–23)	7 (4%, 2–8)	15% (10–20) <0·0001	72 (19%, 15–23)	5 (3%, 1–6)	16% (12–21) <0·0001	87 (23%, 19–28)	13 (7%, 4–12)	16% (10–22) <0·0001
Complete response	0	0		2 (<1%)	0		2 (<1%)	0	
Partial response	72 (19%)	7 (4%)		70 (18%)	5 (3%)		85 (23%)	13 (7%)	
Stable disease	192 (51%)	103 (56%)		195 (51%)	111 (60%)		181 (48%)	96 (52%)	
Progressive disease	98 (26%)	57 (31%)		98 (26%)	51 (28%)		95 (25%)	58 (32%)	
Not evaluable [b]	1 (<1%)	1 (<1%)		0	3 (2%)		1 (<1%)	1 (<1%)	
Not done [c]	13 (3%)	15 (8%)		15 (4%)	15 (8%)		13 (3%)	15 (8%)	
Disease control[¶]	264 (70%, 65–75)	110 (60%, 53–67)		267 (70%, 65–75)	116 (63%, 55–70)		268 (71%, 66–76)	109 (60%, 52–67)	
Median duration of response, months (95% CI)	NE (NE–NE)	9·8 (2·8–NE)		NE (NE–NE)	NE (2·5–NE)		NE (8·2–NE)	6·6 (2·6–NE)	

Data are n (%, 95% CI) or n (%), unless stated otherwise. 95% CI was calculated using the Clopper-Pearson method.

[*], Full analysis set, including the subjects who have measurable lesions at baseline, and who receive at least one dose of the study drug. Patients who had measurable disease according to Response Evaluation Criteria in Solid Tumours, version 1.1 (RECIST version 1.1) by independent radiological review committee (IRRC) and investigator, and according to hepatocellular carcinoma-specific modified RECIST (mRECIST) by IRRC.

[†], confirmed objective response was defined as a response (complete or partial) confirmed by two consecutive tumour assessments with at least a 28-day interval; [a], between-group difference on the confirmed response of the patients; *P* value was calculated using the stratified Miettinen & Nurminen method; Randomisation was performed using an interactive web response system with the stratification factors included macrovascular invasion and/or extrahepatic metastasis (yes *vs* no), alpha-fetoprotein level (< 400 *vs* ≥400 ng/mL), and ECOG PS (0 *vs* 1).

[¶], disease control included the complete response, PR, and stable disease.

[b], inadequate data for evaluation; [c], the imaging assessment was not performed; NE, not evaluable.

***Table S13*: Subsequent therapies in any study treatment in the Intention-to Treat Population**

	Sintilimab–bevacizumab biosimilar (n=380)	**Sorafenib (n=191)**
Systemic therapy other than PD-(L)1 inhibitors	83 (22%)	76 (40%)
Local regional therapy	39 (10%)	25 (13%)
Radiation Therapy	9 (2%)	3 (2%)
Surgery	8 (2%)	3 (2%)
PD-(L)1 inhibitors	9 (2%)	40 (21%)

Data are n (%).

***Table S14*: Patients of quality of life assessment in any study treatment**

Visits	Sintilimab–bevacizumab biosimilar (n=380)	Sorafenib (n=191)
Baseline	380 (100%)	185 (97%)
Week 6	341 (90%)	153 (80%)
Week 12	285 (75%)	109 (57%)
Week 18	248 (65%)	74 (39%)
Week 24	200 (53%)	52 (27%)
Week 30	184 (48%)	38 (20%)
Week 36	128 (34%)	23 (12%)
Week 42	86 (23%)	14 (7%)
Week 48	53 (14%)	7 (4%)
Week 60	9 (2%)	1 (1%)
Safety Visit	80 (21%)	46 (24%)
The last visit	360 (95%)	168 (88%)

Data are n (%).

Phase 2 Safety run-in part results

Table S15: Baseline characteristics of patients in phase 2 part

	Sintilimab–bevacizumab biosimilar (n=24)
Age, years	
Median (range)	53 (43–71)
Sex	
Male	20 (83%)
Female	4 (17%)
ECOG PS	
0	11 (46%)
1	13 (54%)
Child–Pugh classification	
A (5–6)	23 (96%)
B (7)	1 (4%)
Barcelona Clinic Liver Cancer stage	
B	5 (21%)
C	19 (79%)
Alpha-fetoprotein	
<400 ng/mL	14 (58%)
≥400 ng/mL	10 (42%)
Extrahepatic metastasis	18 (75%)
Hepatitis B virus infection	24 (100%)
Hepatitis C virus infection	1 (4%)
Prior local therapy for HCC at least once	24 (100%)

Ablation	11 (46%)
Surgery	12 (50%)
Radiation therapy	2 (8%)
TACE	19 (79%)

Data are n (%) or median (range). ECOG PS=Eastern Cooperative Oncology Group Performance Status; TACE=Transcatheter arterial chemoembolization

***Table S16*: Summary of events in the safety population in phase 2**

	Sintilimab–bevacizumab biosimilar (n =24)
TEAEs (all Grades)	24 (100%)
Treatment-related AEs (all Grades)	22 (92%)
Grade 3-4 TEAE	9 (38%)
Treatment-related Grade 3-4 AE	7 (29%)
Serious AE	7 (29%)
Treatment-related serious AE	6 (25%)
TEAE leading to death	0
Treatment-related AEs leading to death	0
TEAE leading to interruption of any study treatment	11 (46%)
TEAE leading to discontinuation of any study treatment	2 (8%)

Data are n (%). AE=adverse event; TEAE=treatment-emergent adverse event.

***Table S17*: Treatment-emergent adverse events leading to discontinuation of any study treatment in phase 2**

MedDRA Preferred Term	Sintilimab–bevacizumab biosimilar (n=24)
Any treatment-emergent adverse events leading to discontinuation	2 (8%)
Laboratory examination	1 (4%)
Aspartate aminotransferase increased	1 (4%)
Disease of hepatobiliary system	1 (4%)
Immune mediated hepatitis	1 (4%)

Data are n (%).

Table S18: Treatment-related adverse events in phase 2

MedDRA Preferred Term	Sintilimab–bevacizumab biosimilar (n=24)	
	Any grade	Grade 3-4
Any treatment-related adverse event	22 (92%)	7 (29%)
Proteinuria	10 (42%)	3 (13%)
Platelet count decreased	8 (33%)	0
Alanine aminotransferase increased	7 (29%)	0
Aspartate aminotransferase increased	7 (29%)	0
Hypertension	6 (25%)	0
White blood cell count decreased	5 (21%)	1 (4%)
Blood bilirubin increased	5 (21%)	0
Hypothyroidism	5 (21%)	0
Pyrexia	4 (17%)	0
Abnormal liver function	4 (17%)	2 (8%)
Gamma-glutamyltransferase increased	3 (13%)	0

Neutrophil count decreased	3 (13%)	0
Urobilinogen urine increased	3 (13%)	0
Blood urine present	3 (13%)	0
Detection of proteinuria	3 (13%)	0
Elevation of blood pressure	3 (13%)	0
Hyperthyroidism	3 (13%)	1 (4%)
Hypoproteinaemia	3 (13%)	0

Data are n (%). Any grade treatment-related adverse events occurring in ≥10% of patients, and grade 3, 4 treatment-related adverse events are listed.

***Table S19*: Treatment-emergent serious adverse events related to either study drug by SOC and PT in phase 2**

MedDRA Preferred Term	Sintilimab–bevacizumab (N=24)
Serious adverse events related to either study drug	6 (25%)
Hepatobiliary disorders	4 (17%)
Hepatic function abnormal	3 (13%)
Immune-mediated hepatitis	1 (4%)
Endocrine disorders	1 (4%)
Hyperthyroidism	1 (4%)
Respiratory, thoracic and mediastinal disorders	1 (4%)
Immune-mediated pneumonitis	1 (4%)
Gastrointestinal disorders	1 (4%)
Duodenal ulcer	1 (4%)

Data are n (%).

***Table S20*: Adverse events of special interest in phase 2**

MedDRA Preferred Term	Sintilimab–bevacizumab biosimilar (n=24)	
	Any grade	Grade 3-4
Proteinuria	18 (78%)	4 (17%)
Platelet count decreased	12 (50%)	0
Hypertension	11 (46%)	0
Hemorrhage	5 (21%)	0
Hypothyroidism	5 (21%)	0
Hyperthyroidism	4 (17%)	1 (4%)
Skin reaction	3 (13%)	0
Hepatitis	2 (4%)	1 (4%)
Myocarditis	1 (4%)	0
Colitis	1 (4%)	0
Immune-mediated lung disease	1 (4%)	0

Data are n (%).

***Table S21*: Sintilimab ADA results in phase 2**

Visit	Baseline		
	Negative	Positive	Total
Baseline, n (%)	22 (92)	2 (8)	24 (100)
Post-baseline			
n [a]	22	2	24
Treatment-emergent ADA positive, n (%) [b]	0	0	0

[a], number of ADA-evaluable patients with baseline and at least one post-baseline ADA measurements was used as denominator of treatment-emergent ADA positive. [b], included the patients with ADA-negative at baseline and tested as ADA-positive in at least one visit after baseline; or ADA-positive at baseline, and the ADA titer was 4 folds or more than that at baseline. ADA=antidrug antibody.

***Table S22*: Efficacy results in phase 2**

	Sintilimab–bevacizumab biosimilar (n=24)
Confirmed overall tumour response assessed by the investigator per RECIST 1.1	
Complete response (CR)	0
Partial response (PR)	6 (25%)
Stable disease (SD)	14 (58%)
Progressive Disease (PD)	4 (17%)
Not evaluable (NE)	0
Not done (ND)	0
Confirmed ORR (CR + PR)	6 (25%)
95% CI	(10–47)
Confirmed DCR (CR + PR + SD)	20 (83%)
95% CI	(63–95)
Kaplan-Meier estimate of DoR (months)	
Median DoR	12.6
95% CI	(3.2–NE)
Kaplan-Meier estimate of TTP (months)	
Median TTP	8.2
95% CI	(5.5–14.0)
Kaplan-Meier estimate of TTR (months)	
Median TTR	2.9
95% CI	(1.4–6.8)
PFS assessed by the investigator per RECIST 1.1	
Events	16 (67%)

Progressive Disease	15 (63%)
Death	1 (4%)
Kaplan-Meier estimate of PFS (months)	
Median PFS	8.2
95% CI	(5.5–14.0)
PFS rate	
6-month (95% CI)	61% (38–78)
9-month (95% CI)	42% (22–61)
OS	
Events (deaths)	5 (21%)
Censored	19 (79%)
Alive at the time of data cutoff	19 (79%)
Lost to follow-up	0
Withdrawal of informed consent	0
Kaplan-Meier estimate of OS (months)	
Median OS	NR
95% CI	(NR–NR)
Survival rate	
9-month (95% CI)	88% (66–96)
12-month (95% CI)	83% (61–93)

RECIST=Response Evaluation Criteria in Solid Tumours; ORR=Objective Response Rate; CI = confidence interval; DCR=Disease Control Rate; DOR=Duration of Response; TTP=Time to Progression; TTR=Time to Response; PFS=Progression Free Survival; Overall Survival=OS; NE=not evaluable. NR=not reached

CLINICAL STUDY PROTOCOL

Study Title:	A Randomized, Open-Labeled, Multi-Center Study on Efficacy and Safety of IBI308 in Combination with IBI305 vs. Sorafenib as First-Line Treatment for Advanced Hepatocellular Carcinoma (ORIENT-32)
Protocol No.:	CIBI338B301
Version Date/No.:	Apr. 20, 2020/Version 3.0
Product:	Recombinant Fully Human Anti-PD-1 Monoclonal Antibody (IBI308) Recombinant Anti-VEGF Humanized Monoclonal Antibody (IBI305)
Study Phase:	Phases II/III
Sponsor:	Innovent Biologics (Suzhou) Co., Ltd. No. 168 Dongping Street, Suzhou Industrial Park, Jiangsu, China
Sponsor Contact:	Wang Yan, Senior Director of Medical Science Mobile phone: (+86) 18036092918 E-mail: yan.wang@innoventbio.com

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PROTOCOL SYNOPSIS

SYNOPSIS

Protocol No.	CIBI338B301
Sponsor	Innovent Biologics (Suzhou) Co., Ltd.
Study Drugs	IBI308 (trade name: Tyvyt®) IBI305 (generic name: bevacizumab injection)
Active Ingredients	IBI308: recombinant fully human anti-PD-1 monoclonal antibody IBI305: recombinant anti-VEGF humanized monoclonal antibody
Study Title	A randomized, open-labeled, multi-center study on efficacy and safety of IBI308 in combination with IBI305 vs. sorafenib as first-line treatment for advanced hepatocellular carcinoma (ORIENT-32)
Study Phase	Phases II/III
Study Objectives	**Phase II single-arm study (safety run-in period):** Primary objective: • To evaluate the safety and tolerability of IBI308 in combination with IBI305 as the first-line treatment for patients with advanced hepatocellular carcinoma (HCC). Secondary objectives: • To evaluate the preliminary efficacy of IBI308 in combination with IBI305 as the first-line treatment for patients with advanced HCC by the investigator based on Response Evaluation Criteria in Solid Tumors (RECIST V1.1). The endpoints include: - Objective response rate (ORR) - Disease control rate (DCR) - Duration of response (DOR) - Time to progression (TTP) - Time to response (TTR) - Progression free survival (PFS) - Overall survival (OS) • To evaluate the pharmacokinetic (PK) characteristics of IBI308 and IBI305 in patients with advanced HCC; • To evaluate the immunogenicity of IBI308 and IBI305 in patients with advanced HCC.

	Phase III randomized, controlled study: Primary objective: • To compare OS of IBI308 in combination with IBI305 vs. sorafenib as the first-line treatment for advanced HCC; • To compare and evaluate (by the independent radiological review committee (IRRC) as per RECIST V1.1) PFS of IBI308 in combination with IBI305 vs. sorafenib as the first-line treatment of advanced HCC. Secondary objectives: • To compare PFS determined by the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group; • To compare ORR determined by IRRC and the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group; • To compare TTP determined by IRRC and the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group; • To compare DOR determined by IRRC and the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group; • To compare DCR determined by IRRC and the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group; • To compare TTR determined by IRRC and the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group; • To compare PFS, ORR, TTP, DOR, DCR, and TTR determined by IRRC as per modified RECIST (mRECIST) for HCC in IBI308 + IBI305 group vs. sorafenib group; • To compare the safety of IBI308 in combination with IBI305 vs. sorafenib as the first-line treatment for advanced HCC; • To compare quality of life after IBI308 in combination with IBI305 vs. sorafenib as the first-line treatment for advanced HCC, using European Organisation for Research and Treatment of Cancer Quality-of-Life Questionnaire-Core 30 (EORTC QLQ-C30) and European Organisation for Research and Treatment of Cancer Quality-of-Life Questionnaire—hepatocellular carcinoma 18 (EORTC QLQ-HCC18); • To evaluate the immunogenicity of IBI308 in combination with IBI305 as the first-line treatment for advanced HCC; Exploratory objectives: • To explore the relationship between the potential biomarkers with efficacy predictability in tissues and efficacy: (1) Expression level of PD-L1; (2) Detection of RNA in tissues; • To evaluate the population pharmacokinetic characteristics of IBI308 + IBI305 as the first-line treatment for advanced HCC.

Study Design	This is a randomized, controlled, open-label, multi-center study evaluating the efficacy and safety of IBI308 in combination with IBI305 vs. sorafenib in locally advanced or metastatic HCC patients who have not received systemic treatment, who are unsuitable for radical surgical resection or local treatment, or who experienced disease progression after surgical resection or local treatment. The study is composed of 2 sections. Section I is a phase II single-arm study (safety run-in period). A total of 20 patients with advanced HCC will be enrolled to receive the first-line treatment, IBI308 (200 mg, IV, day 1, Q3W) + IBI305 (15 mg/kg, IV, day 1, Q3W). The treatment will continue until disease progression, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring the termination of treatment stipulated in the protocol, whichever occurs first. The patients in the study will receive IBI308 + IBI305 for a maximum of 24 months. The safety and tolerability of combination treatment will be evaluated and ORR, DCR, DOR, TTP, TTR, PFS, and OS of IBI308 in combination with IBI305 as the first-line treatment for patients with advanced HCC will be assessed based on the evaluation by the investigator as per RECIST V1.1. iRECIST is only used 4 to 6 weeks after the first disease progression in patients for confirmation, thereby excluding certain patients with pseudo-progression. Based on the preliminary safety and efficacy data of phase II single-arm study, a phase III, randomized, controlled study will be carried out following the discussion and decision by the safety assessment committee. Section II is a phase III, randomized, controlled study. The study is scheduled to enroll a total of about 546 patients, who will be randomly assigned to treatment group and control group in a 2:1 ratio. The stratification factors include macrovascular invasion and/or extrahepatic metastasis (yes vs. no), baseline AFP (< 400 ng/mL vs. ≥ 400 ng/mL), and the score of ECOG performance status (0 vs. 1). The subjects in treatment group will receive IBI308 (200 mg, IV, day 1, Q3W) in combination with IBI305 (15 mg/kg, IV, day 1, and Q3W) and those in control group will receive sorafenib (400 mg, PO, BID) consecutively. The treatment will continue until disease progression, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring the termination of treatment stipulated in the protocol, whichever occurs first. After the investigator's first evaluation of imaging progression in treatment group according to RECIST V1.1, disease progression should be confirmed by IRRC within 3 weeks. Due to the possibility of pseudo-progression in certain patients, the confirmation of progression based on iRECIST after 4 to 6 weeks is required for patients subjected to disease progression for the first time. Whether drug treatment will be continued or not while awaiting imaging confirmation is to be determined by the investigator's evaluation, and iRECIST will be adopted for the evaluation and recording of efficacy, until immune confirmed progressive disease (iCPD) based on iRECIST, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring termination of treatment stipulated in the protocol, whichever occurs first. The patients in the study will receive IBI308 + IBI305 for a maximum of 24 months.

	The subjects in control group will receive sorafenib (400 mg, BID, PO) consecutively until disease progression, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring the termination of treatment stipulated in the protocol, whichever occurs first. In addition, the optimal replacement treatment will be given to patients according to the investigator's comprehensive judgment and the follow-up treatment methods will be recorded. After the investigator's first evaluation of imaging progression according to RECIST V1.1, disease progression should be confirmed by IRRC within 3 weeks. Whether drug treatment will be continued or not while awaiting imaging confirmation is to be determined by the investigator. Although some patients have reached PD based on RECIST V1.1 or iCPD based on iRECIST (applicable to the treatment group), these patients can still be given the treatment based on the investigator's judgment under the premise that the patients agree to continue the treatment and have signed the informed consent forms if the clinical status of the patients is stable with no adverse events (AEs) requiring termination of the study drugs. The treatment can be continued until the occurrence of any of the conditions, including no benefit or onset of intolerable toxicity assessed by the investigator, initiation of a new anti-tumor therapy, withdrawal of ICF by the patient, and loss to follow-up, whichever occurs first. The primary endpoints of phase III study are OS and PFS. OS refers to the time from randomization to death for any reason. PFS refers to the time from randomization to the first onset of disease progression or death for any reason (whichever occurs first). An interim analysis will be performed during phase III trial and the results and report will be submitted to the Independent Data Monitoring Committee (iDMC) who will make an analysis of primary efficacy endpoints (OS and PFS) and safety endpoints, determine the treatment efficacy based on estimated effective boundaries, and provide advice to the sponsor on whether the trial data can be submitted in advance. Before the interim analysis, iDMC charter will be finalized and approved by iDMC and the sponsor. The responsibilities of iDMC members and related procedures will be defined in iDMC charter.
Inclusion Criteria	1. Histologically/cytologically confirmed HCC or cirrhosis meeting the clinical diagnostic criteria of HCC by American Association for the Study of Liver Diseases (AASLD). 2. Ages of ≥ 18 years old. 3. ECOG PS of 0 or 1. 4. Barcelona Clinic Liver Cancer (BCLC) stage of C. Stage B for those unsuitable for radical surgery and/or local treatment. 5. Absence of systemic anti-tumor treatment for HCC before the first dose. 6. At least one measurable lesion according to RECIST V1.1, or measurable lesions showing definite progression following the local treatment based on RECIST V1.1. 7. Child-Pugh score of ≤ 7.

	8. Sufficient organ and bone marrow functions, with the laboratory test values within 7 days before the enrollment meeting the following requirements (no blood components, cell growth factors, albumin, and other drugs via intravenous or subcutaneous administrations are allowed for correction treatment within the first 14 days after the laboratory test results are obtained). The specific information is as follows: 1) Routine blood test: absolute neutrophil count (ANC) ≥ 1.5 × 10^9/L; platelet count (PLT) ≥ 75 × 10^9/L; hemoglobin (HGB) ≥ 9.0 g/dL. 2) Hepatic function: total bilirubin (TBIL) ≤ 2 × upper limit of normal (ULN), alanine aminotransferase (ALT) and aspartate aminotransferase (AST) ≤ 5 × ULN; serum albumin ≥ 28 g/L; alkaline phosphatase (ALP) ≤ 5 × ULN. 3) Renal function: serum creatinine (Cr) ≤ 1.5 × ULN or clearance of creatinine (CCr) ≥ 50 mL/min (Cockcroft-Gault formula); urinalysis results showing urine protein < 2+; patients whose baseline urinalysis results show urine protein ≥ 2+ should undergo 24-h urine collection and 24-h urine protein quantitation test result should be < 1 g. 4) Blood coagulation function: international normalized ratio (INR) and activated partial thromboplastin time (APTT) ≤ 1.5 × ULN. 9. Estimated survival ≥ 12 weeks. 10. Female patients of childbearing age or male patients with female sexual partners of childbearing age should take effective contraceptive measures throughout the treatment and 6 months after the last dose. 11. Patients who have signed ICF and are able to perform follow-up visits and relevant procedures required in the protocol.
Exclusion Criteria	1. Histologically/cytologically confirmed fibrolamellar hepatocellular carcinoma, sarcomatoid hepatocellular carcinoma, and cholangiocarcinoma. 2. History of hepatic encephalopathy or liver transplantation. 3. Symptomatic pleural effusion, ascites, and pericardial effusion that require drainage. 4. Acute or chronic active hepatitis B or C infection; hepatitis B virus (HBV) DNA > 2000 IU/mL or 10^4 copies/mL; hepatitis C virus (HCV) RNA > 10^3 copies/mL; hepatitis B surface antigen (HbsAg) and anti-HCV antibody positive concurrently. Those who possess the indicators lower than the above criteria after nucleotide antiviral treatment can be enrolled. 5. Presence of metastasis to the central nervous system. 6. Presence of bleeding events from esophageal or gastric varices caused by portal hypertension within the past 6 months. Presence of known severe (G3) varicose veins in endoscopy within 3 months before the first dose. Evidence of portal hypertension (including the finding of splenomegaly in imaging studies) with a high risk of bleeding assessed by the investigator.

	7. Presence of any life-threatening bleeding events within the past 3 months, including the need for transfusion, surgery or local treatment, and continuous medication therapy. 8. Any arterial/venous thromboembolic events within 6 months, including myocardial infarction, unstable angina, cerebrovascular accident or transient cerebral ischemic attack, pulmonary embolism, deep vein thrombosis, or any other history of serious thromboembolism. Presence of implantable venous port or catheter-derived thrombosis, or superficial venous thrombosis, barring stable thrombosis following the conventional anticoagulation treatment. Prophylactic use of low-dose low-molecular-weight heparin (*e.g.*, enoxaparin 40 mg/day) is permitted. 9. Involvement of both the main portal vein and the left and right branches by portal vein tumor thrombus, or of both the main trunk and the superior mesenteric vein concurrently. Presence of tumor thrombus of inferior vena cava. 10. A 10-day consecutive dosing of aspirin (> 325 mg/day) or other drugs, *e.g.*, dipyridamole and clopidogrel, known to inhibit the platelet function within 2 weeks before the first dose. 11. Uncontrolled hypertension (systolic greater than 140 mmHg or diastolic greater than 90 mmHg) after the optimal medical treatment, history of hypertensive crisis or hypertensive encephalopathy. 12. Toxicity (excluding alopecia, events not clinically significant, and asymptomatic laboratory abnormalities) caused by previous therapy that has not yet resolved to grade 0 or 1 (National Cancer Institute Common Terminology Criteria for Adverse Events V5.0 (NCI CTCAE V5.0)) before the first dose of study drugs. 13. Symptomatic congestive cardiac failure (NYHA Class II–IV). Symptomatic or poorly controlled arrhythmia. History of congenital long QT syndrome or corrected QTc > 500 ms (calculated using Fridericia formula) during screening. 14. Serious hemorrhagic tendency or coagulopathy, or currently receiving thrombolytic therapy. 15. History of gastrointestinal perforation and/or fistula, history of bowel obstruction (including incomplete bowel obstruction requiring parenteral nutrition), extensive bowel resection (partial colectomy or extensive small bowel resection accompanied with chronic diarrhea), Crohn's disease, ulcerative colitis, or chronic diarrhea within the past 6 months. 16. Receipt of radiotherapy within 3 weeks before the first dose. For patients receiving radiotherapy 3 weeks before the first dose, all of the following conditions should be met for enrollment: free from radiotherapy-related toxic reactions, glucocorticoids not required, and radiation pneumonitis, radiation hepatitis, and radiation enteritis excluded. 17. History or current experience of pulmonary fibrosis and such lung diseases as interstitial pneumonia, pneumoconiosis, drug-related pneumonia, and severely impaired lung function. 18. Active tuberculosis (TB), currently receiving anti-tuberculosis treatment or received such treatment within 1 year before the first dose.

	19. Human immunodeficiency virus (HIV) infected (HIV 1/2 antibody positive) and known syphilis infection requiring treatment. 20. Active or poorly clinically controlled serious infections. Severe infections within 4 weeks before the first dose, including but not limited to hospitalization caused by infection, bacteremia, or severe pneumonia complication. 21. Presence of active autoimmune diseases requiring systemic treatment (*e.g.*, use of disease-modifying drugs, corticosteroids, or immunosuppressants) within 2 years before the first dose. Use of alternative therapies (*e.g.*, thyroxine, insulin, or physiological corticosteroids for adrenal or pituitary insufficiency) is permitted. Known history of primary immunodeficiency diseases. For patients with positive autoimmune antibodies only, they should be assessed by the investigator to determine whether they had an autoimmune disease. 22. Receipt of immunosuppressants within 4 weeks before the first dose, excluding local glucocorticoids administered by nasal, inhaled, or other topical routes, or systemic glucocorticoids of physiological doses (no more than 10 mg/day of prednisone or equivalents), while the temporary use of glucocorticoids for preventing allergies or treating dyspneic symptoms of such diseases as asthma and chronic obstructive pulmonary disease is permitted. 23. Receipt of a live attenuated vaccine within 4 weeks before the first dose or planned to receive a live attenuated vaccine during the study. 24. Receipt of major surgery (craniotomy, thoracotomy, or laparotomy) within 4 weeks before the first dose or having unhealed wounds, ulcers, or fractures. Receipt of tissue biopsy or other minor surgeries within 7 days before the first dose, barring venipuncture and catheterization for intravenous infusion. 25. Receipt of local treatment for liver cancer within 4 weeks before the first dose. 26. Receipt of systemic treatment with traditional Chinese medicines with cancer indications or immunomodulators (including thymosin, interferon, and interleukin, barring local use for controlling pleural fluid or ascites) within 2 weeks before the first dose. 27. Uncontrolled/uncorrectable metabolic disorders, other non-malignant organ diseases, systemic diseases, or cancer-related secondary diseases with the potential to cause a relatively high medical risk and/or survival evaluation uncertainties unsuitable for subject enrollment as judged by the investigator; other circumstances unsuitable for subject enrollment as judged by the investigator. 28. Other malignancies diagnosed within 5 years before the first dose, excluding radically treated basal cell carcinoma of skin, squamous cell carcinoma of skin, and/or radically resected carcinoma *in situ*. If other malignancies were diagnosed over 5 years pre-dose, the liver lesions should still be subjected to pathological or cytological diagnosis even if they meet the clinical diagnostic criteria for HCC by AASLD; individuals with confirmed HCC can be enrolled.

	29. Previous receipt of any antibody treatment involving anti-PD-1, anti-PD-L1/L2, or anti-CTLA4 or other immunotherapies. Previous receipt of anti-VEGF and/or VEGFR, RAF, MEK, PDGFR, and FGFR targeted therapy. 30. Known hypersensitivity to any ingredients in IBI308 and IBI305 preparations or previous severe hypersensitivity reactions to other monoclonal antibodies. 31. Receipt of treatment in other clinical trials within 4 weeks before the first dose. 32. Pregnant or breastfeeding female patients. 33. Acute or chronic diseases, psychiatric disorders, or laboratory abnormalities that may lead to the following consequences: increased study participation or drug-related risks, or interference with interpreting trial results, and considered ineligible for participating in the trial by the investigator.
Study Drugs, Dosage, and Route of Administration	• IBI308 (200 mg) will be administered via intravenous infusion on day 1 of each 3-week cycle. Patients will be allowed to undergo treatment (a maximum of 24-month) until disease progression, development of intolerable toxicity, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other circumstances for which termination of treatment is required as per the protocol. • IBI305 will be administered via intravenous infusion at 15 mg/kg on day 1 of each 3-week cycle. Patients will be allowed to undergo treatment (a maximum of 24-month) until disease progression, development of intolerable toxicity, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other circumstances for which termination of treatment is required as per the protocol. • Firstly, IBI308 (200 mg) will be given via intravenous infusion for 60 (±15) min. IBI305 will be administered intravenously at 15 mg/kg at least 5 min later. • In the treatment of IBI308 in combination with IBI305, the other drug can still be given alone when either drug treatment is interrupted or terminated.
Control Drug, Dosage, and Route of Administration	• Sorafenib (400 mg, BID, PO) will be given consecutively until disease progression, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring termination of treatment stipulated in the protocol.
Evaluation Criteria	**Efficacy evaluation:** **Phase II single-arm study, safety run-in period:** • ORR, DCR, TTP, DOR, TTR, PFS, and OS determined by the investigator post-dose will be evaluated as per RECIST V1.1. **Phase III randomized, controlled study:** • Post-dose PFS, ORR, TTP, DOR, DCR, and TTR determined by IRRC and the investigator will be evaluated and OS of two groups will be assessed as per RECIST V1.1. • Post-dose ORR, PFS, TTP, DOR, DCR, and TTR determined by IRRC will be evaluated as per mRECIST.

	Safety evaluation (phases II/III studies): The data of phase II and phase III studies will be analyzed separately. • The incidence, relationship with study drugs, and severity level of all adverse events (AEs), treatment-emergent adverse events (TEAEs), treatment related adverse events (TRAEs), adverse events of special interest (AESIs), and serious adverse events (SAEs) will be evaluated. • The changes in vital signs, physical examination results, and laboratory test results before, during, and after the treatment will be evaluated. **Immunogenicity evaluation (phase II/III studies):** • Patients in phase II single-arm study and phase III controlled study will be subjected to anti-drug antibody (ADA) testing (IBI308 and IBI305) and ADA-positive serum samples will then be subjected to neutralizing antibody (NAb) testing. **PK evaluation (phase II/III studies):** • PK characteristics in humans will be described and main PK parameters, including but not limited to area under curve (AUC), C_{max}, CL, V, and $t_{1/2}$, will be determined. • The evaluation will include but are not limited to the descriptive statistical analysis of trough concentrations of IBI308 and IBI305 in cycles 1/3/11. • The population pharmacokinetic characteristics of IBI308 + IBI305 as the first-line treatment for advanced HCC will be evaluated. **Quality of life evaluation (phase III study):** • Quality of life and health status will be compared between IBI308 in combination with IBI305 vs. placebo, as per EORTC QLQ-C30 and EORTC QLQ-HCC18. **Biomarker evaluation (phase III study):** • The relationship of potential biomarkers with efficacy predictability in tissues to efficacy will be explored 1) Expression level of PD-L1: The correlation between the expression level of PD-L1 on the surfaces of tumor cells and efficacy will be reviewed and the values of PD-L1 as the biomarker with IBI308 efficacy predictability in liver cancer will be explored; 2) RNA in tissues will be detected, the relationship between RNA expression level in tumor tissues and efficacy will be retrospectively analyzed, and the values of RNA being a predictive biomarker in liver cancer.

Statistical Methods	**Sample size estimation:** **Phase II single-arm study, safety run-in period:** A total of 20 patients with advanced HCC are scheduled to be enrolled. The safety and tolerability of treatment will be evaluated. Only descriptive analysis will be performed. By referring to the safety data as reported in similar foreign products in studies, TRAEs at grade 3 or above as per CTCAE were observed in 17/68 patients receiving atezolizumab + bevacizumab (incidence rate, 25%; 95% confidence interval, 15–37%). It is predicted in the phase II design of this study that the safety profile of IBI308 in combination with IBI305 will be similar to that of products of the same kind and it is proposed to enroll a total of 20 patients for preliminary safety assessment. Assuming that observation shows over 8 (40%) of the 20 patients receiving IBI308 + IBI305 have TRAEs at grade 3 or above as per CTCAE, there is about 90% power in inferring that the true incidence of such AEs exceeds 29%, namely, the incidence of TRAEs at grade 3 or above as per CTCAE arising from IBI308 + IBI305 group may be higher than estimation. Since the actual number of patients receiving screening was more than the planned enrollment number, a total of 24 patients were eventually enrolled in phase II clinical study. **Phase III randomized, controlled study:** The sample size of this study will be determined according to the calculation of the number of deaths as required to prove efficacy via OS and the number of patients as required to prove efficacy via PFS. To detect OS improvement at the two-sided significance level of 0.048 using the log-rank test, about 408 deaths can offer approximately 89% power assuming HR = 0.714 (median OS in sorafenib group is 11.5 months vs. 16.1 months in IBI308 + IBI305 group). In the final OS analysis, the minimum detectable difference (MDD) of the OS-based statistical test is corresponding to an HR of 0.810 (median OS improvement of about 2.7 months). The dropout rates are expected to stand at 0.5% per month in IBI308 + IBI305 group vs. 0.8% per month in sorafenib group. Assuming that the period of enrollment is 18 months, a total of about 546 patients will be randomized in a 2:1 ratio to two groups, IBI308 + IBI305 group (treatment group) and sorafenib group (control group). A total of 408 deaths will be expected to be observed at 44 months after the enrollment of the first patient. (In the actual study, if the actual number of enrolled subjects is different from that in the protocol due to unforeseen reasons, such as the enrollment speed is faster than the requirements of the protocol, the end date of screening should be determined in advance to ensure that the number of actually enrolled subjects does not exceed that in anticipation by about 10%, i.e., a maximum of 600 subjects can be enrolled, including 400 subjects in treatment group and 200 subjects in control group). In this study, an interim analysis of OS endpoint will be carried out when 265 (65%) cases of OS events are observed. The cut-off value of the interim analysis approximates the O'brien-Fleming boundary according to the Lan-Demets method. The interim analysis has a two-sided level of significance $\alpha = 0.0102$. Based on the previously assumed enrollment speed and the monthly dropout rate of each group, the interim analysis of OS is expected to start approximately 25 months after the enrollment of the first patient and the primary analysis of

	PFS endpoint will be performed concurrently. Assuming that the median PFS is 8 months in IBI308 + IBI305 group and 4 months in sorafenib group and HR is 0.5 for PFS, it is estimated that about 400 cases of PFS events will be observed for interim analysis of OS. Therefore, at the two-sided level of significance α = 0.002, there will be a power of more than 99% based on 400 PFS events. **Hypothesis for superiority:** No formal statistical hypothesis test will be performed in the phase II section of this study. In the phase III section, the superiority test will be carried out for IBI308 + IBI305 vs. sorafenib. The test will be performed on the dual efficacy endpoints of OS and PFS assessed by IRRC as per RECIST V1.1. With respect to the endpoints of OS and PFS: null hypothesis H0: HR ≥ 1; alternative hypothesis Ha: HR < 1. The Bonferroni correction method is used to adjust α to control the overall type I error rate at a two-sided level of significance α = 0.05, i.e., to test OS at the two-sided level of significance α = 0.048, and to test PFS at the two-sided level of significance α = 0.002. If any of the efficacy endpoints of PFS and OS is statistically significant, the superiority will be established statistically. First, the test of PFS will be performed. If the test result is statistically significant, the two-sided level of significance α = 0.002 allocated to PFS is recovered to the OS hypothesis test (i.e., the overall OS hypothesis will be tested at the significance level α = 0.05). If the test results of PFS are not statistically significant, OS will be tested at the original significance level α = 0.048. **Interim analysis:** In the phase III section of this study, OS will be subjected to OS interim analysis and OS final analysis. In this study, an interim analysis of OS will be performed in about 65% of OS events (265 cases). With respect to the efficacy endpoint of OS, the test level will approximate the O'brien-Fleming boundary according to the Lan-Demets method. The interim analysis has a two-sided level of significance α = 0.0102 while the final analysis a two-sided level of significance α = 0.0448. MDD in the interim analysis is corresponding to an HR of 0.716. MDD in the final analysis is corresponding to an HR of 0.810. In the interim analysis of OS, the primary analysis of PFS endpoint (about 400 events observed) will also be carried out first. In PFS test, at the two-sided level of significance α = 0.002, MDD of PFS is corresponding to an HR of about 0.719. If the number of OS events is less than 220 when 400 PFS events have occurred, two interim analyses will be performed for OS events. The first interim analysis will be performed after 400 PFS events occur and the second interim analysis performed when 265 OS events are observed; otherwise, only 1 interim analysis will be carried out only when 265 OS events are observed. The α cut-off value of each OS analysis is allocated to approximate the O'brien-Fleming boundary according to the Lan-Demets method and adjusted according to the actual number of OS events that occur to ensure that the OS analysis is a two-sided test with an overall significance level of α = 0.048.

	Analysis of primary efficacy endpoints: **Phase II single-arm study, safety run-in period:** There are no primary efficacy endpoints. **Phase III randomized, controlled study:** The main analysis of inter-group comparison of PFS will be performed using the stratified log-rank test, the median PFS and its 95% CI will be estimated using the Kaplan-Meier method, and the survival curves will be plotted. HR and corresponding 95% CI will be estimated by a stratified Cox proportional hazards model with the randomization stratification factor. The main analysis of inter-group OS comparison will be performed using the stratified log-rank test, the median OS and its 95% CI will be estimated using the Kaplan-Meier method, and the survival curves will be plotted. HR and corresponding 95% CI will be estimated by a stratified Cox proportional hazards model with the randomization stratification factor. For inter-group OS comparison, the auxiliary analysis will include the situation of other anti-tumor treatment after disease progression (presence or absence). The subsequent anti-tumor treatment of special interest refers to other anti-PD-1/PD-L1 monoclonal antibody treatment. The auxiliary analysis will include, but not limited to, censoring patients who receive the subsequent anti-PD-1/PD-L1 monoclonal antibody as anti-tumor treatment and the analysis will be conducted using Rank Preserving Structural Failure Time Model and two-stage model. **Analysis of secondary efficacy endpoints:** **Phase II single-arm study, safety run-in period:** Each patient's efficacy endpoints will be presented in the form of table. **Phase III randomized, controlled study:** For the main analysis of inter-group ORR and DCR comparison, CMH method with stratification factors will be used for superiority test; Miettinen & Nurminen method with stratification factors will be employed to estimate the inter-group rate differences in ORR and DCR as well as their CIs. TTP, DOR, and TTR will be analyzed using the method adopted in the main analysis of OS. **Safety data (phase II/III studies):** The safety data will be analyzed during phase II and phase III studies, respectively. All AEs will be coded and classified by Medical Dictionary for Regulatory Activities (MedDRA) and their incidence and severity summarized. Measurement values and changes from baseline of vital signs, physical examination, laboratory test, and 12-lead ECG will be summarized using descriptive statistics. **Immunogenicity data (phase II/III studies):** The immunogenicity data will be analyzed during phase II and phase III studies, respectively. The positive rates of ADA and NAb will be summarized.

	PK data (phase II/III studies): PK data will be analyzed during phase II and phase III studies, respectively. PK parameters including but not limited to AUC, C_{max}, CL, V and $t_{1/2}$, and trough concentrations in cycles 1/3/11 will be analyzed using descriptive statistics. The population pharmacokinetic characteristics of IBI308 + IBI305 as the first-line treatment for advanced HCC will be analyzed using descriptive statistics. **Quality of life endpoints (phase III study):** The quality of life endpoints will be analyzed during phase III study. The results of EORTC QLQ-C30 and EORTC QLQ-HCC18 will be analyzed using descriptive statistics. **Biomarkers (phase III study):** • The expression level and the distribution of biomarkers in tissues with the potential of predicting efficacy will be analyzed using descriptive statistics and the potential relationship between the following endpoints and the efficacy will be explored: 1) Expression level of PD-L1; 2) Detection of RNA in tissues.

STUDY FLOW CHART

Overall design:

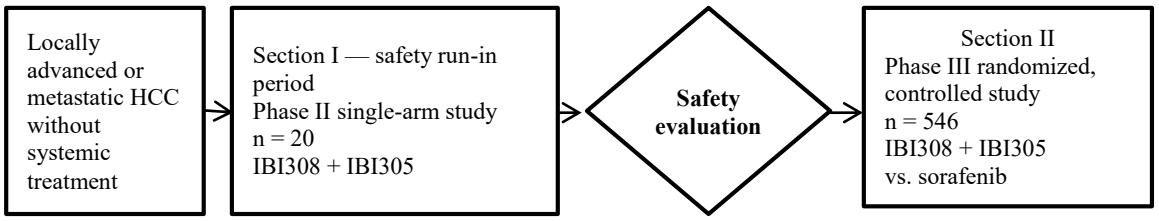

Figure 1. Master flow chart of study design

Section I: phase II, single-arm study (safety run-in period)

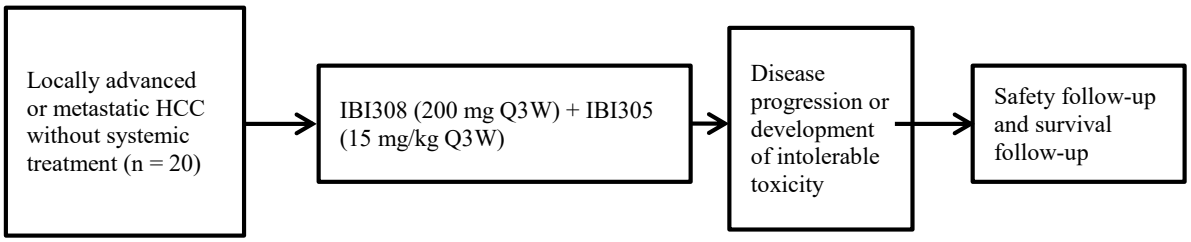

Figure 2. Flow chart of phase II single-arm study

Section II: phase III randomized, controlled study

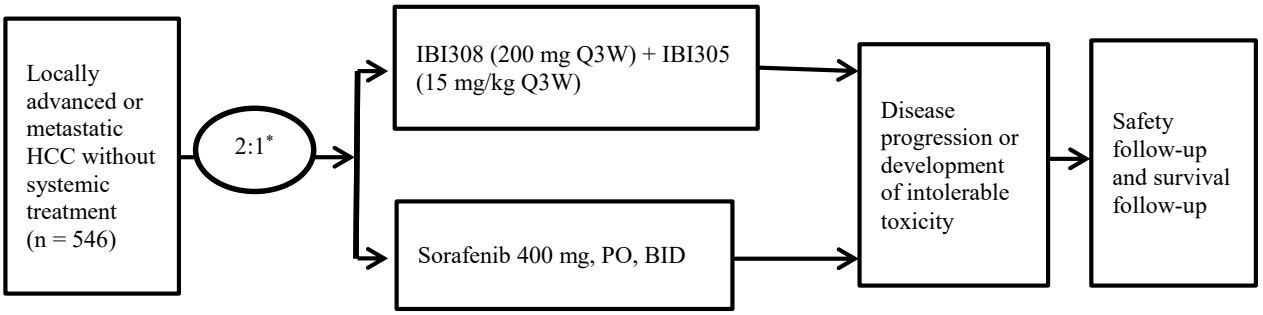

Figure 3. Flow chart of phase III randomized, controlled study

*** Stratification factors:**

- Macrovascular invasion and/or extrahepatic metastasis (yes vs. no)
- Baseline AFP (< 400 ng/mL vs. ≥ 400 ng/mL)
- ECOG PS score (0 vs. 1)

Visit schedule

Table 1. Visit schedule (phase II single-arm study, safety run-in period)

Period	Screening period	Treatment period (21 days = 1 cycle)					Safety visit	Survival follow-up
		C1D1	C2D1	C3D1	C4D1	Cycle 5 until end-of-treatment Day 1 of each cycle	30 days after the last dose	Every 60 days after safety follow-up
Day	–28 to –1	1	22	43	64			
Time window (day)	NA		±3	±3	±3	±3	±3	±7
Standard trial procedures								
Written informed consent form [1]	X							
Inclusion/exclusion criteria	X							
Demographics/medical history/tumor history [2]	X							
Vital signs [3]	X	X	X	X	X	X	X	
Weight/height [4]	X	X	X	X	X	X	X	
Physical examination	X		X	X	X	X	X	
ECOG PS score	X		X	X	X	X	X	
12-lead ECG [5]	X		X	X	X	X	X	
Myocardial test [5]	X		X	X	X	X	X	
Routine blood test/blood chemistry/urinalysis [6]	X		X	X	X	X	X	
AFP [6]	X		X	X	X	X	X	
Coagulation function [7]	X		X	X	X	X	X	

Period	Screening period	Treatment period (21 days = 1 cycle)					Safety visit	Survival follow-up
		C1D1	C2D1	C3D1	C4D1	Cycle 5 until end-of-treatment Day 1 of each cycle	30 days after the last dose	Every 60 days after safety follow-up
Day	–28 to –1	1	22	43	64			
Time window (day)	NA		±3	±3	±3	±3	±3	±7
Pregnancy test [8]	X						X	
Thyroid function [9]	X		X	X	X	X	X	
HIV, HBV, and HCV [10]	X							
HCV-RNA [11]	X		X	X	X	X	X	
HBV-DNA [12]	X		X	X	X	X	X	
AE evaluation [13]	X	X	X	X	X	X	X	
Concomitant medication [14]	X	X	X	X	X	X	X	
Survival condition		X						X
Subsequent anti-tumor therapy							X	X
Efficacy evaluation								
Tumor imaging evaluation [15]	X			X		X		
Study drug infusion [16]								
IBI308		X	X	X	X	X		
IBI305		X	X	X	X	X		
PK and immunogenicity								
Immunogenicity [17]		X	X		X	X	X	
PK [18]		X	X		X	X		

Note:

1. The informed consent form (ICF) should be signed by subjects before any study procedures. The test and examination results from the study site before the informed consent is obtained can be used for screening evaluation and such tests and examinations are not required to be performed anew.
2. The tumor history includes disease stage, date of diagnosis, and previous anti-tumor treatment (*e.g.*, systemic treatment, radiotherapy, surgery, and local treatment).
3. Vital signs include body temperature, pulse, respiratory rate, and blood pressure (sitting or supine position). On the day of study treatment (IBI308 + IBI305), vital signs should be measured within 60 min before each infusion. The vital signs will be measured at the end of IBI305 infusion and within 2 (±1) h after the infusion. Should clinical indications appear, the vital signs will be measured during and after each infusion of IBI308 as well.
4. Height is measured during the screening period only.
5. Time points of 12-lead ECG examination are as follows: The examinations will be conducted during the screening period, before dosing of study drugs in each cycle from cycle 2, during the safety visit, and when clinically necessary.

   Myocardial test includes at least such related tests as myocardial enzyme spectra of troponin, creatine kinase isoenzyme (CK-MB), and creatine kinase (CK). The test will be conducted within 7 days before the first dosing of study drugs during screening, within 3 days before each dosing of study drugs from cycle 2, during safety follow-up, and when clinically necessary. For subjects with clinically significant abnormalities in ECG or relevant medical history, signs, and symptoms, tests for other myocardial markers, such as brain natriuretic peptide (BNP), can be carried out as judged by the investigator. Tests will be conducted at each study site.
6. Routine blood tests include red blood cell count (RBC), HGB, WBC, PLT, and WBC differentials [lymphocyte count (LYM) and ANC]. Blood chemistry includes hepatic function [TBIL, DBIL, CK, ALT, AST, γ-glutamine transferase (γ-GT), alkaline phosphatase (ALP), albumin (ALB), total protein (TP), lactate dehydrogenase (LDH)], renal function [urea (UREA) and Cr], electrolytes (Na, K, Cl, Mg, Ca, and P), amylase, and blood glucose (GLU). Urinalysis includes pH, urine white blood cell (UWBC), UPRO, urine red blood cell (URBC), and urine glucose (UGLU). Routine blood test, blood chemistry, urinalysis, and AFP will be conducted or tested within 7 days before the first dosing of study drugs during the screening period, within 3 days before each dosing of study drugs from cycle 2, during the safety follow-up, and when clinically necessary. Tests will be conducted at each study site.
7. Coagulation function tests include INR and APTT. The test will be conducted within 7 days before the first dosing of study drugs during screening, within 3 days before each dosing of study drugs from cycle 2, during safety follow-up, and when clinically necessary. Tests will be conducted at each study site.
8. Women of child-bearing age should undergo urine or serum pregnancy test within 3 days before the first dosing and during the safety visit. If the urine pregnancy test is not conclusive, serum pregnancy test will be performed. The conclusion should be based on the serum pregnancy test. Tests will be conducted at each study site.

9. The test will be conducted within 28 days before the first dosing of study drugs, within 7 days before the dosing of study drugs from cycle 2, and during the safety follow-up. Free triiodothyronine (FT3), free thyroxine (FT4), and thyroid-stimulating hormone (TSH) will be tested during the screening period. Only TSH will be tested from cycle 2 and indicators for other thyroid functions can be tested as judged by the investigator. Tests will be conducted at each study site.
10. HIV antibody, five items on hepatitis B [HBsAg, HBsAb, HBcAb, hepatitis B e antigen (HBeAg), and hepatitis B e antibody (HBeAb)], and HCV antibody will be tested during the screening period. The tests will be conducted at each study site during the screening period.
11. For patients positive for HCV antibody, their HCV-RNA load should be determined during the screening period, within 7 days before each cycle from cycle 2, and during the safety follow-up, or can be determined earlier should clinical indications appear.
12. For patients positive for HBsAg and/or HBcAb, their HBV-DNA load should be determined during the screening period, within 7 days before each cycle from cycle 2, and during the safety follow-up, or can be determined earlier should clinical indications appear.
13. All AEs, including SAEs, that occur from the signing of ICF to 30 days after the last dose are collected, regardless of whether they are observed by the investigator or self-reported by patients. From day 30 to day 90 after the last dose, the investigator should report all SAEs and AEs related to study drugs or procedures. After 90 days from the last dose, the investigator should report any SAEs that are considered related to study drugs or procedures. Should new anti-tumor treatment is initiated within 90 days after the last dose, only SAEs related to study drugs will be recorded subsequently. The safety assessment of AEs and laboratory tests will be carried out by CTCAE V5.0.
14. Concomitant medication: Concomitant medication includes any prescription or over-the-counter drugs. Any drugs used by patients within 28 days before the treatment should be recorded at the time of screening. The changes in current medication or medication used after the previous medication records should be recorded during the follow-up visits.
15. RECIST V1.1 is mainly used for this study and the tumor imaging evaluation will be performed by the investigator. iRECIST is only used for confirmation 4 to 6 weeks after the first disease progression in patients, thereby excluding certain patients with pseudo-progression. Tumor imaging evaluations usually include contrast-enhanced CT or MRI and the sites for evaluation must include chest, abdomen, and pelvic cavity. The dual-phase imaging for liver must be obtained to obtain high-quality arterial phase imaging of the whole liver during the first scanning and high-quality portal vein phase imaging of the whole liver during the second scanning, should enhanced CT scanning be not contraindicated. Contrast-enhanced MRI or CT of the brain should also be performed at the baseline for patients with suspected signs and symptoms of CNS metastasis during the screening period. Individuals with metastasis to CNS cannot be enrolled. For patients without metastatic lesions before enrollment, there is no need for routine cranial imaging examinations afterward. The same patient should receive the same type of imaging examination during the study. If there are clinical indications, other imaging examinations, such as bone scan and neck CT scan, should also be performed for evaluation. The baseline evaluation will be performed within 28 days before the first dose. From the first infusion, the tumor imaging evaluation is performed once every 6 weeks (±7 days) until week 48 and once every 12 weeks (±7 days) thereafter. For patients in whom complete or partial remission is recorded by imaging for the first time, imaging studies must be repeated to confirm the remission in 4–6 weeks

after the first remission is observed. When patients are clinically stable, the imaging studies can be arranged in the next study cycle (6/12 weeks ± 7 days). For patients in whom imaging progression is recorded for the first time, IRRC should confirm the progression within 3 weeks, repeat the imaging studies in 4–6 weeks after the first finding of progression to confirm progression, and evaluate the efficacy as per iRECIST. See Section 7.3 for details.

16. The patients will receive IBI308 (200 mg, day 1, Q3W) + IBI305 (15 mg/kg, day 1, Q3W). The study treatment will continue until disease progression, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring the termination of treatment stipulated in the protocol, whichever occurs first. The IBI308 + IBI305 treatment will last 24 months at maximum.
17. Immunogenicity samples will be collected within 1 h before IBI308 infusion in cycles 1/2/4/8/12/16, every 8 cycles (cycle 24, 32...) thereafter, and during the safety follow-up visits. If an IBI308 and/or IBI305 infusion-related reaction occurs, blood samples should be taken after the start of the event with the least delay, end of event, and around 30 days after the event for immunogenicity analysis.
18. The PK sampling time points for patients are shown in Table 2.

Table 2. PK sampling schedule (involving all patients enrolled in phase II study and the first 6 patients in designated site for phase III study)

Period	Screening period	Treatment period												Safety follow-up [2]
		Cycle 1 (21 days) [2]									Cycle 2 and thereafter [2]			
Day	–28 to –1	1					2	3	8	15	Day 1 (±3)			30 days after the last dose
Sampling time [1]		Within 1 h before IBI308 infusion	During infusion	Immediately at the end of IBI305 infusion	After IBI305 infusion	After the start of IBI308 infusion					–1 h	During infusion	Immediately at the end of infusion	
					1 h	6 h	24 h	48 h	168 h	336 h				
Sampling time window		-1h		+5min	±5min	±15min	±1h	±2h	±8h	±12h			+5min	
IBI308			X									X		
IBI305			X									X		
PK [2]		X		X	X	X	X	X	X	X	X [2]		X [2]	

Note:

1. Samples should be collected from the opposite arm of the infusion site.
2. PK sampling for patients in the designated site (involving all patients enrolled in phase II study and the first 6 patients in the designated site for phase III study). Cycle 1 (intensive sampling): within 1 h before IBI308 infusion, immediately after IBI305 infusion (+5 min), 1 h ± 5 min after IBI305 infusion, and 6 h ± 15 min, 24 h ± 1 h, 48 h ± 2 h, 168 h ± 8 h (day 8), and 336 h ± 12 h (day 15) after the start of IBI308 infusion; should the infusion be delayed on day 1 of cycle 2 due to AEs or other reasons, an additional sampling at 504 h ± 24 h (day 22) will be required in cycle 1. Cycle 2: within 1 h before IBI308 infusion; cycle 4 (intensive sampling): within 1 h before IBI308 infusion, immediately after IBI305 infusion (+5 min), 1 h ± 5 min after IBI305 infusion, and 6 h ± 15 min, 24 h ± 1 h, 48 h ± 2 h, 168 h ± 8 h (day 8), 336 h ± 12 h (day 15), and 504 h ± 24 h (day 22, namely, within 1 h before the infusion in cycle 5) after the start of IBI308 infusion; cycles 6/8/12: within 1 h before IBI308 infusion.

Table 3. Visit schedule (phase III study)

Period	Screening period	Treatment period (21 days = 1 cycle)					Safety visit	Survival follow-up
		C1D1	C2D1	C3D1	C4D1	Cycle 5 until end-of-treatment Day 1 of each cycle	30 days after the last dose	Every 60 days after safety follow-up
Day	-28~-1	1	22	43	64			
Time window (day)	NA		±3	±3	±3	±3	±3	±7
Standard trial procedures								
Written informed consent form [1]	X							
Inclusion/exclusion criteria	X							
Demographics/medical history/tumor history [2]	X							
Vital signs [3]	X	X	X	X	X	X	X	
Weight/height [4]	X	X	X	X	X	X	X	
Physical examination	X		X	X	X	X	X	
ECOG PS score	X		X	X	X	X	X	
12-lead ECG [5]	X		X	X	X	X	X	
Myocardial test [5]	X		X	X	X	X	X	
Routine blood test/blood chemistry/urinalysis [6]	X		X	X	X	X	X	
AFP [6]	X		X	X	X	X	X	
Coagulation function [7]	X		X	X	X	X	X	

Period	Screening period	Treatment period (21 days = 1 cycle)					Safety visit	Survival follow-up
		C1D1	C2D1	C3D1	C4D1	Cycle 5 until end-of-treatment Day 1 of each cycle	30 days after the last dose	Every 60 days after safety follow-up
Day	-28~-1	1	22	43	64			
Time window (day)	NA		±3	±3	±3	±3	±3	±7
Pregnancy test [8]	X						X	
Thyroid function [9]	X		X	X	X	X	X	
HIV, HBV, and HCV [10]	X							
HCV-RNA [11]	X		X	X	X	X	X	
HBV-DNA [12]	X		X	X	X	X	X	
AE evaluation [13]	X	X	X	X	X	X	X	
Concomitant medication [14]	X	X	X	X	X	X	X	
Survival condition		X						X
Subsequent anti-tumor therapy							X	X
Efficacy evaluation								
Tumor imaging evaluation [15]	X			X		X		
Study drug infusion [16]								
IBI308		X	X	X	X	X		
IBI305		X	X	X	X	X		
Sorafenib		X	X	X	X	X		

Period	Screening period	Treatment period (21 days = 1 cycle)					Safety visit	Survival follow-up
		C1D1	C2D1	C3D1	C4D1	Cycle 5 until end-of-treatment Day 1 of each cycle	30 days after the last dose	Every 60 days after safety follow-up
Day	-28~-1	1	22	43	64			
Time window (day)	NA		±3	±3	±3	±3	±3	±7
PK and immunogenicity								
Immunogenicity [17]		X	X		X	X	X	
PK [18]		X	X		X	X		
Quality of life evaluation								
EORTC QLQ-C30 and EORTC QLQ-HCC18 [19]		X		X		X	X	
Biomarker exploration								
Collection of archived tumor tissue samples [20]	X							

Note:

1. The informed consent form (ICF) should be signed by subjects before any study procedures. The test and examination results from the study site before the informed consent is obtained can be used for screening evaluation and such tests and examinations are not required to be performed anew.
2. The tumor history includes disease stage, date of diagnosis, and previous anti-tumor treatment (*e.g.*, systemic treatment, radiotherapy, surgery, and local treatment).
3. Vital signs include body temperature, pulse, respiratory rate, and blood pressure (sitting or supine position). For patients randomized to receive the combination therapy, vital signs should be measured within 60 min before each infusion on the day of study treatment (IBI308 + IBI305). The vital signs will be measured at the end of IBI305 infusion and within 2 (±1) h after the infusion. Should clinical indications appear, the vital signs will be measured during and after each infusion of IBI308 as well.

4. Height is measured during the screening period only.
5. Time points of 12-lead ECG examination are as follows: The examinations will be conducted during the screening period, before dosing of study drugs in each cycle from cycle 2, during the safety visit, and when clinically necessary.

   Myocardial test includes at least such related tests as myocardial enzyme spectra of troponin, creatine kinase isoenzyme (CK-MB), and creatine kinase (CK). The test will be conducted within 7 days before the first dosing of study drugs during screening, within 3 days before each dosing of study drugs from cycle 2, during safety follow-up, and when clinically necessary. For subjects with clinically significant abnormalities in ECG or relevant medical history, signs, and symptoms, tests for other myocardial markers, such as BNP, can be carried out as judged by the investigator. Tests will be conducted at each study site.
6. Routine blood tests include red blood cell count (RBC), HGB, WBC, PLT, and WBC differentials [lymphocyte count (LYM) and ANC]. Blood chemistry includes hepatic function [TBIL, DBIL, CK, ALT, AST, γ-glutamine transferase (γ-GT), alkaline phosphatase (ALP), albumin (ALB), total protein (TP), lactate dehydrogenase (LDH)], renal function [urea (UREA) and Cr], electrolytes (Na, K, Cl, Mg, Ca, and P), amylase, and blood glucose (GLU). Urinalysis includes pH, urine white blood cell (UWBC), UPRO, urine red blood cell (URBC), and urine glucose (UGLU). Routine blood test, blood chemistry, urinalysis, and AFP will be conducted or tested within 7 days before the first dosing of study drugs during the screening period, within 3 days before each dosing of study drugs from cycle 2, during the safety follow-up, and when clinically necessary. Tests will be conducted at each study site.
7. Coagulation function tests include INR and APTT. The test will be conducted within 7 days before the first dosing of study drugs during screening, within 3 days before each dosing of study drugs from cycle 2, during safety follow-up, and when clinically necessary. Tests will be conducted at each study site.
8. Women of child-bearing age should undergo urine or serum pregnancy test within 3 days before the first dosing and during the safety visit. If the urine pregnancy test is not conclusive, serum pregnancy test will be performed. The conclusion should be based on the serum pregnancy test. Tests will be conducted at each study site.
9. The test will be conducted within 28 days before the first dosing of study drugs, within 7 days before the dosing of study drugs from cycle 2, and during the safety follow-up. Free triiodothyronine (FT3), free thyroxine (FT4), and thyroid-stimulating hormone (TSH) will be tested during the screening period. Only TSH will be tested from cycle 2 and indicators for other thyroid functions can be tested as judged by the investigator. Tests will be conducted at each study site.
10. HIV antibody, five items on hepatitis B [HBsAg, HBsAb, HBcAb, hepatitis B e antigen (HBeAg), and hepatitis B e antibody (HBeAb)], and HCV antibody will be tested during the screening period. The tests will be conducted at each study site during the screening period.
11. For patients positive for HCV antibody, their HCV-RNA load should be determined during the screening period, within 7 days before each cycle from cycle 2, and during the safety follow-up, or can be determined earlier should clinical indications appear.
12. For patients positive for HBsAg and/or HBcAb, their HBV-DNA load should be determined during the screening period, within 7 days before each cycle from cycle 2, and during the safety follow-up, or can be determined earlier should clinical indications appear.

13. All AEs, including SAEs, that occur from the signing of ICF to 30 days after the last dose are collected, regardless of whether they are observed by the investigator or self-reported by patients. From day 30 to day 90 after the last dose, the investigator should report all SAEs and AEs related to study drugs or procedures. After 90 days from the last dose, the investigator should report any SAEs that are considered related to study drugs or procedures. Should new anti-tumor treatment is initiated within 90 days after the last dose, only SAEs related to study drugs will be recorded subsequently. The safety assessment of AEs and laboratory tests will be carried out by CTCAE V5.0.
14. Concomitant medication: Concomitant medication includes any prescription or over-the-counter drugs. Any drugs used by patients within 28 days before the treatment should be recorded at the time of screening. The changes in current medication or medication used after the previous medication records should be recorded during the follow-up visits.
15. RECIST V1.1 is mainly used for this study and the tumor imaging evaluation will be performed by IRRC and the investigator. The efficacy assessment of mRECIST will be carried out by IRRC. iRECIST is only used for confirmation 4 to 6 weeks after the first disease progression in patients from the treatment group, thereby excluding certain patients with pseudo-progression. Tumor imaging evaluations usually include contrast-enhanced CT or MRI and the sites for evaluation must include chest, abdomen, and pelvic cavity. The dual-phase imaging for liver must be obtained to obtain high-quality arterial phase imaging of the whole liver during the first scanning and high-quality portal vein phase imaging of the whole liver during the second scanning, should enhanced CT scanning be not contraindicated. Contrast-enhanced MRI or CT of the brain should also be performed at the baseline for patients with suspected signs and symptoms of CNS metastasis during the screening period. Individuals with metastasis to CNS cannot be enrolled. For patients without metastatic lesions before randomization, there is no need for routine cranial imaging examinations afterward. The same patient should receive the same type of imaging examination during the study. If there are clinical indications, other imaging examinations, such as bone scan and neck CT scan, should also be performed for evaluation. The baseline evaluation will be performed within 28 days before randomization. From randomization, the tumor imaging evaluation is performed once every 6 weeks (±7 days) until week 48 and once every 12 weeks (±7 days) thereafter. For patients in whom imaging progression is recorded for the first time, IRRC should confirm the progression within 3 weeks and repeat the imaging studies in 4–6 weeks after the first finding of progression to confirm the progression. See Section 7.3 for details.
16. Patients will receive IBI308 (200 mg, day 1) + IBI305 (15 mg/kg, day 1, Q3W) or sorafenib (400 mg, PO, BID) consecutively. The study treatment will continue until disease progression, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring the termination of treatment stipulated in the protocol, whichever occurs first. The IBI308 + IBI305 treatment will last 24 months at maximum.
17. Immunogenicity samples will be collected within 1 h before IBI308 infusion in cycles 1/2/4/8/12/16, every 8 cycles (cycle 24, 32...) thereafter, and during the safety follow-up visits. If an IBI308 and/or IBI305 infusion-related reaction occurs, blood samples should be taken after the start of the event with the least delay, end of event, and around 30 days after the event for immunogenicity analysis.

18. The PK sampling time points of the first 6 patients in the designated site are shown in Table 2. Note: Should more than 20 individuals are enrolled for infusion in phase II study, the number of individuals subjected to intensive sampling in phase III study can be reduced following the clinical pharmacological evaluation. The PK sampling points for other patients in the treatment group (except the first 6 patients in the designated site) are within 1 h before IBI308 infusion and immediately after IBI305 infusion (+5 min) in cycle 1 and within 1 h before IBI308 infusion in cycles 2/4/12. If a patient withdraws from the treatment between cycles 5–11, a PK sampling will be carried out in withdrawal visit.
19. The assessment of quality of life, involving EORTC QLQ-C30 and EORTC QLQ-HCC18, will be performed on the day of the first infusion (before the infusion), all imaging studies, and safety follow-up visits. A quality of life questionnaire should also be filled in as required in the meantime should an unscheduled imaging evaluation be performed.
20. For subjects with HCC pathological diagnosis results compliant with the criteria and being assigned to the treatment group, their tumor tissues conforming to the analysis requirements can be offered for the detection of biomarkers *e.g.*, PD-L1 expression (for IBI308 + IBI305 group only). The detection will take place at the central laboratory. Eligible archived tumor tissues: It is recommended that the surgical time is within half a year and the time of sectioning within 3 months (the time of surgery performed should not be over 1 year and the time of sectioning over half a year in principle) with at least 10 unstained sections (4–5 μm in thickness; containing a tissue area of 25 mm^2; tumor cells accounting for about 20% or more).

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LIST OF ABBREVIATIONS AND DEFINITIONS

Abbreviation	Definition
AE	Adverse event
AESI	Adverse event of special interest
AFP	Alpha-fetal protein
ALB	Albumin
ALP	Alkaline phosphatase
ALT	Alanine aminotransferase
ANC	Absolute neutrophil count
APTT	Activated partial thromboplastin time
AST	Aspartate aminotransferase
AUC	Area under curve
BCLC	Barcelona Clinic Liver Cancer
β-HCG	β human chorionic gonadotropin
BNP	Type B natriuretic peptide
CK	Creatine kinase
CK-MB	Creatine kinase isoenzymes
CRA	Clinical research associate
CRO	Contract research organization
CR	Complete remission
CT	Computed tomography
CTCAE	Common Terminology Criteria for Adverse Events
Cr	Creatinine
DCR	Disease control rate
DOR	Duration of response
EC	Ethics committee
ECOG PS	Eastern Cooperative Oncology Group Performance Status
eCRF	Electronic case report form
EDC	Electronic data capture
FAS	Full analysis set
FBG	Fasting blood glucose
FT3	Free triiodothyronine
FT4	Free thyroxine
GCP	Good Clinical Practice
γ-GT	γ-glutamyltransferase

Abbreviation	Definition
HBcAb	Hepatitis B core antibody
HBeAb	Hepatitis B e antibody
HBeAg	Hepatitis B e antigen
HBsAb	Hepatitis B surface antibody
HBsAg	Hepatitis B surface antigen
HBV	Hepatitis B virus
HCC	Hepatocellular carcinoma
HCV	Hepatitis C virus
HIV	Human immunodeficiency virus
HGB	Hemoglobin
iBOR	Immunity best overall response
ICF	Informed consent form
iCPD	Immunity confirmed progression disease
iCR	Immunity complete response
INR	International normalized ratio
iPR	Immunity partial response
irAE	Immunity related adverse event
iRECIST	Immune Response Evaluation Criteria in Solid Tumors
iSD	Immunity stable disease
IRRC	Independent radiological review committee
MRI	Magnetic resonance imaging
mRECIST	Modified RECIST
NAb	Neutralizing antibody
NYHA	New York Heart Association
ORR	Objective response rate
OS	Overall survival
PD	Pharmacodynamics
PD-1	Programmed cell death 1
PD-L1	Programmed cell death ligand 1
PD-L2	Programmed cell death ligand 2
PFS	Progression free survival
PK	Pharmacokinetics
PLT	Platelet
PPS	Per-protocol set
PR	Partial remission

Abbreviation	Definition
PT	Preferred Term
RBC	Red blood cell
RECIST	Response Evaluation Criteria in Solid Tumors
SAE	Serious adverse event
SAP	Statistical analysis plan
SOC	System Organ Class
SD	Stable disease
SS	Safety set
TBIL	Total bilirubin
TEAE	Treatment emergent adverse event
TP	Total protein
TRAE	Treatment related adverse event
TSH	Thyroid stimulating hormone
TTP	Time to progression
TTR	Time to response
ULN	Upper limit of normal value
UGLU	Urine glucose
UPRO	Urine protein
URBC	Urine red blood cell
UREA	Urea
UWBC	Urinary white blood cell
VEGF	Vascular endothelial growth factor
WBC	White blood cell

1 BACKGROUND

1.1 Disease Background

The incidence and mortality of liver cancer are increasing in spite of the continued decrease in the overall incidence and mortality of cancers. The incidence of liver cancer is much higher in males than in females. Liver cancer is the second leading cause of cancer-related deaths in men in developing countries and the sixth leading cause of cancer-related deaths in developed countries. It was estimated that there were 0.782 million new liver cancer cases and 0.745 million new deaths related to liver cancer worldwide in 2012, about half of which occurred in China. The most common primary liver malignancy is hepatocellular carcinoma (HCC, 70–90%) [1].

HCC occurs as a result of hepatitis B virus (HBV) and hepatitis C virus (HCV) infections, alcohol, and diet-related factors. Most HCC patients in China are suffered from such a disease due to chronic HBV infection leading to hepatic cirrhosis that progresses into HCC eventually. The majority of HCC has already been at the locally advanced or metastatic stage when it is observed and is thus unsuitable for radical treatment. These patients possess a poor prognosis with a median survival of merely 7.9 months under supportive treatment [2]. For patients from the Asia-Pacific region with HCC mainly resulted from viral hepatitis, their median survival stands at only 4.2 months [3]. Therefore, effective treatment options are urgently needed to meet the demand of patients with advanced HCC both at home and abroad.

Due to its heterogeneity and etiological diversity, HCC shows high resistance to conventional systemic chemotherapy. At present, only FOLFOX4 (fluorouracil, leucovorin, and oxaliplatin) regimen shows the trend to improve the overall survival (OS) through optimizing progression-free survival (PFS) and objective response rate (ORR) of patients with advanced HCC, as compared with the adriamycin-based chemotherapy, and such a chemotherapy regimen may offer the clinical benefits to certain patients [4]. In 2007, FDA approved the multi-target kinase inhibitor, sorafenib, for the treatment of inoperable patients with advanced HCC, with the targets including VEGFR1, VEGFR2, VEGFR3, PDGFR-β, c-kit, FLT-3, and RET. It is the first systemic drug and treatment regimen with proven significant survival benefits to advanced HCC patients and is the standard first-line treatment regimen currently as well [2]. However, the median OS and PFS in the sorafenib group were 10.7 months and 5.5 months, respectively, the effective rate stood at only 2%, while the rate of AEs was as high as 80%. In the REFLECT study, the efficacy of lenvatinib vs. sorafenib as the first-line treatment for inoperable patients with advanced liver cancer was compared and the results showed that OS in the lenvatinib group was not inferior to that in the sorafenib group (13.6 months vs. 12.3 months; HR 0.92) and PFS (7.4 months vs. 3.7 months; HR 0.66, $P < 0.0001$), time to progression (8.9 months vs. 3.7 months; HR 0.63, $P < 0.0001$), and ORR (18.8% vs. 6.5%; $P < 0.0001$) were superior to those in

the sorafenib group [5]. On Aug. 16, 2018, lenvatinib was approved by the U.S. FDA as the first-line treatment for HCC non-resectable. At present, the second-line treatment after failure of sorafenib treatment mainly includes regorafenib and ramucirumab targeting VEGFR2 in patients with α-fetoprotein (AFP) > 400 ng/mL. In the RESORCE study [6], the efficacy of regorafenib vs. placebo for the treatment of patients with the progression of advanced HCC after sorafenib treatment was compared and the results showed that OS (10.6 months vs 7.8 months; HR 0.63, P < 0.0001), PFS (3.1 months vs. 1.5 months; HR 0.46, P < 0.0001), and ORR (11% vs. 4%; one-sided P = 0.0047) improved by regorafenib were also superior to those in the placebo group. On Dec. 12, 2017, regorafenib was approved by CFDA (now NMPA) for the treatment of HCC patients who have previously been treated with sorafenib. In the REACH-2 study [7], the efficacy of ramucirumab vs. placebo in HCC patients with baseline AFP of 400 ng/mL or greater who were intolerant of sorafenib or who experienced disease progression after sorafenib administration was compared and the results showed that the ramucirumab group was superior to the placebo group with respect to OS (8.5 months vs. 7.3 months; HR 0.71, P = 0.0199) and PFS (2.8 months vs. 1.6 months; HR 0.452, P < 0.0001), and there was no difference between the two groups in ORR (4.6% vs. 1.1%; P = 0.1697). Currently, ramucirumab has not been approved as the second-line treatment for advanced HCC. Therefore, the main targets for the treatment of advanced HCC are focused on the VEGF/VEGFR2 pathway. As another important drug for this pathway, the anti-VEGF antibody bevacizumab (monotherapy or in combination with erlotinib/chemotherapy) also exhibits mild clinical activity in advanced HCC; however, there is still a paucity of sufficient data to support the use of bevacizumab in advanced HCC [8,9]. In another randomized, open-label, phase II study comparing bevacizumab + erlotinib vs. sorafenib as the first-line treatment for patients with advanced HCC, patients received the treatment with bevacizumab (10 mg/kg, Q2W) + erlotinib (150 mg, QD) (n = 47) or sorafenib (400 mg, BID) (n = 43). The primary endpoint was OS and the secondary endpoints included event-free survival (EFS) and ORR. The results showed that the median OS was identical for the two groups (8.55 months for both), while the median EFS was 4.37 months in the combination therapy group vs. 2.76 months in the sorafenib group (HR = 0.67). Specifically, patients with Child-Pugh Class A were subjected to subgroup analysis, which showed a median OS of 11.4 months (the combination therapy group, n = 39) vs. 10.26 months (the sorafenib group, n = 38) (HR = 0.88). The results suggested no difference in efficacy between bevacizumab + erlotinib and sorafenib [9].

Now, the immune checkpoint inhibitors also have begun to exhibit the therapeutic values in advanced HCC. Based on the CheckMate-040 study, FDA expedited the approval of the anti-programmed death receptor 1 (PD-1) monoclonal antibody nivolumab in 2017 for the treatment of advanced HCC patients previously treated with sorafenib. CheckMate-040 was an open-labeled, single-arm, dose-escalation/expansion phase I/II study. A total of 214 patients were included in the 3 mg/kg/2-week dose expansion cohort, including individuals without hepatitis

virus infection who are naive, intolerable, or resistant to sorafenib and those with hepatitis B or C virus infections. The patients treated with nivolumab had ORR of 20%, disease control rate of 64%, duration of remission of 17 months, and PFS of 4 months, while the median OS was not reached; for the dose-escalation group, the median OS was 15 months and the adequate safety profile was noted in the meantime [10]. The KEYNOTE-224 study was an open-labeled, single-arm, phase II study that enrolled a total of 104 patients who had failed in the sorafenib treatment or who could not tolerate such a treatment to receive pembrolizumab. The results showed objective effective rate of 17% (18/104), PFS of 4.9 months, and OS of 12.9 months [11].

In 2018, a phase Ib trial involving a PD-L1 monoclonal antibody in combination with an anti-VEGF monoclonal antibody as the first-line treatment for advanced liver cancer (NCT02715531) was reported by ASCO. In this study, patients with unresectable or metastatic HCC without previously systemic treatment were enrolled to undergo the treatment with anti-PD-L1 monoclonal antibody atezolizumab (1200 mg) + bevacizumab (15 mg/kg, Q3W) until no clinical benefits were seen or intolerable toxicities developed. The primary endpoint of the study was to evaluate the safety and tolerability of the combination therapy. The secondary endpoints of the study included ORR, PFS, DOR, and TTP as per RECIST V1.1. Specifically, safety evaluation results were obtained from 26 patients. The incidence of TRAEs was (81%, 21/26) and that of TRAEs at grade 3–4 was (35%, 9/26), of which hypertension was commonly seen (19%, 5/26). In the 21 patients with efficacy evaluation results (followed up for at least 16 weeks with a median follow-up of 8.3 months), 13 (62%) achieved confirmed partial remission (PR), which was not related to the cause of HCC, regions (Asia or the U.S.), the baseline AFP (≥ or <400 ng/mL), and the extrahepatic metastasis. Median PFS, DOR, TTP, and OS were not reached with the 6-month rates of PFS and OS being 65% and 86%, respectively. Atezolizumab in combination with bevacizumab was well tolerated in advanced HCC and no new toxicity was identified [12]. An effective rate as high as 62% indicated that the two antibodies work in a synergistic manner, to which the U.S. FDA granted a breakthrough designation. The phase III expansion study corresponding to this phase Ib study is currently underway.

In 2018, a phase Ib trial involving pembrolizumab in combination with lenvatinib for the treatment of unresectable HCC was reported by ASCO, with an unconfirmed ORR being 42.3% (11/26) and PFS being 9.69 months as per the modified RECIST (mRECIST) for HCC [13]. In addition, the ASCO also reported in 2018 the results of PD-1 monoclonal antibody SHR-1210 in combination with apatinib for the treatment of patients with advanced HCC failing the standard treatment. The confirmed ORR stood at 38.9% (7/18), the PFS reached 7.2 months, and good safety and tolerance profiles were seen [14].

Overall, for the treatment of advanced HCC currently, neither the multi-target kinase inhibitor targeting the VEGF/VEGFR2 pathway, nor the immune checkpoint inhibitor, such as the

anti-PD-1/PD-L1 monoclonal antibody, can yield a satisfactory ORR or ideal improvement in OS as a monotherapy. While, the anti-PD-1/PD-L1 monoclonal antibody in combination with the anti-angiogenesis targeted drug (anti-VEGF monoclonal antibody) for treating advanced HCC has shown preliminary good safety and encouraging efficacy, and thus, further explorations are merited.

1.2 Study Drugs

1.2.1 IBI308

1.2.1.1 Mechanism of action

Sintilimab (R&D code, IBI308) is a recombinant fully humanized IgG4 PD-1 monoclonal antibody developed by Innovent Biologics, Inc. (Suzhou) (the sponsor).

After PD-1 expressed on activated T cells binds to programmed cell death receptor ligand 1 (PD-L1)/programmed cell death receptor ligand 2 (PD-L2) on antigen-presenting cells or tumor cells, the resulting signals inhibit proliferation of T cells and secretion of active cytokines and reduce the function of T cells. Therefore, tumor cells can escape the killing effect of T cells. IBI308 binds to PD-1 and blocks the binding of PD-1 to PD-L1 and PD-L2, removing the immunosuppressive effects, activating the function of T cells, and enhancing the immune surveillance and killing capabilities of T cells against tumors, which exerts anti-tumor immune responses. Multiple nonclinical studies *in vitro* have validated the effects of IBI308 on PD-1 pathway blockade and IBI308 was subjected to the clinical study phase in Sep. 2016.

1.2.1.2 Clinical study results

After the clinical trial approval of IBI308 was obtained in Sep. 2016, more than 10 clinical trials have been conducted in advanced cancer patients in China.

The CIBI308A101-Ib study was a multi-center, phase Ib clinical study involving patients with advanced solid tumors in China. Cohort B mainly included advanced HCC patients who had undergone such second-line treatment and above as transcatheter arterial chemoembolization (TACE), sorafenib, and systemic chemotherapy. Specifically, 19 patients received IBI308 as monotherapy and had imaging evaluation results; PR, SD, and PD were achieved in 3, 6, and 10 patients, respectively, and ORR and disease control rate reached 15.7% and 47.4%, respectively. Patients showed good safety profile and were tolerated well.

The IBI308 efficacy data are primarily derived from a pivotal, single-arm phase II clinical trial (ORIENT-1) involving patients with classical Hodgkin's lymphoma (cHL). With Oct. 16, 2018 as the data cut-off date, the analysis was carried out after the last subject completing 12 months of observation. The primary efficacy endpoint was assessed by IRRC. The objective response rate in the full analysis set (FAS) (n = 75) was 85.3% (95% CI; lower limit: 75.3%) based on

IWG 2007 Criteria and 78.7% (95% CI; lower limit: 67.7%) based on Lugano 2014 Criteria for Assessing PET/CT. The objective response rate in FAS, as assessed by the investigator, was 84.0% (95% CI; lower limit: 73.7%) based on IWG 2007 Criteria and 81.3% (95% CI; lower limit: 70.7%) based on Lugano 2014 Criteria for Assessing PET/CT. As assessed by IRRC, the median time to response was 42.5 days in FAS and the median DOR and PFS have not been reached. Relapsed/refractory cHL is a malignancy with poor prognosis. IBI308 monotherapy is effective in treating this population. It was, therefore, approved for marketing on Dec. 24, 2018. And its first indication is relapsed or refractory cHL after the failure of at least second-line systemic chemotherapy. The relevant results have been published. Multiple phase III clinical trials are being carried out.

Data in the safety summary are derived from the 5 IBI308 studies, including CIBI308B201. As of Aug. 31, 2018, a total of 540 subjects received at least 1 dose of IBI308, and these patients are included in the safety analysis. Among the 540 subjects, 34.4% had NSCLC, 19.8% esophageal cancer, and 17.8% Hodgkin's lymphoma. A total of 71.3% of subjects (385/540) experienced IBI308 treatment-related adverse events (TRAEs), among which the most common AEs (≥ 10%) included fever (13.1%), increased AST (10.6%), and increased ALT (10.2%). A total of 15.7% (85/540) subjects had TRAEs at grade 3 or greater, among which the most common AEs (≥ 1%) were increased lipase (2.4%), lung infections (2.0%), pneumonitis (1.7%), and decreased platelet count (1.1%). For detailed safety data, refer to the Investigator's Brochure of IBI308.

1.2.2 IBI305

1.2.2.1 Mechanism of action

As a recombinant anti-VEGF humanized monoclonal antibody injection, IBI305, the biosimilar, is developed by the sponsor with the generic name of bevacizumab injection that specifically binds to human VEGF-A, inhibits the binding of VEGF-A to VEGF-R1 and VEGF-R2, blocks signaling pathways such as PI3K-Akt/PKB and Ras-Raf-MEK-ERK, impedes the growth, proliferation, and migration of vascular endothelial cells and angiogenesis, decreases the vascular permeability, obstructs blood supply to tumor tissues, inhibits the proliferation and metastasis of tumor cells, and induces the apoptosis of tumor cells, thereby generating anti-tumor effects.

According to the various study results of CMC, pharmacodynamics, and pharmacokinetics concerning animals, IBI305 is highly similar to bevacizumab, and therefore, it may also share similar clinical efficacy and safety. Then, the corresponding clinical study is carried out.

1.2.2.2 Clinical study results

In the phase I clinical study, a total of 100 patients were randomized to IBI305 group and bevacizumab group in a 1:1 ratio. The results showed that the trends of the drug concentration-time curves of IBI305 and bevacizumab were consistent with no significant difference. The PK parameters were not significantly different between IBI305 group and bevacizumab group. The results of bioequivalence analysis showed that the main PK parameters [Area under drug concentration-time curve from 0 to infinity ($AUC_{0-\infty}$), area under drug concentration-time curve from 0 to t h (AUC_{0-t}), and C_{max}] of IBI305 group and bevacizumab group were within the acceptable range of equivalence (80–125%), indicating that IBI305 and bevacizumab are bioequivalent.

The most common treatment emergent adverse events (TEAEs) related to IBI305 in the study included hypertriglyceridemia, increased ALT, hyperglycemia, increased AST, increased white blood cell (WBC) count, increased neutrophil count, decreased WBC count, positive urine red blood cells (URBC), sinus bradycardia, decreased neutrophil count, headache, and intermediate insomnia. Among them, the severity levels of hypertriglyceridemia, AST increased, and neutrophil count decreased were grade 3.

The clinical study on efficacy was randomized, double-blind, and multi-center. A total of 450 previously untreated patient with advanced or recurrent non-squamous non-small cell lung cancer were enrolled with 224 in IBI305 group and 226 in bevacizumab group. The trial aimed to compare the efficacy and safety of IBI305 in combination with paclitaxel/carboplatin vs. the RLD bevacizumab in combination with paclitaxel/carboplatin.

With respect to efficacy, the IRRC-evaluated ORRs in FAS as per RECIST v1.1 were 44.3% in IBI305 group and 46.4% in bevacizumab group. The ORR ratio between the two groups (including stratification factor) was 0.95 with 90% CI of 0.803–1.135 falling within the predetermined margin (0.75, 1/0.75 ≈ 1.33) specified in the statistical analysis plan, demonstrating the equivalence hypothesis between the two groups. In FAS, 90.5% and 87.7% in DCR, 7.3 months and 6.9 months in median TTR, and 8.4 months and 8.3 months in median PFS are shown in IBI305 group and bevacizumab group with no significant differences.

In terms of safety for IBI305 group and bevacizumab group, the incidences of TEAE were both 100% and no statistically significant differences ($P > 0.05$) between the two groups were noted in the incidence of grade 3 or greater TEAE (84.4% vs. 89.8%), treatment-emergent SAE (TESAE) (33.5% vs. 37.6%), significant TEAE (96.4% vs. 98.2%), TEAE leading to death (2.7% vs. 2.2%), TEAE leading to mAb-treatment termination (12.5% vs. 9.3%), TEAE leading to mAb-treatment interruption (44.2% vs. 45.1%), and AE of specific interest (AESI) (53.6% vs. 62.4%).

In SS, significant AEs with incidences ≥ 10% in IBI305 group included white blood cell count decreased (71.9%), neutrophil count decreased (70.5%), platelet count decreased (50.4%), anemia (23.7%), constipation (19.6%), nausea (13.8%), hypertension (11.2%), hypoesthesia (11.2%), AST increased (10.7%), upper respiratory tract infection (10.3%), and ALT increased (10.3%). Significant AEs with incidences ≥ 10% in bevacizumab group included white blood cell count decreased (79.2%), neutrophil count decreased (74.8%), platelet count decreased (43.4%), anemia (23.9%), nausea (23.0%), constipation (22.1%), pyrexia (14.6%), diarrhea (13.3%), hypertension (11.1%), hypoesthesia (10.6%), and vomiting (10.6%). For detailed safety data, refer to the Investigator's Brochure of IBI305.

The results of clinical PK and efficacy comparability studies between IBI305 and bevacizumab showed that the two shared comparable PK characteristics, efficacy, safety, and immunogenicity. Therefore, they are expected to possess similar clinical efficacy and safety, and the clinical trials can be carried out based on the clinical study data of the bevacizumab injection.

1.3 Risk/Benefit Assessment

1.3.1 Potential risks

Given the mechanism of action underlying IBI308 and the clinical safety information on products with similar mechanisms, the main AEs during this clinical trial will possibly be the immune-mediated inflammatory resulted from the activation of immune system, *e.g.* pneumonitis, colitis, hepatitis, renal insufficiency, and endocrine events. According to the available clinical data, anti-PD-1 monoclonal antibodies are well-tolerated despite of high incidence of adverse reactions. Treatment termination due to adverse reactions only occurs in a small number of patients and most events resolve after appropriate interventions. Early symptoms of immune-related adverse events (irAEs) vary. Therefore, the investigator must closely monitor early signs and symptoms of irAEs during the trial, make correct judgments timely, adjust the dose according to Section 5.4 in the protocol, and provide effective treatment measures to reduce the subject risks. Meanwhile, subjects with autoimmune diseases should be excluded from the trial to avoid the original disease aggravation due to the activation of immune system.

IBI305 is a recombinant anti-vascular endothelial growth factor (VEGF) humanized monoclonal antibody developed based on bevacizumab. Throughout the development process, a comprehensive comparison was carried out with bevacizumab and IBI305 complies with the national regulations on the definition of biosimilars. The two drugs are identical in amino acid sequences and comparable in CMC, pharmacodynamics, characteristics of preclinical toxicity, and clinical pharmacokinetic and safety characteristics. Marketed for more than 1 decade, bevacizumab injection possesses the efficacy that has been fully validated in the treatment of various tumors and has become one of the most common anti-tumor drugs. The most common

AEs of bevacizumab include hypertension, fatigue or weakness, and diarrhea and abdominal pain. Therefore, it is estimated that IBI305 will share comparable safety characteristics as bevacizumab and can be administered as per the safety management measures of bevacizumab.

IMpower150 was a multi-center, open-labeled, randomized, controlled study evaluating the efficacy and safety of atezolizumab and bevacizumab in combination with chemotherapy (paclitaxel and carboplatin) in patients with stage IV non-squamous non-small cell lung cancer who had never received chemotherapy. A total of 1202 patients were enrolled and assigned to three groups (A, B, and C) with patients in group B receiving atezolizumab in combination with bevacizumab and chemotherapy and patients in group C receiving bevacizumab in combination with chemotherapy. The safety data of IMpower150 showed that in group B and group C, the incidences of TRAEs at grades 1–2 were 35.9% and 45.4% while those at grades 3–4 were 55.7% and 47.7%, respectively, with no significant differences. 77.4% of irAEs were at grades 1–2 [15]. The safety data of the phase Ib study involving atezolizumab in combination with bevacizumab as the first-line treatment for advanced HCC showed that the incidence of TRAEs was (81%, 21/26) and that of TRAEs at grades 3–4 was (35%, 9/26), of which hypertension (19%, 5/26) was the most common event; no new toxic reactions were identified. Based on the above study results, it is expected that IBI308 in combination with IBI305 will not lead to additional risks to patients.

1.3.2 Potential benefits

The results of basic studies showed that the T cell-mediated tumor cell killing effect of anti-PD-1 monoclonal antibody can be strengthened by bevacizumab reversing VEGF-mediated tumor immunosuppression. The initiation and activation of T cells is promoted and the vascularization of tumor normalized by the maturation of dendritic cells; the tumor immune microenvironment is regulated by down-regulating myeloid-derived suppressor cells (MDSCs) and regulatory T cells (Treg) clusters [14]. The above mechanisms suggest that the anti-VEGF monoclonal antibody can strengthen the tumor cell killing effects of anti-PD-1/PD-L1 monoclonal antibody.

In the first-line treatment of advanced lung cancer, clinical study IMpower150 confirmed the above-mentioned conclusions. The anti-PD-L1 monoclonal antibody and bevacizumab in combination with chemotherapy group were compared with the bevacizumab in combination with chemotherapy group with PFS of 8.3 months vs. 6.8 months (HR = 0.62, $P < 0.001$) and median OS of 19.2 months vs. 14.7 months (HR = 0.78, and $P = 0.02$), reaching the common primary endpoints of the study.

The phase Ib trial involving atezolizumab in combination with bevacizumab as the first-line treatment for advanced liver cancer (NCT02715531) further validated the above conclusions in HCC patients. Among the 21 patients with efficacy evaluation results, confirmed PR was achieved in 13 individuals (62%). The report results of the ESMO conference in 2018 showed

that atezolizumab + bevacizumab exhibited good efficacy in patients with advanced HCC. Among the 73 evaluable patients, complete remission (CR) was achieved in 1 patient and partial remission (PR) in 30 patients as assessed by the investigator as per RECIST1.1, with a confirmed overall ORR of 32%; median PFS was 14.9 months, 6-month PFS was attained in 65% of patients, and OS was not reached. Sustained response was realized in 52% of patients for more than 6 months and in 26% of patients for more than 12 months [16]. Atezolizumab in combination with bevacizumab was well tolerated in advanced HCC and no new toxicity was identified. The phase III expansion study IMbrave150 corresponding to this phase Ib study is underway at present (NCT03434379).

Therefore, IBI308 in combination with IBI305 may exhibit synergistic anti-tumor effects and may confer benefits to patients with advanced HCC.

1.3.3 Assessment of potential risks and benefits

Based on the above analysis, the toxic reactions of IBI308 in combination with IBI305 are controllable and may offer the maximum benefits to patients with advanced HCC. This study possesses controllable risks and predictable efficacy benefits.

2 STUDY OBJECTIVES

2.1 Primary Objectives

Phase II single-arm study, safety run-in period:

- To evaluate the safety and tolerability of IBI308 in combination with IBI305 as the first-line treatment for patients with advanced HCC;

Phase III randomized, controlled study:

- To compare the overall survival (OS) of IBI308 in combination with IBI305 vs. sorafenib as the first-line treatment for advanced HCC;
- To compare PFS (assessed by IRRC as per RECIST V1.1) of IBI308 in combination with IBI305 vs. sorafenib as the first-line treatment for advanced HCC.

2.2 Secondary Objectives

Phase II single-arm study, safety run-in period:

- To evaluate the preliminary efficacy (assessed by the investigator as per RECIST V1.1) of IBI308 in combination with IBI305 as the first-line treatment for patients with advanced HCC. The endpoints include ORR, DCR, DOR, TTP, TTR, PFS, and OS.
- To evaluate the PK characteristics of IBI308 and IBI305 in patients with advanced HCC.
- To evaluate the immunogenicity of IBI308 and IBI305 in patients with advanced HCC.

Phase III randomized, controlled study:

- To compare PFS determined by the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group;
- To compare ORR determined by IRRC and the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group;
- To compare TTP determined by IRRC and the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group;
- To compare DOR determined by IRRC and the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group;
- To compare DCR determined by IRRC and the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group;
- To compare TTR determined by IRRC and the investigator as per RECIST V1.1 in IBI308 + IBI305 group vs. sorafenib group;
- To compare PFS, ORR, TTP, DOR, DCR, and TTR determined by IRRC as per modified RECIST (mRECIST) in IBI308+ IBI305 group vs. sorafenib group;
- To compare the safety of IBI308 in combination with IBI305 vs. sorafenib as the first-line treatment for advanced HCC;
- To compare quality of life after IBI308 in combination with IBI305 vs. sorafenib as the first-line treatment for advanced HCC, using European Organisation for Research and Treatment of Cancer Quality-of-Life Questionnaire-Core 30 (EORTC QLQ-C30) and European Organisation for Research and Treatment of Cancer Quality-of-Life Questionnaire—hepatocellular carcinoma 18 (EORTC QLQ-HCC18);
- To evaluate the immunogenicity of IBI308 in combination with IBI305 as the first-line treatment for advanced HCC.

2.3 Exploratory Objectives

Phase III

- To explore the relationship between the potential biomarkers with efficacy predictability in tissues and efficacy:
  1) Expression level of PD-L1;
  2) Detection of RNA in tissues;
- To evaluate the population pharmacokinetic characteristics of IBI308 + IBI305 as the first-line treatment for advanced HCC.

3 STUDY DESIGN

3.1 Overall Design

This is a randomized, controlled, open-label, multi-center study evaluating the treatment of IBI308 in combination with IBI305 in previously-untreated Chinese patients with advanced HCC. The efficacy and safety of such a combination regimen will be evaluated.

The study is composed of 2 sections. Section I is a phase II single-arm study (safety run-in period). Section II is a phase III, randomized, controlled study.

Section I: phase II, single-arm study (safety run-in period). A total of 20 patients with advanced HCC will be enrolled to receive the first-line treatment, IBI308 (200 mg, IV, day 1, Q3W) + IBI305 (15 mg/kg, IV, day 1, Q3W). The treatment will continue until disease progression, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring the termination of treatment stipulated in the protocol, whichever occurs first. Due to the possibility of pseudo-progression in certain patients, the confirmation of progression based on iRECIST after 4 to 6 weeks is required for patients with the first time of imaging progression assessed by the investigator as per RECIST V1.1. Whether drug treatment will be continued or not while awaiting imaging confirmation is to be determined by the investigator's evaluation, and iRECIST will be adopted for the evaluation and recording of efficacy, until iCPD as per iRECIST, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring termination of treatment stipulated in the protocol, whichever occurs first. The patients in the study will receive IBI308 + IBI305 for a maximum of 24 months. The judgment of iCPD is defined as: 1. The lesions of previously undetermined progressive disease (iUPD) further deteriorate, *e.g.*, progressed target lesions are increase by at least 5 mm in diameter anew, new target lesions as found previously are increased by at least 5 mm in diameter, any degree of enlargement of target lesions is observed, or progressed non-target lesions further progress. 2. Non-iUPD lesions show progressions conforming to RECIST V1.1. For individuals whose iUPD cannot be determined, the reasons why disease progression is not or cannot be confirmed should be recorded. See Appendix 5 for details.

The safety and tolerability of combination treatment will be evaluated and ORR, DCR, DOR, TTP, TTR, PFS, and OS of IBI308 in combination with IBI305 as the first-line treatment for patients with advanced HCC will be assessed based on the evaluation by the investigator as per RECIST V1.1. Based on the preliminary safety and efficacy data of phase II single-arm study, a phase III, randomized, controlled study will be carried out following the discussion and decision by the Safety Evaluation Committee.

By referring to the safety data as reported in similar foreign products in studies, TRAEs at grade 3 or above as per CTCAE were observed in 17/68 patients receiving atezolizumab + bevacizumab (incidence rate, 25%; 95% confidence interval, 15–37%). Therefore, it is predicted in the phase II design of this study that the safety profile of IBI308 in combination with IBI305 will be similar to that of products of the same kind and it is proposed to enroll a total of 20 patients for preliminary safety assessment. Assuming that observation shows over 8 (40%) of the 20 patients receiving IBI308 + IBI305 have TRAEs at grade 3 or above as per CTCAE, there is about 90% power in inferring that the true incidence of such AEs exceeds 29%, namely, the incidence of TRAEs at grade 3 or above as per CTCAE arising from IBI308 + IBI305 group may be higher than estimation.

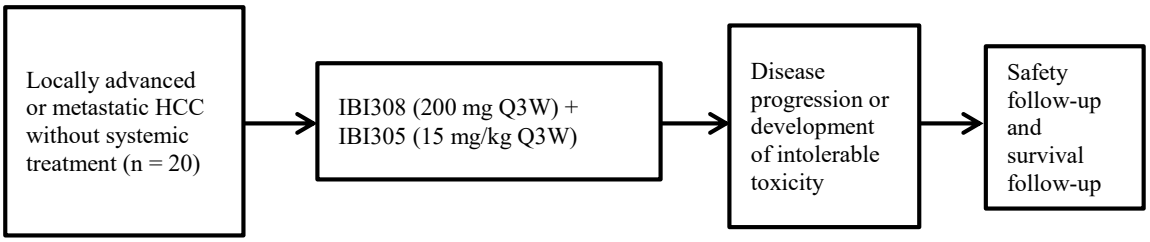

Figure 4. Flow chart of phase II single-arm study

Section II is a phase III, randomized, controlled study. About 546 patients with locally advanced or metastatic HCC who have not received systemic anti-tumor therapy and who are unsuitable for radical surgery/local treatment or show disease progression after the above treatment will be enrolled. After the signing of informed consent form, patients eligible in the screening period will be randomly assigned to the treatment group (IBI308 in combination with IBI305) or the control group (sorafenib) in a 2:1 ratio. The stratification factors include macrovascular invasion and/or extrahepatic metastasis (presence vs. absence), baseline AFP (< 400 ng/mL vs. ≥ 400 ng/mL), and the score of ECOG performance status (0 vs. 1).

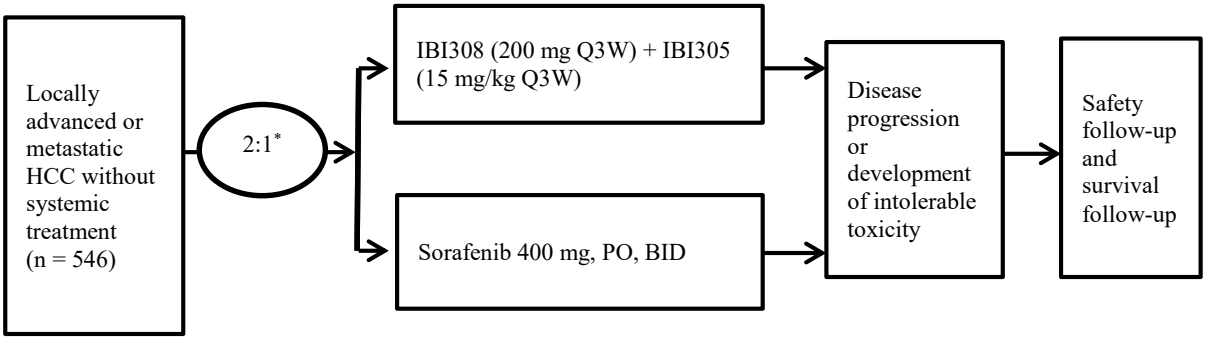

Figure 5. Flow chart of phase III randomized, controlled study

The patients in the treatment group will receive IBI308 (200 mg, IV, day 1, Q3W) + IBI305 (15 mg/kg, IV, day 1, Q3W). The study treatment will continue for 24 months at maximum or until disease progression, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring the termination of treatment stipulated in the protocol, whichever occurs first.

After the investigator's first evaluation of imaging progression in treatment group according to RECIST V1.1, disease progression should be confirmed by IRRC within 3 weeks. Due to the possibility of pseudo-progression in certain patients, the confirmation of progression based on iRECIST after 4 to 6 weeks is required for patients subjected to disease progression for the first time. Whether drug treatment will be continued or not while awaiting imaging confirmation is to be determined by the investigator's evaluation, and iRECIST will be adopted for the evaluation and recording of efficacy, until iCPD as per iRECIST, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring termination of treatment stipulated in the protocol, whichever occurs first. The judgment of iCPD is defined as: 1. The lesions of previously undetermined progressive disease (iUPD) further deteriorate, *e.g.*, progressed target lesions are increase by at least 5 mm in diameter anew, new target lesions as found previously are increased by at least 5 mm in diameter, any degree of enlargement of target lesions is observed, or progressed non-target lesions further progress. 2. Non-iUPD lesions show progressions conforming to RECIST V1.1. See Appendix 5 for details.

For individuals whose iUPD cannot be determined, the reasons why disease progression is not or cannot be confirmed should be recorded. Pseudo-progression confirmed by IRRC and subsequent PR and CR in imaging studies must be recorded in the study records and ORR, PFS, and DOR will be subjected to correction analysis in the final efficacy analysis.

The subjects in control group will receive sorafenib (400 mg, BID, PO) consecutively until disease progression, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring the termination of treatment stipulated in the protocol, whichever occurs first. In addition, the optimal replacement treatment will be given to patients according to the investigator's comprehensive judgment and the follow-up treatment methods will be recorded. After the investigator's first evaluation of imaging progression according to RECIST V1.1, disease progression should be confirmed by IRRC within 3 weeks. Whether drug treatment will be continued or not while awaiting imaging confirmation is to be determined by the investigator.

Although some patients have reached PD based on RECIST V1.1 or iCPD based on iRECIST (applicable to the treatment group), these patients can still be given the treatment based on the investigator's judgment under the premise that the patients agree to continue the treatment and

have signed the informed consent forms if the clinical status of the patients is stable with no adverse events (AEs) requiring termination of the study drugs. The treatment can be continued until the occurrence of any of the conditions, including no benefit or onset of intolerable toxicity assessed by the investigator, initiation of a new anti-tumor therapy, withdrawal of ICF by the patient, and loss to follow-up, whichever occurs first.

3.2 Rationale for Study Design

3.2.1 Rationale for selection of phase II, single-arm study as safety run-in period

In the international phase Ib trial involving PD-L1 monoclonal antibody in combination with anti-VEGF monoclonal antibody as the first-line treatment for advanced liver cancer (NCT02715531), no TRAE at grade 5 was observed, no unexpected safety signal was noted, the overall safety was controllable, and the combination therapy was well tolerated. The open-labeled, randomized phase III study involving atezolizumab in combination with bevacizumab as the first-line treatment for HCC is currently ongoing (NCT03434379). A comparison was carried out between atezolizumab (1200 mg, IV, day 1, Q3W) in combination with bevacizumab (15 mg/kg, IV, day 1, Q3W) vs. sorafenib (400 mg, PO, BID). Due to the limited data with respect to these two types of drugs in combination, it is necessary to explore the safety and tolerability of IBI308 + IBI305 for advanced liver cancer to ensure patient safety and to provide support for subsequent studies. Therefore, it is necessary to conduct an exploratory phase II study as the safety run-in period before phase III study.

3.2.2 Rationale for selection of open-labeled study design for phase III study

Due to the relatively huge differences in the characteristics of adverse reactions arising from the study treatment between the two groups, both patients and the investigator can still make the groups out easily even if blinding is performed. In addition, a double-blind/double-dummy design is required to ensure blinding. On the one hand, additional burdens will be brought by the infusion of IBI308 and placebo to patients in the sorafenib treatment group; on the other hand, it is difficult to make the dummy sorafenib preparation and keep it identical to sorafenib in appearance. Therefore, this study is designed as open-labeled. In addition, this study will be assessed by IRRC with respect to such endpoints as ORR and OS to ensure the effectiveness of data in this open-labeled study.

3.2.3 Rationale for selection of sorafenib as control group for phase III study

Patients in the control group will receive sorafenib. Since 2007, sorafenib has been approved as the first-line standard treatment globally for patients with advanced HCC. Two randomized phase III trials involving Western and Eastern populations confirmed the survival benefits of sorafenib over placebo [2, 3]. Sorafenib is the initial standard of care for patients with advanced HCC as recommended in the U.S. NCCN guidelines and "Diagnosis and Treatment Norms for Primary

Liver Cancer" in China.

Although lenvatinib was approved by the U.S. FDA and NMPA, China on Aug. 17, 2018 and Sep. 4, 2018, respectively, both domestic investigators and patients are very limited in experience in the use of lenvatinib. Therefore, sorafenib is still an appropriate option as a comparator.

3.2.4 Principle for selection of IBI305 in combination with IBI308

Currently, the therapeutic target of advanced HCC is mainly the VEGF/VEGFR2 pathway. The anti-VEGF monoclonal antibodies have shown mild anti-tumor activity in advanced HCC and the anti-PD-1 monoclonal antibodies as the immune checkpoint inhibitors have also shown adequate prospects in treating HCC. The results of basic studies showed that anti-VEGF monoclonal antibodies can promote the normalization of tumor vessels, reverse VEGF-mediated tumor immunosuppression, and regulate the tumor immune microenvironment by down-regulating MDSCs and Treg, thereby promoting the T cell-mediated tumor cell killing effects of anti-PD-1 monoclonal antibodies [17]. Therefore, anti-PD-1 monoclonal antibody in combination with anti-VEGF monoclonal antibody may serve as an effective solution. In 2018, the ASCO reported the results of a series of trials involving anti-PD-1/PD-L1 monoclonal antibodies in combination with anti-angiogenic drugs for the treatment of advanced HCC [12–14]. Therefore, we believe that IBI305 in combination with IBI308 represents a reasonable therapy as the first-line treatment for advanced HCC.

3.2.5 Rationale for selection of exploratory biomarkers in phase III study

Biomarkers serve as an important means to achieve precision treatment. In the process of propelling the immune system to kill the tumor cells *in vivo*, immunotherapy is correlated with tumor cells, immune system, and tumor microenvironment to a certain extent. With tissues as the base for evaluation, this study will explore the relationship between biomarkers and immunotherapy. The specific study is as follows:

- To explore the relationship between the potential biomarkers with efficacy predictability in tissues and efficacy:

1) Expression level of PD-L1: Tumor cells activate the immune suppression system by expressing PD-L1, thereby achieving the goal of immune escape. IBI308 nullifies immunosuppression by blocking the binding of PD-1 to PD-L1, thus achieving the tumor-killing effects. Therefore, PD-L1 expressed on the surface of tumor cells may serve as an important biomarker for indicating the efficacy of IBI308. Both the clinical studies of KEYNOTE-024 and KEYNOTE-042 have shown a certain correlation between expression level of PD-L1 and efficacy of immunosuppressants. By using the tissue samples from patients before administration, this study retrospectively investigated the correlation between expression level of PD-L1 on the surface of tumor cells and IBI308 immunotherapy.

2) RNA detection in tissues: It has been reported that the expression level or abnormal expression of some immune-related genes in liver cancer is correlated with efficacy or post-recovery situation to a certain extent [18]. In this study, we aim to, at the level of RNA, explore the correlation between expression level of RNA in tumor microenvironment and efficacy.

3.3 Independent Data Monitoring Committee

The independent Data Monitoring Committee (iDMC) is established in the study to conduct one (or two, if applicable) interim analysis/analyses of primary efficacy endpoints (OS, PFS) and safety endpoints in phase III study as scheduled in the study protocol. The data analysis report required by iDMC will be provided by an independent statistician; iDMC will determine the treatment efficacy based on estimated effective boundaries in the study and provide suggestions to the sponsor about the submission of study data in advance. Refer to the iDMC charter for the detailed members, responsibilities, and procedures of iDMC.

3.4 Study Termination

Phase II:

The study will be terminated with the permission of the sponsor after the discussion and decision by the Safety Evaluation Committee due to the occurrence of events concerning major safety and tolerability.

Phase III:

Study termination is defined as the time point at which data collection is terminated and final analysis of this study is carried out. The main analysis of efficacy and safety will be conducted when the predetermined number of deaths is reached. The study will continue until death, loss to follow-up, or withdrawal from the study of the last patient, or the sponsor decides to terminate.

Phases II/III:

This study may be interrupted or terminated if sufficient grounds are offered. The permission by the sponsor should be obtained before the interruption or termination of the study. The party interrupting or terminating the study should provide the written notification to patients, the investigator, funding agencies, and regulatory authorities, and document the reasons for interruption or termination. If the study is prematurely terminated or interrupted, the principal investigator must notify patients, the ethics committee (EC), and the sponsor immediately, and provide the reasons for termination or interruption. Where applicable, the investigator should contact patients and inform them of the changes in visit schedule.

Reasons for study termination or interruption include but are not limited to:

- Unexpected, significant, or unacceptable risks to patients are identified;
- The efficacy rationale indicates the necessity of study termination and interruption;
- It fails to meet protocol requirements for compliance;
- Data is incomplete and/or insufficient for evaluation;
- It is confirmed that the primary endpoint is met.

The study may be resumed only if the safety, protocol compliance, and data quality issues have been addressed and the requirements of the sponsor, EC, and/or NMPA are met.

4 STUDY POPULATION

4.1 Inclusion Criteria

1. Histologically/cytologically confirmed HCC or cirrhosis meeting the clinical diagnostic criteria of HCC by American Association for the Study of Liver Diseases (AASLD).
2. Ages of ≥ 18 years old.
3. ECOG PS of 0 or 1.
4. Barcelona Clinic Liver Cancer (BCLC) stage of C. Stage B for those unsuitable for radical surgery and/or local treatment.
5. Absence of systemic anti-tumor treatment for HCC before the first dose.
6. At least one measurable lesion according to RECIST V1.1, or measurable lesions showing definite progression following the local treatment based on RECIST V1.1.
7. Child-Pugh score of ≤ 7.
8. Sufficient organ and bone marrow functions, with the laboratory test values within 7 days before the enrollment meeting the following requirements (no blood components, cell growth factors, albumin, and other drugs via intravenous or subcutaneous administrations are allowed for correction treatment within the first 14 days after the laboratory test results are obtained). The specific information is as follows:
   1) Routine blood test: Absolute neutrophil count (ANC) ≥ 1.5×10^9/L; platelet count (PLT) ≥ 75×10^9/L; hemoglobin (HGB) ≥ 9.0 g/dL.
   2) Hepatic function: total bilirubin (TBIL) ≤ 2 × upper limit of normal (ULN), alanine aminotransferase (ALT) and aspartate aminotransferase (AST) ≤ 5 × ULN; serum albumin ≥ 28 g/L; alkaline phosphatase (ALP) ≤ 5 × ULN.

3) Renal function: serum creatinine (Cr) ≤ 1.5 × ULN or clearance of creatinine (CCr) ≥ 50 mL/min (Cockcroft-Gault formula); urinalysis results showing urine protein < 2+; patients whose baseline urinalysis results show urine protein ≥ 2+ should undergo 24-h urine collection and 24-h urine protein quantitation test result should be < 1 g.

4) Blood coagulation function: international normalized ratio (INR) and activated partial thromboplastin time (APTT) ≤ 1.5 × ULN.

9. Expected survival of ≥ 12 weeks.
10. Female patients of childbearing age or male patients with female sexual partners of childbearing age should take effective contraceptive measures throughout the treatment and 6 months after the last dose.
11. Patients who have signed ICF and are able to perform follow-up visits and relevant procedures required in the protocol.

4.2 Exclusion Criteria

1. Histologically/cytologically confirmed fibrolamellar hepatocellular carcinoma, sarcomatoid hepatocellular carcinoma, and cholangiocarcinoma.
2. History of hepatic encephalopathy or liver transplantation.
3. Symptomatic pleural effusion, ascites, and pericardial effusion that require drainage.
4. Acute or chronic active hepatitis B or C infection; hepatitis B virus (HBV) DNA > 2000 IU/mL or 10^4 copies/mL; hepatitis C virus (HCV) RNA > 10^3 copies/mL; hepatitis B surface antigen (HbsAg) and anti-HCV antibody positive concurrently. Those who possess the indicators lower than the above criteria after nucleotide antiviral treatment can be enrolled.
5. Presence of metastasis to the central nervous system.
6. Presence of bleeding events from esophageal or gastric varices caused by portal hypertension within the past 6 months. Presence of known severe (G3) varicose veins in endoscopy within 3 months before the first dose. Evidence of portal hypertension (including the finding of splenomegaly in imaging studies) with a high risk of bleeding assessed by the investigator.
7. Presence of any life-threatening bleeding events within the past 3 months, including the need for transfusion, surgery or local treatment, and continuous medication therapy.
8. Any arterial/venous thromboembolic events within 6 months, including myocardial infarction, unstable angina, cerebrovascular accident or transient cerebral ischemic attack, pulmonary embolism, deep vein thrombosis, or any other history of serious thromboembolism. Presence of implantable venous port or catheter-derived thrombosis, or

superficial venous thrombosis, barring stable thrombosis following the conventional anticoagulation treatment. Prophylactic use of low-dose low-molecular-weight heparin (*e.g.*, enoxaparin 40 mg/day) is permitted.

9. Involvement of both the main portal vein and the left and right branches by portal vein tumor thrombus, or of both the main trunk and the superior mesenteric vein concurrently. Presence of tumor thrombus of inferior vena cava.
10. A 10-day consecutive dosing of aspirin (> 325 mg/day) or other drugs, *e.g.*, dipyridamole and clopidogrel, known to inhibit the platelet function within 2 weeks before the first dose.
11. Uncontrolled hypertension (systolic greater than 140 mmHg or diastolic greater than 90 mmHg) after the optimal medical treatment, history of hypertensive crisis or hypertensive encephalopathy.
12. Toxicity (excluding alopecia, events not clinically significant, and asymptomatic laboratory abnormalities) caused by previous therapy that has not yet resolved to grade 0 or 1 (National Cancer Institute Common Terminology Criteria for Adverse Events V5.0 (NCI CTCAE V5.0)) before the first dose of study drugs.
13. Symptomatic congestive cardiac failure (NYHA Class II–IV). Symptomatic or poorly controlled arrhythmia. History of congenital long QT syndrome or corrected Qtc > 500 ms (calculated using Fridericia formula) during screening.
14. Serious hemorrhagic tendency or coagulopathy, or currently receiving thrombolytic therapy.
15. History of gastrointestinal perforation and/or fistula, history of bowel obstruction (including incomplete bowel obstruction requiring parenteral nutrition), extensive bowel resection (partial colectomy or extensive small bowel resection accompanied with chronic diarrhea), Crohn's disease, ulcerative colitis, or chronic diarrhea within the past 6 months.
16. Receipt of radiotherapy within 3 weeks before the first dose. For patients receiving radiotherapy 3 weeks before the first dose, all of the following conditions should be met for enrollment: free from radiotherapy-related toxic reactions, glucocorticoids not required, and radiation pneumonitis, radiation hepatitis, and radiation enteritis excluded.
17. History or current experience of pulmonary fibrosis and such lung diseases as interstitial pneumonia, pneumoconiosis, drug-related pneumonia, and severely impaired lung function.
18. Active tuberculosis (TB), currently receiving anti-tuberculosis treatment or received such treatment within 1 year before the first dose.
19. Human immunodeficiency virus (HIV) infected (HIV 1/2 antibody positive) and known syphilis infection requiring treatment.

20. Active or poorly clinically controlled serious infections. Severe infections within 4 weeks before the first dose, including but not limited to hospitalization caused by infection, bacteremia, or severe pneumonia complication.
21. Presence of active autoimmune diseases requiring systemic treatment (*e.g.*, use of disease-modifying drugs, corticosteroids, or immunosuppressants) within 2 years before the first dose. Use of alternative therapies (*e.g.*, thyroxine, insulin, or physiological corticosteroids for adrenal or pituitary insufficiency) is permitted. Known history of primary immunodeficiency diseases. For patients with positive autoimmune antibodies only, they should be assessed by the investigator to determine whether they had an autoimmune disease.
22. Receipt of immunosuppressants within 4 weeks before the first dose, excluding local glucocorticoids administered by nasal, inhaled, or other topical routes, or systemic glucocorticoids of physiological doses (no more than 10 mg/day of prednisone or equivalents), while the temporary use of glucocorticoids for preventing allergies or treating dyspneic symptoms of such diseases as asthma and chronic obstructive pulmonary disease is permitted.
23. Receipt of a live attenuated vaccine within 4 weeks before the first dose or planned to receive a live attenuated vaccine during the study.
24. Receipt of major surgery (craniotomy, thoracotomy, or laparotomy) within 4 weeks before the first dose or having unhealed wounds, ulcers, or fractures. Receipt of tissue biopsy or other minor surgeries within 7 days before the first dose, barring venipuncture and catheterization for intravenous infusion.
25. Receipt of local treatment for liver cancer within 4 weeks before the first dose.
26. Receipt of systemic treatment with traditional Chinese medicines with cancer indications or immunomodulators (including thymosin, interferon, and interleukin, barring local use for controlling pleural fluid or ascites) within 2 weeks before the first dose.
27. Uncontrolled/uncorrectable metabolic disorders, other non-malignant organ diseases, systemic diseases, or cancer-related secondary diseases with the potential to cause a relatively high medical risk and/or survival evaluation uncertainties unsuitable for subject enrollment as judged by the investigator; other circumstances unsuitable for subject enrollment as judged by the investigator.

28. Other malignancies diagnosed within 5 years before the first dose, excluding radically treated basal cell carcinoma of skin, squamous cell carcinoma of skin, and/or radically resected carcinoma *in situ*. If other malignancies were diagnosed over 5 years pre-dose, the liver lesions should still be subjected to pathological or cytological diagnosis even if they meet the clinical diagnostic criteria for HCC by AASLD; individuals with confirmed HCC can be enrolled.
29. Previous receipt of any antibody treatment involving anti-PD-1, anti-PD-L1/L2, or anti-CTLA4 or other immunotherapies. Previous receipt of anti-VEGF and/or VEGFR, RAF, MEK, PDGFR, and FGFR targeted therapy.
30. Known hypersensitivity to any ingredients in IBI308 and IBI305 preparations or previous severe hypersensitivity reactions to other monoclonal antibodies.
31. Receipt of treatment in other clinical trials within 4 weeks before the first dose.
32. Pregnant or breastfeeding female patients.
33. Acute or chronic diseases, psychiatric disorders, or laboratory abnormalities that may lead to the following consequences: increased study participation or drug-related risks, or interference with interpreting trial results, and considered ineligible for participating in the trial by the investigator.

4.3 Restrictions and Precautions

Women of childbearing potential who are sexually active with male partners who have not undergone surgical sterilization must use at least 1 acceptable and effective method of contraception (see Table 4) starting from the screening, and agree to continue the method(s) during the 90 days since the last dose of study treatment. Besides, these subjects should discuss with a responsible physician about the discontinuation of contraception after this time point. Periodic abstinence, contraceptive rhythm method, and withdrawal method are not the acceptable forms of contraception.

Women of childbearing potential are defined as females who have not undergone surgical sterilization (bilateral tubal ligation, bilateral salpingectomy, or panhysterectomy) or are not postmenopausal (defined as 12 months of amenorrhea without alternative medical causes).

Menopause is defined as 12 months of amenorrhea of a woman without any other medical reasons. Age requirements are as follows:

- Females > 50 years old who have at least 12 months of amenorrhea after stopping hormone replacement therapy, and luteinizing hormone and follicle stimulating hormone levels within the postmenopausal range, are considered menopausal.

- Females ≤ 50 years old who have at least 12 months of amenorrhea after stopping hormone replacement therapy, radiation-induced ovariectomy and the time from the last menorrhea > 1 year, chemotherapy-induced amenorrhea and the time from the last menorrhea > 1 year, or surgical sterilization (bilateral ovariectomy or hysterectomy), are considered menopausal.

Table 4. Effective methods of contraception (one of the followings must be used)

Barrier methods	Intrauterine devices (IUDs)	Hormonal methods
Male condom with spermicide	Copper-T IUD	Implant
Cervical cap with spermicide Diaphragm with spermicide	Progesterone-T IUD[a] Levonorgestrel-releasing intrauterine system (e.g. Mirena®)[a]	Hormone injection Combined oral contraceptive pill Low-dose oral contraceptive pill Contraceptive patch

[a]Also considered as a hormonal method.

4.4 Treatment Termination and Withdrawal of Subjects

4.4.1 Treatment termination of patients

Treatment termination is not the same as withdrawal from the study.

Since some data on clinical events after treatment termination may be very important to the study, the information must be collected until the patient's last scheduled visit, even if the treatment has already been terminated.

The treatment can be terminated by a patient at any time for any reason, or determined by the investigator in the case of any AEs. In addition, the investigator or sponsor may terminate the treatment of a patient if the patient is not suitable for treatment or violates study protocol, or for administrative and/or other safety reasons.

A patient must terminate the treatment in the case of any one of the followings, but can continue to be monitored during the study:

- Treatment termination required by the patient or his/her legal representative;
- Occurrence of AEs or circumstances that require termination due to protocol-specified reasons;
- Onset of another malignant tumor that requires active treatment;
- The patient needs to receive a new anti-tumor treatment;
- Onset of a concurrent disease that interferes further treatment;
- Patient withdrawal determined by the investigator;
- Positive serum pregnancy test results of the patient;

- Poor compliance of the patient;
- Unnecessary risks of study drug continuation to the subject determined by the investigator and/or sponsor according to the subject's clinical status or condition;
- The maximum duration for treatment of the study drugs is reached.

For patients who terminate the treatment but continue to be monitored in the study, all the visits and procedures as listed in the visit forms (if applicable), including the safety follow-up and survival follow-up should be completed.

4.4.2 Patient withdrawal

A patient has the right to withdraw from the study at any time for any reason. A patient must withdraw from the study if the subject or his/her legal representative withdraws the ICF. A patient who withdraws from the study will no longer receive the treatment and protocol-specified follow-up visits. However, the investigator should make every effort to persuade him/her to complete safety follow-up, imaging follow-up evaluation, and survival follow-up.

The reasons for patient withdrawal should be documented in the electronic case report forms (eCRFs). A patient who has signed the ICF but not received study interventions can be replaced. A patient who has signed the ICF and received any study interventions cannot be replaced after withdrawal or treatment termination.

4.4.3 Loss to follow-up

The following actions must be taken if a patient does not return to the study site for a scheduled visit:

- The study site should try to contact with the patient, reschedule the missed visit, reiterate the importance of complying with the schedule of visits, and confirm whether he/she is willing and/or should resume to participate in the study.
- Before a patient is considered lost to follow-up, the investigator or designee should make every effort to recontact the patient (at least 2 phone calls should be made; if the patient is still out of contact, a letter should be mailed to the patient's recently updated address). These attempts to contact with the patient should be documented in the patient's medical records or study documents. Before the end of this study, the investigator still needs to reach out to the patients as much as possible.

The patient is considered lost to the follow-up if he/she is still out of contact.

5 STUDY DRUGS AND OTHER TREATMENT

5.1 Treatment Regimen

5.1.1 Treatment regimen

The study drugs in the study are defined as IBI308, IBI305, and sorafenib. The first dose of study treatment should be started on the day of randomization (day 1 of cycle 1), and no later than 72 h after the randomization. The sponsor should be notified if the first dose is not administered after 72 h. Every effort should be made to start the study treatment on the day of randomization. As determined by the investigator and for administrative reasons, the study treatment can be administered within 3 days after the scheduled day of administration for the rest treatment cycles. Treatment can be delayed for up to 1 week if the administration day is on a holiday.

The treatment regimens for this study are shown in **Table 5**.

Table 5. Treatment regimens and administration sequence

Treatment regimen	Treatment sequence
IBI308 + IBI305 (Phases II/III)	The drugs are administered on the first day of each cycle, with 3 weeks as a cycle. Firstly, IBI308 (200 mg) will be given via intravenous infusion for 60 (±15) min. IBI305 will be administered intravenously at 15 mg/kg at least 5 min later. If no infusion-related reactions occur during the infusion, the two drugs will still be administered at an interval of at least 5 min in the subsequent cycles. If infusion-related reactions occur, the investigator will determine whether anti-allergic drugs should be given as preventive administration before the infusion, and determine the interval between the administration of the two drugs. Administration method of IBI305: The first infusion lasts 90 (± 15) min, and if the first infusion is well tolerated, then the second infusion can be completed within 60 (± 10) min. The subsequent infusion can be completed within over 30 min if the 60-minute infusion is well tolerated.
Sorafenib (Phase III)	400 mg, PO, BID, continuous treatment.

5.2 Description of Study Drugs

5.2.1 Description of IBI308

Major active ingredient: recombinant fully human anti-PD-1 monoclonal antibody

Excipients: 140 mmol/L mannitol, 25 mmol/L histidine, 20 mmol/L dihydrate sodium citrate, 50 mmol/L sodium chloride, 0.02 mmol/L disodium edetate (ethylenediaminetetraacetic acid disodium salt), and 0.2 mg/mL polysorbate 80, pH 6.0

Strength: 10 mL: 100 mg

Appearance: clear, colorless liquid, no foreign matters, flocs, or precipitation

Storage condition and shelf life: stored and transported at 2–8 °C and away from sunlight. The shelf life is 24 months

Manufacturer: Innovent Biologics (Suzhou) Co., Ltd.

5.2.2 Description of IBI305

Major active ingredient: recombinant anti-VEGF humanized monoclonal antibody

Excipients: 1.64 mg/mL sodium acetate, 50 mg/mL sorbitol, and 2.0 mg/mL polysorbate 80, pH 5.2 (adjusted by glacial acetic acid)

Strength: 4 mL: 100 mg

Appearance: clear, colorless liquid, no foreign matters, flocs, or precipitation

Storage condition and shelf life: The shelf life of the finished product is tentatively set to 24 months when stored at 2–8 °C away from light. This product should be used immediately after being diluted with 0.9% sodium chloride injection (if the preparation solution is exposed at room temperature during transfer or under other conditions in an environment not exceeding 30 ± 2 °C, and if the total time until the infusion is completed does not exceed 4 h, there is no impact). After preparation, if it cannot be used immediately, it should not be stored at 2–8 °C for more than 24 h under normal conditions.

Manufacturer: Innovent Biologics (Suzhou) Co., Ltd.

5.2.3 Description of sorafenib

Sorafenib tablets will be supplied by the sponsor in the dosage form of round, biconvex, and red film-coated tablets, with the crossed word "Bayer" engraved on one side and "200" on the other. The content of sorafenib toluene sulfonate in each tablet is equivalent to 200 mg sorafenib. Sorafenib is packaged in boxes, with each box containing 60 sorafenib tablets. The drug box should be sealed and stored below 25 °C.

For details, see the local prescribing information on sorafenib.

5.3 Study-Related Drug Preparation and Infusion (or Administration)

5.3.1 Preparation and infusion process of IBI308

1. Completely withdraw the contents of two vials of IBI308 injection and transfer them into a 100 mL IV infusion bag containing 0.9% (w/v) sterile normal saline solution, then document the start time of the preparation.
2. Gently invert the IV bag to mix the solution, ensuring the uniformity of the contents. Do not vigorously shake it to avoid foam.

3. Administer with an IV infusion line equipped with a 0.2–1.2 μm in-line filter [suggested infusion time is 60 (±15) min]. Document the start and stop time of infusion.
4. 0.9% normal saline will be used to flush the infusion tube after the administration.

Note: Before preparation, make sure that the IBI308 injection is clear without any quality issues such as turbidity or precipitation. Make sure that the time from withdrawing IBI308 from the first vial to the end of infusion is no more than 24 h (storage conditions for the prepared solution is 2–8 °C in the fridge). Avoid mixing with other drugs. Do not administer as an IV push.

Table 6. First and subsequent infusions of IBI308

First infusion	Subsequent infusions
• No prophylactic administration is allowed. • The vital signs should be measured within 60 min before the infusion (pulse, breathing rate, blood pressure, and body temperature). • The infusion of IBI308 should be completed within 60 (±15) min. • If there are clinical indications, the vital signs should be measured every 15 (±5) min during the infusion, and measured again at 30 (±10) min after the infusion. • The patients should be informed of the possibility of delayed symptoms after the infusion, and should be required to contact their physicians when these symptoms occur.	• If a patient develops an infusion-related reaction arising from any of previous infusions, prophylactic administration can be performed using antihistamines, antipyretics, and/or analgesics during the subsequent administration as per the judgment of the investigator. • The vital signs should be measured within 60 min before the infusion. • The infusion of IBI308 should be completed within 60 (±15) min. • If a patient develops infusion-related reactions arising from the previous infusion or clinical signs, his/her vital signs should be measured in the process of the infusion and at 30 (±10) min after the infusion.

5.3.2 Preparation and infusion process of IBI305

The same IV tube as that for IBI308 will be used for administration.

1. Calculate the dosing volume at 15 mg/kg and record the time that the solution is prepared. The calculation formula is as follows:

Dosing volume V (mL) = 15 mg/kg × body weight (kg) ÷ 25 mg/mL.

2. Draw V mL of normal saline from the 250 mL 0.9% (weight/volume) sterile normal saline intravenous infusion bag, and inject a total of V mL of IBI305 into the same sterile normal saline intravenous infusion bag.

3. Gently invert the IV bag to mix the solution, ensuring the uniformity of the contents. Do not vigorously shake it to avoid foam.
4. Record the start and end time points of administration.
5. 0.9% normal saline will be used to flush the infusion tube after the administration.

Precautions: Before preparation, make sure that the IBI305 injection is clear without any quality issues such as turbidity and precipitation; if the preparation solution is exposed at room temperature during transfer or under other conditions in an environment not exceeding 30 °C ± 2 °C, and if the total time until the infusion is completed does not exceed 4 h, there is no impact. If it cannot be used immediately, it should not be stored at 2–8 °C for more than 24 h under normal conditions; avoid mixing with other drugs; do not administer as an intravenous push.

Table 7. First and subsequent infusions of IBI305

First infusion	Subsequent infusions
• No prophylactic administration is allowed before IBI305 infusion. • The vital signs should be recorded within 60 min before the infusion (pulse, breathing rate, blood pressure, and body temperature). • The infusion of IBI305 should be completed within 90 (±15) min. • The vital signs should be recorded at the end of the infusion (±5 min) and at 2 (±1) h after the infusion. • Patients should be informed of the possibility of delayed symptoms after the infusion, and should be required to contact their study physicians when these symptoms occur.	• If a patient develops an infusion-related reaction arising from any of previous infusions, prophylactic administration can be performed using antihistamines, antipyretics, and/or analgesics during the subsequent administration as per the judgment of the investigator. • The vital signs should be recorded within 60 min before the infusion. • If a patient can tolerate the previous infusion with no infusion-related reactions, then the infusion of IBI305 can be completed within 60 (±10) min, or if a patient develops infusion-related reactions arising from the previous infusion, then infusion can be completed within 90 (±15) min. The infusion of IBI305 can be completed within over 30 min should the 60-minute infusion be well tolerated. • The vital signs should be recorded at the end of the infusion and at 2 (±1) h after the infusion.

5.3.3 Sorafenib administration (phase III study)

400 mg sorafenib (2 × 200 mg tablets), PO, BID, from the first day of cycle 1 treatment. The tablets will be taken with approximately 240 mL (1 cup or 8 ounces) of water under fasting state (at least 1 h before the meal or 2 h after the meal). The drug will be administered once in the morning and evening, respectively. The patients need to record their medication in a timely manner in the patients' medication diary as provided by the sponsor.

To manage the toxicity, a temporary interruption of treatment or dose adjustment may be required. Any toxicity that is or that is likely to be related to sorafenib treatment will be managed in accordance with the standard medical practice. Refer to Table 8 for the dose delay and adjustment recommendations for skin toxicity arising from sorafenib, and Table 9 for dose adjustment recommendations for hand-foot skin toxicity. Management of suspected adverse drug reactions may warrant the temporary interruption of sorafenib treatment and/or lowering of the dose of sorafenib. When dose reduction is warranted, the dose of sorafenib may be reduced to 400 mg per day. If dose reduction is still warranted, then the dose of sorafenib may be reduced to 400 mg once every other day. If further dose reduction is required, patients should terminate the study treatment. After the dose reduction, if a patient has received the drug at the stable dose for 3 weeks or longer and is free from other toxicities requiring dose adjustment, then the dose of sorafenib may be adjusted according to the investigator's judgment. The doses for all dose adjustments are as follows:

Dose level 1: 400 mg (2 × 200 mg), PO, BID

Dose level 2: 400 mg (2 × 200 mg), PO, QD

Dose level 3: 400 mg (2 × 200 mg), PO, QOD

Table 8. Dose delay and adjustment recommendations for skin toxicity arising from sorafenib

Grade	Dose delay	Dose adjustments
Hematological toxicity		
Grade 0–2	On-time treatment	Unchanged
Grade 3	On-time treatment	Lower the dose level by one
Grade 4	Delayed* until recovery to ≤ grade 2	Lower the dose level by one
Non-hematological toxicity (except for skin toxicity)†		
Grade 0–2	On-time treatment	Unchanged
Grade 3	Delayed* until recovery to ≤ grade 2	Lower the dose level by one‡
Grade 4	Case by case	Case by case

*If no recovery is seen after a 30-day delay, then the treatment will be terminated, unless patients have clinical benefits from the previous treatment

†Exclude nausea/vomiting without the pretreatment, or diarrhea

‡The treatment will be terminated if over two dose levels are lower

Table 9. Recommendations on dose adjustment of sorafenib after the development of hand-foot skin reactions

Grade	Occurrence of events	Dose adjustment of sorafenib
Grade 1: Numbness, dysesthesia, paresthesia, tingling, painless swelling, erythema, or discomfort of hands or feet, but which do not interfere with the normal activities of patients.	Any occurrence	Continue the treatment and consider using typical treatments to relieve the symptoms.
Grade 2: Painful erythema and swelling of hands or feet, and/or discomfort affecting the normal activities of patients.	First occurrence	Continue the treatment and consider using typical treatments to relieve the symptoms. Interrupt sorafenib treatment until alleviation of toxicity to grade 0/1.
	No improvement within 7 days or 2nd or 3rd occurrence	Interrupt sorafenib treatment until alleviation of toxicity to grade 0/1.
		The dose will be lowered to 400 mg per day when treatment is restarted.
	The 4th occurrence	Terminate the sorafenib treatment.
Grade 3: Wet peeling of hands or feet, formation of ulcers, blistering or severe pain, or severe discomfort discouraging patients from working or performing daily activities.	The 1st or 2nd occurrence	Interrupt sorafenib treatment until alleviation of toxicity to grade 0/1.
		The dose will be lowered to 400 mg per day when treatment is restarted.
	The 3th occurrence	Terminate the sorafenib treatment.

For sorafenib-related AEs that are not listed in the table, refer to the guidelines applicable for the local prescribing information (if available).

5.4 Dose Modification of Study Drugs

5.4.1 General principle

- The patient's hematologic, hepatic, and renal function must meet the requirements for administration before each dose of the study drugs on day 1, and all toxic reactions related to the study drug must resolve to CTCAE V5.0 grade 0 or 1 or the baseline levels (excluding alopecia, fatigue, non-clinical significant laboratory abnormalities, and asymptomatic laboratory abnormalities).
- All dose modifications should be documented, including the reasons and actions taken.

5.4.2 Dose modification of IBI308

The dose of IBI308 should not be modified throughout the study. The dose modification of IBI308 is shown in Table 10. For the handling of immune-related adverse events which are not listed in the table, refer to "IBI308 Prescribing Information" and "Immune-Related Adverse Event Management Manual" provided by the sponsor.

Table 10. Treatment adjustment recommendations for IBI308

Immune-related adverse reactions	Severity level	Adjustments
Pneumonia	Grade 2	Treatment withholding until adverse reaction returns to grade 0–1
	Grade 3 or 4 or persistent Grade 2	Termination
Diarrhea and colitis	Grade 2 or 3	Treatment withholding until adverse reaction returns to grade 0–1
	Grade 4	Termination
Hepatitis	Grade 2, AST or ALT within 3–5 × ULN, or TBIL within 1.5–3 × ULN	Treatment withholding until adverse reaction returns to grade 0–1
	Grade 3 or 4, AST or ALT > 5× ULN, or TBIL > 3× ULN	Termination *
Nephritis	Grade 2 or 3 increased creatinine	Treatment withholding until adverse reaction returns to grade 0–1
	Grade 4 increased creatinine	Termination
Endocrine disorders	Symptomatic grade 2 or 3 hypothyroidism, grade 2 or 3 hyperthyroidism, grade 2 or 3 hypophysitis, grade 2 adrenal insufficiency Grade 3 hyperglycemia or type I diabetes	Treatment withholding until adverse reaction returns to grade 0–1
	Grade 4 hypothyroidism Grade 4 hyperthyroidism Grade 4 hypophysitis Grade 3 or 4 adrenal insufficiency Grade 4 hyperglycemia or type I diabetes	Termination
Skin reactions	Grade 3	Treatment withholding until adverse reaction returns to grade 0–1
	Grade 4 Stevens-Johnson syndrome (SJS) or toxic epidermal necrolysis (TEN)	Termination
Thrombocytopenia	Grade 3	Treatment withholding until adverse reaction returns to grade 0–1
	Grade 4	Termination
Other immune-related adverse reactions	Grade 3 or 4 hyperamylasemia or increased lipase Grade 2 or 3 pancreatitis Grade 2 myocarditis* Other Grade 2 or 3 immune-related adverse reactions (first occurrence)	Treatment withholding until recovered to grade 0–1
	Grade 4 pancreatitis or relapsed pancreatitis of all grades Grade 3 or 4 myocarditis** Grade 3 or 4 encephalitis Grade 4 other immune-related adverse reactions, first occurrence	Termination

Immune-related adverse reactions	Severity level	Adjustments
Relapsed or persistent adverse reactions	Relapsed grade 3 or 4 (other than endocrine disorders) Grade 2 or 3 adverse reactions that have not recovered to grade 0–1 within 12 weeks from the last dose (other than endocrine disorders) Corticosteroid failing to decrease to ≤ 10 mg/day of prednisone or equivalent within 12 weeks after the last dose	Termination
Infusion reaction	Grade 2	Infusion rate reduction or treatment interruption; considering treatment continuation with close monitoring after symptoms resolved
	Grade 3 or 4	Termination

Note 1: The severity levels of adverse reactions are graded as per the National Cancer Institute Common Terminology Criteria for Adverse Events version 4.03 (NCI-CTCAE V4.03).

*For immune-related hepatitis, the dose of drugs can be adjusted by referring to the domestic and international immune checkpoint inhibitors-related toxicity management guidelines.

**The safety of resuming treatment after myocarditis returns to grade 0–1 in severity is yet unclear.

Note 2: For more detailed drug information on IBI308 (Tyvyt®), refer to the "IBI308 Prescribing Information". Or the dose of drugs can be adjusted by referring to the domestic and international guidelines for the management of immune checkpoint inhibitors-related toxicity.

If the treatment with IBI308 is terminated temporarily or permanently due to the intolerable toxicity during the combination therapy period, the patients can continue to undergo IBI305 treatment according to the investigator's judgment, until disease progression, development of intolerable toxicity, initiation of a new anti-tumor treatment, withdrawal of informed consent, loss to follow-up or death, termination decided by the investigator or sponsor, or completion of the 24-month treatment by the patients (whichever occurs first).

The study drug is allowed to be interrupted for up to 12 weeks. If the symptoms do not resolve within 12 weeks and the treatment cannot be resumed, the patient must terminate IBI308 treatment. Except for the following two cases:

- IBI308 interruption > 12 weeks due to glucocorticoid reduction for irAEs. Consult the sponsor's medical monitor before resuming IBI308. Tumor imaging evaluation for efficacy should not be affected by treatment interruption and should be performed as scheduled.
- IBI308 interruption > 12 weeks due to AEs unrelated to IBI308. Consult the sponsor's medical monitor before resuming IBI308. Tumor imaging evaluation for efficacy should not be affected by treatment interruption and should be performed as scheduled.

When IBI308 is resumed, the relevant AEs need to be decreased to 0–1 or the baseline level, and the ECOG PS score needs to be 0–1.

5.4.2.1 Management of IBI308 infusion-related reaction

IBI308 may lead to severe or life-threatening infusion-related reactions, including severe hypersensitivity or allergic reactions. Signs and symptoms usually occur during or after the drug infusion and usually resolve within 24 h after the infusion completion. See Table 11 for the guidelines for the management of IBI308 infusion-related reactions.

Table 11. Guidelines for the management of IBI308 infusion-related reactions

CTCAE grading	Treatments	Premedications for subsequent infusions
Grade 1 Mild reaction; infusion interruption not indicated; intervention not indicated	According to the patient's medical indications, monitor the vital signs closely until the patient is stable as determined by the investigator.	N/A
Grade 2 Treatment or infusion interruption required, but responds promptly to timely symptomatic treatment (e.g. antihistamines, non-steroidal anti-inflammatory drugs [NSAIDS], anesthetics, and IV fluids); prophylactic medications indicated for ≤ 24 h	Stop the infusion and monitor symptoms. Other appropriate treatments include but are not limited to: IV fluids Antihistamines NSAIDS Anesthetics According to the patient's medical indications, monitor the vital signs closely until the patient is stable as determined by the investigator. If symptoms resolve within 1 h after the interruption of the infusion, then the infusion will be resumed at 50% of the original infusion rate (e.g. from 100 mL/h to 50 mL/h). Otherwise, interrupt the treatment until symptoms resolve. Pre-medications should be given for subsequent infusions. If toxicities at grade 2 occur despite adequate pre-medications, the study drugs should be terminated.	The following pre-medications are recommended within 1.5 h (± 30 min) before IBI308 infusion: Diphenhydramine 50 mg PO (or equivalent antihistamines). Acetaminophen 500–1000 mg PO (or equivalent antipyretics).
Grade 3 or 4 Grade 3: Prolonged (i.e. not rapidly responsive to symptomatic medication and/or brief interruption of infusion); recurrence of symptoms following initial improvement; hospitalization indicated for other clinical sequelae (e.g. renal impairment, pulmonary infiltration) Grade 4: Life threatening; pressors or ventilatory support indicated	Discontinue infusion. Other appropriate treatments include but are not limited to: Epinephrine** IV fluids Antihistamines NSAIDS Anesthetics Oxygen Pressors Corticosteroids According to the patient's medical indications, monitor the vital signs closely	Not applicable.

CTCAE grading	Treatments	Premedications for subsequent infusions
	until the patient is stable as determined by the investigator. Hospitalization may be indicated. **Epinephrine should be used immediately for allergic reactions. The study drugs should be terminated.	
Appropriate first-aid equipment should be provided in the ward and physicians should be available at all times during the administration. For more information, refer to CTCAE V5.0 (http://ctep.cancer.gov).		

5.4.2.2 Other circumstances with dose modification of IBI308 allowed

In addition to treatment-related AEs, the IBI308 treatment may be interrupted due to other circumstances, *e.g.*, medical/surgical events or management reasons which are not related to the study treatment. Patients should restart the study treatment within 3 weeks after the scheduled medication interruption, unless there are other discussion results with the sponsor. The reasons for medication interruption should be recorded in the study record of patients.

5.4.3 Dose modification of IBI305

No dose adjustment for IBI305 is allowed during the study. The dose of IBI305 is calculated according to the patient weight at baseline (before the first dose), and remains unchanged during the study, unless the patient weight variation is ≥ 10% from the baseline. If an infusion-related reaction occurs during a 60-minute infusion, the infusion time should be extended to 90 minutes for all subsequent infusions. Likewise, if an infusion-related reaction occurs during a 30- to 60-minute infusion, the infusion time should be extended to 60 minutes for all subsequent infusions.

If the treatment with IBI305 is interrupted or terminated due to the intolerable toxicity during the combination therapy period, the patients can continue to undergo IBI308 treatment according to the investigator's judgment, until disease progression, development of intolerable toxicity, initiation of a new anti-tumor treatment, withdrawal of informed consent, loss to follow-up or death, termination decided by the investigator or sponsor, or completion of the 24-month treatment by the patients (whichever occurs first).

When a grade 3 or 4 IBI305-related toxicity occurs, the investigator should determine whether IBI305 will be continued according to the following requirements:

1. First occurrence

IBI305 administration should be interrupted until the symptoms return to baseline levels or at least are reduced to ≤ CTCAE grade 1 (except for special circumstances listed below).

Note that when grade 4 thrombocytopenia occurs, IBI305 administration should be interrupted until the symptoms return to baseline levels or at least are reduced to ≤ CTCAE grade 1, since these events increase the risk of hemorrhage.

2. Re-occurrence in re-administration

If grade 3 IBI305-related toxicity occurs again, the investigator should assess the patient risks/benefits of study treatment continuation. If the toxicity re-occurs upon the re-administration of IBI305, IBI305 should be terminated.

If grade 4 IBI305-related toxicity occurs again, IBI305 should be terminated.

Measures should be taken in the following special circumstances (classified based on CTCAE V5.0):

1. Hemorrhage

Patients with grade 3 or 4 hemorrhages should be treated accordingly and terminate the study treatment.

2. Thrombosis/embolism

Patients with arterial thrombosis of any severity grades should terminate the study treatment.

Patients with grade 4 venous thrombosis should terminate the study treatment.

Subjects with grade 3 venous thrombosis should interrupt the study treatment. If the scheduled therapeutic dose of anticoagulant therapy lasts < 2 weeks, the study treatment should be interrupted until the anticoagulant treatment is completed. If the scheduled therapeutic dose of anticoagulant therapy lasts > 2 weeks, IBI305 administration should be for 2 weeks, and the study treatment can be restarted during the anticoagulant therapy if all the following all criteria are met:

- INR is within the target range (usually 2–3) before restarting of study treatment;
- Patients must not have experienced grade 3 or 4 hemorrhage since the enrollment.
- No signs of great vessel invasion or adjacency to great vessels from previous tumor assessments.

Note: Therapeutic dose of anticoagulant therapy is defined as the escalating dose of warfarin or other anticoagulants until INR is maintained at no less than 1.5 (usually 2–3). The anticoagulant dose should be documented in the eCRFs, and the INR of patients receiving anticoagulant therapy should be monitored during the treatment.

3. Hypertension

BP should be measured frequently in patients to monitor the occurrence and exacerbation of hypertension. Patients should remain at resting position for at least 5 minutes before BP measurement.

Definition of hypertension: Pathologically increased BP with repeated measurements persistently over 140/90 mmHg.

The doses of all anti-hypertensive agents used in the patients should be documented during each visit. If the subject remains hypertensive despite treatment termination, BP and antihypertensive drug used should be monitored every 3 months until BP returns to normal or end of study.

4. Posterior reversible encephalopathy syndrome (PRES)

There have been a few reports of subjects with signs and symptoms consistent with PRES after study treatment. This is a rare neurological disease. Signs and symptoms include epilepsy, headache, altered mental status, visual impairment, or cortical blindness, with or without hypertension. Patients with PRES should terminate the study treatment.

5. Proteinuria

Routine urinalysis should be performed before each dose of IBI305 unless a 24-h urinary protein test has already been done.

1) Routine urinalysis should be performed at the time of the first proteinuria if urinary protein:
   - < 2+: Continue study treatment as scheduled, no additional tests indicated.
   - ≥ 2+: Perform a 24-h urinary protein test within 3 days before administration:
     - 24-h urinary protein ≤ 2 g: Continue study treatment as scheduled, and perform routine urinalysis before each scheduled dose.
     - 24-h urinary protein > 2 g: Delay study treatment until 24-h urinary protein ≤ 2 g, and perform a 24-h urinary protein test before each subsequent dose until urinary protein ≤ 1 g/24 h. Interrupt study treatment only when 24-h urinary protein > 2 g.
2) Routine urinalysis should be performed at the time of second and subsequent occurrence of proteinuria if urinary protein:
   - < 3+: Continue study treatment as scheduled, no additional tests indicated.
   - ≥ 3+: Perform 24-h urinary protein test within 3 days before administration:
     - 24-h urinary protein ≤ 2 g: Continue study treatment as scheduled.

- 24-h urinary protein > 2 g: Delay study treatment until 24-h urinary protein ≤ 2 g, and perform 24-h urinary protein test before each subsequent dose until urinary protein ≤ 1 g/24 h. Interrupt study treatment only when 24-h urinary protein > 2 g.

3) Nephrotic syndrome (grade 4): terminate the study treatment.

6. Gastrointestinal perforation

If gastrointestinal perforation occurs, appropriate measures should be taken and the study treatment should be terminated.

7. Wound healing complications

Study treatment should not begin within 28 days of a major surgery, or before the surgical wound has fully healed. If a complication of wound healing occurs during study treatment, the study treatment should be interrupted until the wound is fully healed. If an elective surgery is required, the study treatment should be interrupted.

8. Abdominal abscess or fistula

If abdominal abscess or fistula occurs, the study treatment should be terminated. However, the investigator will determine whether study treatment will be continued if the above AE is resolved.

9. Infusion-related and/or allergic reactions:

Infusion-related reactions after the first dose of IBI305 are uncommon (< 3%), and the incidence of a severe reaction was only 0.2%.

If a mild (grade 1/2) reaction (such as fever, chills, headache, and nausea) occurs, pretreatment before subsequent administration should be performed and infusion time should not be reduced. If the patient is well-tolerated during infusion after pretreatment, the infusion time can be reduced by 30 min (±10 min) for subsequent administration with pretreatment. If an infusion-related AE occurs during a 60-minute infusion, the subsequent infusion should be completed within 90 min (±15 min) with pretreatment. Likewise, if an infusion-related AE occurs during a 30-minute infusion, the subsequent infusion should be completed in 60 min (±10 min) with pretreatment. If a patient has a grade 3 infusion-related reaction, IBI305 should be interrupted and not be restarted on the same day. However, since there lacks the dose adjustment method for grade 3 infusion-related reactions, the investigator may decide to either terminate IBI305 or perform pretreatment, and complete the infusion within 90 min (±15 min). If an adverse reaction still occurs during a 90-minute infusion, the infusion should be continued slowing down and then gradually returned to a 90-minute infusion. If the investigator is uncertain about the handling, IBI305 should be terminated. When the study treatment is restarted, the subject should be closely monitored based on routine clinical practice until the

possible time of adverse reaction has passed. If a patient has a grade 4 infusion-related reaction, IBI305 should be terminated.

An allergic reaction is defined as the vascular collapse or shock (systolic BP < 90 mmHg, unresponsive to rehydration) that occurs within 30 minutes of a study drug infusion caused by an allergy, with or without respiratory distress. Skin reactions include pruritus, urticaria, and angioedema. Patients with allergic reactions related to IBI305 should terminate IBI305.

5.4.4 Dose modification of sorafenib (phase III study)

To manage the toxicity, a temporary interruption of treatment or dose adjustment may be required. Any toxicity that is or that is likely to be related to sorafenib treatment will be managed in accordance with the standard medical practice. Refer to Table 12 for the dose adjustment recommendations for the hand-foot skin toxicity. Management of suspected adverse drug reactions may warrant the temporary interruption of sorafenib treatment and/or lowering of the dose of sorafenib. When dose reduction is warranted, the dose of sorafenib may be reduced to 400 mg per day. If dose reduction is still warranted, then sorafenib dose may be reduced to 400 mg once every other day. After the dose reduction, if a patient has received the drug at the stable dose for 3 weeks or longer and is free from other toxicities requiring dose adjustment, then the dose of sorafenib may be adjusted according to the investigator's judgment.

Table 12. Recommendations on dose adjustment of sorafenib after the development of hand-foot skin reactions

Grade	Occurrence of events	Dose adjustment of sorafenib
Grade 1: Numbness, dysesthesia, paresthesia, tingling, painless swelling, erythema, or discomfort of hands or feet, but which do not interfere with the normal activities of patients.	Any occurrence	Continue the treatment and consider using typical treatments to relieve the symptoms.
Grade 2: Painful erythema and swelling of hands or feet, and/or discomfort affecting the normal activities of patients.	First occurrence	Continue the treatment and consider using typical treatments to relieve the symptoms. Interrupt sorafenib treatment until alleviation of toxicity to grade 0/1.
	No improvement within 7 days or 2nd or 3rd occurrence	Interrupt sorafenib treatment until alleviation of toxicity to grade 0/1.
		The dose will be lowered to 400 mg per day when treatment is restarted.
	The 4th occurrence	Terminate the sorafenib treatment.

Grade	Occurrence of events	Dose adjustment of sorafenib
Grade 3: Wet peeling of hands or feet, formation of ulcers, blistering or severe pain, or severe discomfort discouraging patients from working or performing daily activities.	The 1st or 2nd occurrence	Interrupt sorafenib treatment until alleviation of toxicity to grade 0 or 1.
		The dose will be lowered to 400 mg per day when treatment is restarted.
	The 3th occurrence	Terminate the sorafenib treatment.

For sorafenib-related AEs that are not listed in the table, refer to the guidelines applicable for the local prescribing information (if available).

5.5 Study Drug Compliance

Study treatment will be given at the study sites. Treatment compliance will be monitored by medication dispensing and return records, and medical records of patients.

Formula for sorafenib medication compliance: medication compliance = (number of actually taken tablets/number of tablets should be taken) × 100%.

5.6 Concomitant Treatment

5.6.1 Prohibited concomitant treatment

The following treatments are prohibited throughout the trial:

- Anti-tumor treatment and immunomodulatory treatment: In addition to the study drugs used in this clinical trial, the patients are not allowed to take any other systemic anti-tumor treatment at the same time, including the drugs for molecular targeted therapy, cytotoxic drugs, biological therapy, immunomodulatory drugs (including the Chinese patent medicines), or any other study drugs. Moreover, Chinese patient medicines with anti-tumor indications or Chinese herbal medicines for anti-tumor treatment are also not permitted to be used in the study. Due to the lack of high-level evidence currently to prove the definite anti-tumor effects of any Chinese patent medicines/Chinese herbal medicines, some Chinese patent medicines/Chinese herbal medicines can be used during the study according to the investigator's judgment. In the meantime, patients cannot receive immunomodulatory drugs affecting the judgment of the efficacy of immunotherapy, *e.g.*, thymosin, interleukin, and interferon (barring interleukin 11 for increasing platelets).
- Local treatment for liver cancer, *e.g.*, local ablation, hepatic artery intervention, local radiotherapy, etc.

- Live vaccine inoculation within 30 days before the first administration of the treatment and throughout the trial. Live vaccines include but are not limited to measles, mumps, rubella, chicken pox, yellow fever, rabies, Bacillus Calmette Guerin, and typhoid (oral) vaccines. Seasonal inactivated influenza virus vaccines inoculated via injection are permitted, but live attenuated influenza vaccines are not.
- Corticosteroids are permitted only for topical use (*e.g.*, external skin and inhalation), systemic medication at the physiological dose (≤ 10 mg/day prednisone or drugs at the equivalent dose), for the treatment of AEs arising from potential immune causes, other circumstances as specified in the protocol, and prevention of allergic reactions before the imaging studies or clinical operations.
- Patients receiving treatment with sorafenib: The administration of strong CYP3A inducers is prohibited within 14 days before and during the study (including but not limited to phenytoin, hypericum perforatum, dexamethasone, carbamazepine, rifampicin, rifabutin, and phenobarbital).

5.6.2 Permitted concomitant treatment

The following concomitant treatments are permitted in this study:

- Medications that meet the protocol requirements, as determined by the investigator (e.g. concomitant treatments used for disease-related symptoms and treatment-related AEs).
- Patients who need medications for a long time due to pre-existing diseases, such as hypertensive and diabetes mellitus, can continue the use of drug.
- During the study treatment, local palliative radiotherapy is permitted for isolated lesions (non-liver lesions) which is conducted for the purpose of controlling symptoms and which does not affect the judgment of therapeutic effects. Before the proposed palliative radiotherapy, the investigator should notify the sponsor and the treatment will be started after the review and approval by the medical monitor.
- Supportive care for relieving tumor-related symptoms, such as bisphosphonate treatment for bone metastases.
- For patients with viral hepatitis, reference should be made to the relevant guidelines for the standard antiviral therapy (except biological therapy such as interferon). Prophylactic antiviral therapy is recommended according to the guidelines for HBV carriers during the study. HBV-DNA should be monitored regularly during the trial.
- The temporary use of anti-allergic drugs, glucocorticoids, etc. for the prevention or treatment of allergic reactions is permitted.

- The temporary use of glucocorticoids for the treatment of such dyspneic symptoms as COPD or asthma is permitted.
- The use of colony stimulating factors as per related guidelines is permitted.

5.6.3 Drug interaction

5.6.3.1 Drug interaction of IBI308

There are no data on drug interactions with IBI308 at present.

5.6.3.2 Drug interaction of IBI305

No evidence available has shown clinically significant drug interactions between IBI305 and other drugs.

5.6.3.3 Drug interaction of sorafenib

CYP3A4 inducers: There have been no clinical data that show CYP3A4 inducers affect the drug metabolism of sorafenib. CYP3A4 inducers (*e.g.*, rifampicin, hypericum perforatum, phenytoin, carbamazepine, phenobarbital, and dexamethasone) may quicken the metabolism of sorafenib, therefore reducing the drug concentration of sorafenib.

CYP3A4 inhibitor: The use of ketoconazole, a strong inhibitor of CYP3A4, in healthy male volunteers once daily for 7 consecutive days together with the single dose of 50 mg sorafenib daily will not result in changes in the mean AUC of sorafenib. Therefore, it is very unlikely that CYP3A4 inhibitors affect the metabolism of sorafenib.

CYP2C9 substrate: Warfarin is the substrate of CYP2C9. The impact of sorafenib on warfarin is assessed by comparing the patients administered with sorafenib vs. placebo. As compared with the placebo group, the average PT-INR of patients administered with sorafenib and warfarin is not changed. However, the INR values should be monitored on a regular basis when patients are co-administered with warfarin.

Selective substrates for CYP isoenzyme: Midazolam, dextromethorphan, and omeprazole serve as the substrates of the cytochromes CYP3A4, CYP2D6, and CYP2C19, respectively. Sorafenib in combination with the above three drugs will not change their level of exposure. This indicates that for the cytochrome P450 isoenzyme, sorafenib is neither its inhibitor nor inducer.

UTG1A1 route: Caution is recommended for sorafenib used in combination with drugs which are metabolized/eliminated via the UTG1A1 route (*e.g.*, irinotecan).

QT interval prolongation: Reports have shown that sorafenib can prolong QT/QTc interval, and lead to an increased risk of ventricular arrhythmia. Careful consideration is recommended when sorafenib is used in combination with certain antiarrhythmic agents or other drugs that can prolong the QT interval of patients.

5.7 Management of Study Drugs

5.7.1 Receipt and counting of study drug

The sponsor will provide the study drugs according to the study site's expected enrollment plan. The study drugs will be shipped to the study site via a qualified third-party logistics company. The authorized staff of the study site will sign on the shipping order to confirm the receipt of the drugs.

IBI308, IBI305, and sorafenib will be used in this study only, and will be managed by dedicated personnel as authorized by the investigator only. To fully control the distribution and use of the study drugs, the drugs will be subjected to quantity registration at each visit. All administration/distribution steps for this study will be conducted in the study site. The remaining study drugs will be returned for destruction as per relevant Chinese regulations.

5.7.2 Packaging and label of study drug

The smallest packaging unit is one box. Each box contains 2 vials of IBI308 injection. The smallest packaging unit of IBI305 is a box. Each box contains 1 vial of IBI305 injection.

The package box of the study drug contains the drug name, dosage form, strength, drug code, batch number, expiration date, storage conditions, and sponsor's information. The label on the vial contains the same information as the outer package except for dosage form, precautions, and dosage and administration. The package and vial should be both labeled "use for clinical study only".

5.7.3 Drug return and destruction

The containers, vials, infusion bags, and syringes of used and partially used study drugs can be destroyed or returned on-site according to the appropriate guidelines and operating procedures established by study sites and local agencies.

During the study and upon the completion or termination of the study, all unused or expired study drugs must be returned to the sponsor for destruction. Arrangements for the return of IBI308 were made by the CRA designated by the sponsor.

6 STUDY PROCEDURE

6.1 Enrollment and Randomization of Patient

6.1.1 Enrollment and randomization of patient

Phase II single-arm study:

The investigator will enroll the patients with the following steps:

1. Obtain the ICF signed by the patients before any study-related procedures.
2. Confirm the patients' eligibility by the investigator or trained designee after reviewing the inclusion/exclusion criteria.
3. A total of 20 patients with advanced HCC will be enrolled and receive IBI308 (200 mg, IV, day 1, Q3W) + IBI305 (15 mg/kg, day 1, Q3W) as the first-line treatment. The treatment will continue until disease progression, development of intolerable toxic reactions, initiation of new anti-tumor treatment, withdrawal of informed consent, loss to follow-up, death, or other conditions requiring the termination of treatment stipulated in the protocol, whichever occurs first. The safety and tolerability of combination treatment will be evaluated and ORR, DCR, DOR, TTP, TTR, PFS, and OS of IBI308 in combination with IBI305 as the first-line treatment for patients with advanced HCC will be assessed based on the evaluation by the investigator as per RECIST V1.1. iRECIST is only used for confirmation 4 to 6 weeks after the first disease progression in patients, thereby excluding certain patients with pseudo-progression.

Phase III randomized, controlled study

The investigator will enroll the patients with the following steps:

1. Obtain the ICF signed by the patients before any study-related procedures.
2. Confirm the patients' eligibility by the principal investigator or trained designee after reviewing the inclusion/exclusion criteria.
3. Central randomization is used in this study, the stratification factors including macrovascular invasion and/or extrahepatic spread (yes vs. no), baseline AFP (<400 ng/mL vs. ≥400 ng/mL), and the score of ECOG performance status (0 vs. 1).

Subjects who failed to meet the criteria (screen failures) were re-screened. If re-screening is considered, the investigator must contact the sponsor. Each subject can be re-screened once. The subjects must sign the ICF again and receive a new identification number when they are re-screened.

6.1.2 Procedure for handling enrollment error

The inclusion/exclusion criteria must be followed strictly. If an ineligible patient is enrolled, the sponsor's medical monitor and investigator must discuss whether to allow the patient to continue participating in the study and whether to use the study drugs. If as determined by the investigator, allowing the patient to continue with the study is appropriate medically, which is also agreed with by the sponsor's medical monitor, then the patient will continue participating in the study and receive the study drugs. If as determined by the investigator, allowing the patient to continue with the study is appropriate medically, which is not agreed with by the sponsor's medical monitor, then the patient should not continue participating in the study (regardless of receiving the study drug or not). The investigator must not allow the patient who was accidentally included in this study to continue the study until he has received the formal approval from the sponsor.

6.1.3 Treatment procedures upon irresistible factor

If the visits, drug administration, and imaging evaluations cannot be carried out as planned due to irresistible reasons such as major natural disasters, wars, and public health events, based on the relevant guidelines issued by the regulatory authority (if applicable), the sponsor can arrange corresponding study process for each subject and make detailed records after consultation with the investigator and the drug clinical study site/ethics committee, and under the premise of maximizing the protection of the safety, rights and well-being of the subjects. At the same time, it can be considered to add an analysis on the impact of irresistible factors on the study results.

6.2 Schedule of Study Visits

6.2.1 Screening period (phases II/III)

The following procedures must be completed during the screening (Day -28 to -1) to ensure patient eligibility:

- Signing of the informed consent form
- Confirming the inclusion/exclusion criteria
- Recording demographic data, medical history, and tumor history
- Recording vital signs, height, and weight
- Physical examination
- ECOG PS score
- 12-lead electrocardiogram (ECG)
- Myocardial test

- Count of routine blood test/blood chemistry/urinalysis (within 7 days before the first dose)
- Coagulation function (within 7 days before the first dose)
- AFP (within 7 days before the first dose)
- Pregnancy test (within 3 days before the first dose)
- Thyroid function
- HIV antibody, hepatitis B panel (HBsAg, HBsAb, HBcAb, HBeAg, and HBeAb) and HBV DNA, HCV antibody, and HCV-RNA
- AE evaluation
- Concomitant medication
- Tumor imaging evaluation
- Archived tumor tissue samples (for patients with pathologically/cytologically diagnosed HCC in the treatment group in phase III study only)

6.2.2 Baseline (before day 1 of cycle 1)

- Recording the vital signs
- ECOG PS score
- Immunogenicity sample collection
- PK sample collection
- AE evaluation
- Recording concomitant medication
- Survival condition
- Quality of life evaluation (phase III only)

6.2.3 Visit during treatment

- Recording the weight of patients. If the weight of a patient fluctuates by less than 10% from the baseline (the day when the first dose of study treatment was given), the baseline weight will be used to calculate the dose. Otherwise use the current weight to calculate the dose
- Physical examination
- ECOG PS score

- 12-lead electrocardiogram (ECG)
- Myocardial test
- Count of routine blood test/blood chemistry/urinalysis
- AFP
- Coagulation function
- Thyroid function
- HBV DNA and/or HCV RNA (if applicable)
- Immunogenicity blood sample collection
- PK sample collection
- AE evaluation
- Recording concomitant medication
- Survival condition
- Tumor imaging evaluation
- Study drug infusion
- Quality of life evaluation (phase III only)

6.2.4 Safety visit

A safety follow-up will be carried out within 30 days (± 3 days) after the last dose or before the start of a new anti-tumor treatment, whichever occurs first. If the time interval between the safety visit and the previous visit is ≤ 7 days, there is no need to repeat the examination. Including the following procedures:

- Recording the vital signs and weight
- Physical examination
- ECOG PS score
- 12-lead electrocardiogram (ECG)
- Myocardial test
- Count of routine blood test/blood chemistry/urinalysis
- Coagulation function
- AFP

- Pregnancy test
- Thyroid function test
- HBV DNA and/or HCV RNA (if applicable)
- Immunogenicity sample collection
- AE evaluation
- Concomitant medication
- Survival condition
- Quality of life evaluation (phase III)
- Subsequent anti-tumor therapy (if applicable)

6.2.5 Survival follow-up

A survival follow-up is carried out every 60 ± 7 days after the safety follow-up. If there is no safety follow-up, the survival follow-up will be carried out every 60 ± 7 days after the last dose. For patients for whom tumor imaging assessments are still required as per the protocol, tumor assessments should continue by following the time points for imaging evaluations as stipulated in the protocol during the follow-up stage of survival.

6.3 Safety Evaluation Committee

The Safety Evaluation Committee will be composed of the sponsor and the investigator. The safety and tolerability of the phase II, single-arm study will be assessed and discussed during the trial. Whether the phase III study can be conducted should be confirmed after the discussion by the Safety Evaluation Committee.

The reasons, including but not limited to the following ones during the trial process, sudden onset of major safety incidents, major adjustments in the doses of study drugs, and major adjustments in the protocol, can trigger the discussion within the Safety Evaluation Committee.

7 STUDY EVALUATION

7.1 Informed Consent Form and Screening Log

A written informed consent form should be obtained from the patients to participate in the study before conducting any study-related operations (including screening and evaluation). The informed consent forms from the enrolled patients and those who are not enrolled subsequently will be kept by the study site.

All screening assessments should be completed to confirm that the patients meet all the inclusion criteria and do not meet any of the exclusion criteria before the enrollment. The investigator will

keep the screening log to record the detailed information of all patients who have undergone screening, and to confirm the eligibility for enrollment or to record the reasons for the failure of screening as appropriate.

7.2 Medical History, Tumor History, Concomitant Medication, and Demographic Data

The medical history at the baseline will be recorded, including major clinical and chronic diseases, history of liver diseases, surgical procedures, fertility status, alcohol and drug abuse, etc. The tumor history includes date of diagnosis, pathological results, tumor stage, and previous anti-tumor treatment (*e.g.*, systemic treatment, radiotherapy, surgery, local treatment, etc., including the initiation and completion dates, optimal efficacy, reasons for termination, and date of recurrence or progression). And information on whether infiltration of large vessels and/or extrahepatic spread occurs will be collected. In addition, all drugs taken by the patients with 28 days before the study treatment will be recorded.

Tumor history will include the assessment of previous surgery, radiotherapy, and drug treatment; the imaging results before the study may be collected for investigators' assessment.

The demographic data will include age, sex, and the self-reported race/ethnicity.

7.3 Efficacy Evaluation

7.3.1 Tumor imaging and disease assessments

RECIST V1.1 will be mainly used for this study, and tumor imaging assessments will be performed by the investigator (phase II) and both the investigator and IRRC (phase III). The efficacy assessment of mRECIST will be carried out by IRRC (phase III). iRECIST is used for patients in the immunotherapy group only. At 4–6 weeks after the first disease progression, the disease progression will be confirmed to rule out pseudo-progression (unless patients withdraw from the study after the first disease progression). For patients meeting iRECIST with confirmed disease progression, treatment discontinuation is required. Refer to Appendixes 3, 4, and 5 for the evaluation methods.

After the investigator's assessment of the first imaging progression based on RECIST V1.1, IRRC should confirm disease progression within 3 weeks, and whether the study drugs will continue to be used will be determined based on the investigators' comprehensive assessment. Tumor imaging evaluations usually include contrast-enhanced CT or MRI and the sites for evaluation must include chest, abdomen, and pelvic cavity. The dual-phase imaging for liver must be obtained to obtain high-quality arterial phase imaging of the whole liver during the first scanning and high-quality portal vein phase imaging of the whole liver during the second scanning, should enhanced CT scanning be not contraindicated. Contrast-enhanced MRI or CT of the brain should also be performed at the baseline for patients with suspected signs and

symptoms of CNS metastasis. Individuals with metastasis to CNS cannot be enrolled. For patients without metastatic lesions before randomization, there is no need for routine cranial imaging examinations afterward. The same patient should receive the same type of imaging examination during the study. If there are clinical indications, other imaging examinations, such as bone scan and neck CT scan, should also be performed for evaluation.

7.3.2 First tumor imaging

The first tumor imaging upon screening should be performed within 28 days before the randomization. Before the enrollment for the phase II study or randomization for the phase III study, the investigator at the study site will confirm that the patients have the measurable lesions that meet RECIST V1.1.

If the quality and standards of imaging, as the routine clinical practice, meet the requirements of this study, and it is conducted within 28 days before the first administration or randomization, it can be used as the tumor imaging assessment during the screening period.

The imaging method used for evaluation of tumor burden during each visit should be the same as the one used at the baseline (CT/MRI). Other affected sites should be examined based on the signs and symptoms of each patient.

7.3.3 Tumor imaging during study treatment

The planned imaging assessment time points: starting from randomization (from the first dose in the phase II study), every 6 weeks (±7 days) until 48 weeks, and then once every 12 weeks (±7 days). If a patient has clinically unstable conditions, the investigator can perform unscheduled imaging assessments at any time.

For patients in whom imaging progression is recorded for the first time, IRRC should confirm the progression within 3 weeks. The imaging study will be repeated at 4–6 weeks after the first finding of progression to confirm disease progression to rule out pseudo-progression (for IBI308 + IBI305 treatment group in the phases II and III studies only).

Tumor assessments will continue until disease progression, regardless of whether treatment has been terminated (*e.g.*, treatment termination due to toxicity), or until withdrawal of consent, initiation of new anti-tumor treatment, loss to follow-up, death, or study termination decided by the sponsor, whichever occurs first.

Although some patients have reached PD based on RECIST V1.1 or iCPD based on iRECIST (applicable to the treatment group), these patients can still be given continued treatment based on the investigator's judgment under the premise that the patients agree to continue the treatment and have signed the informed consent forms if the clinical status of the patients is stable with no adverse events (AEs) requiring termination of the study drug. The treatment can be continued

until the occurrence of any of the conditions, including no benefit or onset of intolerable toxicity assessed by the investigator, start of a new anti-tumor therapy, withdrawal of the ICFs by the patients, and loss to follow-up, whichever occurs first. The imaging assessment time points are still as follows: One tumor imaging evaluation is performed every 6 weeks (±7 days) until 48 weeks, and then once every 12 weeks (±7 days).

The scheduled tumor imaging should be conducted at the fixed time points by referring to the calendar, and cannot be postponed as a result of treatment delays, etc. If exceptions are present, discussion with and consent from the sponsor are required.

7.3.4 Tumor imaging at the time of study treatment termination and during follow-up

For patients in whom the treatment has terminated, a tumor imaging should be performed at the time of treatment termination. If the time interval between the previous scanning and the treatment termination is not more than 4 weeks, then it is not mandatory for the patients to undergo scanning at the time of treatment termination. For patients in whom treatment is terminated for reasons other than objective disease progression, imaging evaluations should be continued after treatment termination according to the time points as specified in the imaging study plan, until one of the following circumstances occur: Initiation of a new anti-tumor treatment, disease progression, withdrawal of informed consent, loss to follow-up or death, and study termination decided by the sponsor, whichever occurs first.

7.4 Safety Evaluation

7.4.1 AE evaluation

The investigator or qualified designee should evaluate each patient according to the visit schedule to identify the potential new or worsening AEs. Safety evaluations can be performed more frequently if clinically indicated. AEs are graded and documented according to CTCAE V5.0 during the study and follow-up. Toxicities are characterized by severity level, causality, and measures taken for study treatment.

All AEs of unknown causes after exposure to the study treatment must be evaluated to determine whether the event is potentially immune-related.

7.4.2 Laboratory test

The laboratory tests will be completed according to the arrangements in the visit schedule. See Table 13 for the laboratory test items involved in this study.

Table 13. Laboratory test items

Routine blood test	RBC, HGB, WBC, PLT, LYM, and ANC
Blood chemistry	TBIL, DBIL, CK, ALT, AST, γ-GT, ALP, ALB, TP, LDH, UREA, Cr, Na, K, Cl, Mg, Ca, P, amylase, and GLU

Urinalysis	pH, UWBC, UPRO, URBC, and UGLU
Thyroid function	FT3, FT4, and TSH
Coagulation function	APTT and INR
Viral serological test	HBsAg, HBsAb, HBcAb, HBeAg, HBeAb, HBV DNA, HCV antibody, HCV RNA, and HIV antibody
AFP	AFP
Myocardial test	Troponin, CK-MB, CK, BNP (if necessary), etc.

ALB = albumin; ALP = alkaline phosphatase; ALT = alanine aminotransferase; ANC = absolute neutrophil count; APTT = activated partial thromboplastin time; AST = aspartate aminotransferase; CK = creatine kinase; Cr = serum creatinine; DBIL = serum direct bilirubin; FT3 = free triiodothyronine; FT4 = free thyroxine; GLU=glucose; γ-GT = γ-glutamyltransferase; HBcAb = hepatitis B core antibody; HBeAb = hepatitis B e antibody; HBeAg = hepatitis B e antigen; HBsAb = hepatitis B surface antibody; HBsAg = hepatitis B surface antigen; HBV = hepatitis B virus; HCV = hepatitis C virus; HGB = hemoglobin; HIV = human immunodeficiency virus; INR = international normalized ratio; LDH = lactate dehydrogenase; LYM = lymphocyte count; pH = pH; PLT = platelet; TBIL = total bilirubin; TP = total protein; TSH = thyroid stimulating hormone; RBC = red blood cell; UGLU = urine glucose; UPRO = urine protein; URBC = urine red blood cells; UREA = urea; UWBC = urine white blood cells; WBC = white blood cell; CK-MB = creatine kinase isoenzyme; CK = creatine kinase; BNP = B-type brain natriuretic peptide.

7.4.3 Physical examinations and ECOG PS score

The physical examinations and ECOG PS score will be completed according to the arrangements in the visit schedule.

A comprehensive physical examination includes general conditions, respiratory tract, cardiovascular system, abdomen, skin, head and neck (including ears, eyes, nose, and throat), lymph nodes, thyroid, musculoskeletal system (including spine and limbs), genitalia/anus (if necessary), and nervous system. Complete physical examinations are required during the screening period; during the subsequent period of the study, the investigator can conduct targeted physical examinations according to the clinical manifestations of patients.

See Appendix 1 for the ECOG PS score criteria.

7.4.4 Height and weight

Weight will be recorded before every planned dose during the study by referring to the visit schedule. If the weight fluctuation is less than 10% compared to baseline (date of first dose), then use the baseline weight to calculate the dose. Otherwise, use the current weight to calculate the dose of the drug.

7.4.5 Vital signs

The vital signs will be examined according to the arrangements in the study visit schedule. The vital signs include body temperature, pulse, respiratory rate, and blood pressure (sitting or supine

position).

Additional monitoring points of vital signs are allowed based on standard clinical practice or clinical needs.

Additional vital signs record values can be collected in the eCRF when an AE/SAE occurs (if applicable). The time and date of measurement should be documented in the appropriate section of the eCRF.

7.4.6 12-lead electrocardiogram (ECG)

The 12-lead ECG examinations will be completed according to the arrangements in the study visit schedule.

The investigator should evaluate all the 12-lead ECGs as either a clinically significant abnormality or clinically insignificant abnormality. For a clinically significant abnormality, the investigator documented the result as an AE in the eCRF.

7.4.7 Pregnancy test and reporting

Urine or serum β human chorionic gonadotropin (β-HCG) pregnancy test must be performed in women of childbearing potential (refer to Section 4.3 for definition) within 3 days before the first dose of study treatment. If the result of urine β-HCG is positive or inconclusive, then serum β-HCG pregnancy test should be performed. The conclusion should be based on the serum pregnancy test. For the positive serum β-HCG result, the patient is not eligible for participating in the study and his participation will be terminated. A repeated test was performed for the suspected pregnancy during the study.

If a female patient is pregnant during the study, then the study treatment should be terminated immediately.

Pregnancy is not deemed an AE unless the study drug is suspected to be likely to interfere with the efficacy of contraceptives. Congenital malformation/birth defects and spontaneous abortion should be reported and handled as SAEs. Selective abortion without complications should not be handled as an AE. However, even if a patient has interrupted the study, all the pregnancy outcomes (spontaneous abortion, selective termination of pregnancy, ectopic pregnancy, normal delivery, or congenital malformation) should be followed up and recorded.

If a female patient or the spouse of a male patient is pregnant during the study, the investigator should report it to the sponsor within 24 h. The same time limit also applies to the information on pregnancy outcomes (see Section 8.4.2).

7.5 Collection and Analysis of Pharmacokinetic Specimens

Collection of phase II PK samples (for all enrolled patients):

- Cycle 1 (intensive sampling): Within 1 h before IBI308 infusion, immediately after IBI305 infusion (+5 min), 1 h ± 5 min after IBI305 infusion, and 6 h ± 15 min, 24 h ± 1 h, 48 h ± 2 h, 168 h ± 8 h (day 8), 336 h ± 12 h (day 15) after the start of IBI308 infusion. When the dosing on day 1 of cycle 2 is delayed due to an AE or other reasons, blood collection at 504 h ± 24 h (day 22) in cycle 1 will be added;
- Cycle 2: within 1 h before IBI308 infusion;
- Cycle 4 (intensive sampling): Within 1 h before IBI308 infusion, immediately after IBI305 infusion (5 min), 1 h ± 5 min after IBI305 infusion, and 6 h ± 15 min, 24 h ± 1 h, 48 h ± 2 h, 168 h ± 8 h (day 8), 336 h ± 12 h (day 15), 504 h ± 24 h (day 22) (namely within 1 h before the infusion in cycle 5) after the start of IBI308 infusion;
- Cycles 6/8/12: within 1 h before IBI308 infusion.

Collection of PK samples in phase III:

The PK sampling of the first 6 patients in the site will be designated.

- Cycle 1 (intensive sampling): Within 1 h before IBI308 infusion, immediately after IBI305 infusion (5 min), 1 h ± 5 min after IBI305 infusion, and 6 h ± 15 min, 24 h ± 1 h, 48 h ± 2 h, 168 h ± 8 h (day 8), 336 h ± 12 h (day 15) after the start of IBI308 infusion. When the dosing on day 1 of cycle 2 is delayed due to an AE or other reasons, blood collection at 504 h ± 24 h (day 22) in cycle 1 will be added;
- Cycle 2: within 1 h before IBI308 infusion;
- Cycle 4 (intensive sampling): Within 1 h before IBI308 infusion, immediately after IBI305 infusion (5 min), 1 h ± 5 min after IBI305 infusion, and 6 h ± 15 min, 24 h ± 1 h, 48 h ± 2 h, 168 h ± 8 h (day 8), 336 h ± 12 h (day 15), 504 h ± 24 h (day 22) (namely within 1 h before the infusion in cycle 5) after the start of IBI308 infusion;
- Cycles 6/8/12: within 1 h before IBI308 infusion.
- Note: Should more than 20 individuals are enrolled for infusion in phase II study, the number of individuals subjected to intensive sampling in phase III study can be reduced following the clinical pharmacological evaluation.

The PK sampling time points for other patients (barring the first 6 patients in the site as designated) will be:

- Cycle 1: within 1 h before IBI308 infusion and immediately at the end of IBI305 infusion (+5 min);
- Cycles 2/4/12: within 1 h before IBI308 infusion.
- If a patient terminates the treatment early between cycles 5–11, then a PK sampling will be carried out at the time of early termination of visit.

A coagulation-promoting vacutainer will be used to collect 3.5 mL of whole blood. Then, the serum will be separated from the whole blood sample and frozen in aliquots for the PK (IBI308 and IBI305) analysis. Refer to the "Laboratory Manual" provided by the sponsor-designated central laboratory for sampling methods, sample storage, transport, and analysis.

7.6 Collection and Analysis of Immunogenicity Specimens

Immunogenicity samples will be collected within 1 h before IBI308 infusion in cycles 1/2/4/8/12/16, then every 8 cycles (cycle 24, 32, and so on) thereafter, and during the safety follow-up.

If an IBI308 and/or IBI305 infusion-related reaction occurs, blood samples should be taken after the start of the event with the least delay, end of event, and around 30 days after the event for immunogenicity analysis.

The immunogenic samples will be subjected to the anti-drug (IBI308 and IBI305) antibody (ADA) test, and the ADA-positive serum samples will be further subjected to neutralizing antibody (NAb) assay.

For ADA and NAb assays, 7 mL of whole blood will be collected using vacutainers with clot activator. Serum will then be separated, dispensed in aliquots, and frozen.

Refer to the "Laboratory Manual" provided by the sponsor-designated central laboratory for sampling methods, sample storage, transport, and analysis.

7.7 Quality of life evaluation (phase III)

Quality of life evaluation includes the EORTC QLQ-C30 and EORTC QLQ-HCC18 scales, and will be performed on the day of the first administration (before the administration), and at the time of each imaging studies and safety follow-up. A quality of life evaluation will be conducted as required in the meantime should an unscheduled imaging evaluation be performed.

The EORTC QLQ-C30 is a core questionnaire for patients with cancer consisting of 30 items which are divided into 15 scales: 5 functional scales (physical, role, cognitive, emotional, and social function), 3 symptom scales (fatigue, pain, and nausea/vomiting), 1 global health status/quality of life scale, and 6 single item scales.

The EORTC QLQ-HCC18 scale is a subscale which is developed by adding the specific modules on the evaluation of the quality of life of patients with HCC to the EORTC QLQ-C30 quality of life core scale series, serving as a supplement to the EORTC QLQ-C30. The EORTC QLQ-HCC18 includes 8 dimensions: Fatigue, physical changes, jaundice, changes in nutrition, pain, fever, changes in sexual lives, and abdominal distension, all of which form a scale with a total number of 18 items; the 4-level Likert scoring will be used (1–4 points), with 1 point = no, 2 points = somewhat, 3 points = quite, and 4 points = very. The higher the score of symptom dimension is, the worse the quality of life will be.

7.8 Biomarker Specimens Collection and Evaluation (phase III)

7.8.1 Tissue biomarkers

After meeting the pathological inclusion criteria for HCC and being randomized into the treatment group, the patients can provide tumor tissue samples (tissue block or unstained section) meeting the testing requirements pending detection of biomarkers, *e.g.*, PD-L1 expression (for the treatment group in phase III only). The following archived tumor tissues are acceptable:

1) The surgical time should be within half a year from now (1 year at maximum); the recommended time for archiving of sections is within 3 months from now (6 months at maximum);
2) The number of unstained sections should be at least 10, and the sections should be 4–5 μm in thickness (containing 25 mm^2 tissue area, of which the tumor cells account for over 20%).

Refer to the "Laboratory Manual" provided by the sponsor-designated central laboratory for sampling requirements, sample storage, shipment, and analysis of sections.

7.8.2 Storage and destruction of biological samples

Samples will be disposed of or destroyed, pooled and anonymized. Additional analyses of pooled and anonymized samples may be performed to further evaluate and validate the analytical method. Results of these analyses may be published separately from the CSR.

Reproducibility (if performed) will be assessed simultaneously with the biological analysis of the samples. The results of these evaluations will not be published in the clinical study report, but will be presented separately in a biological analysis report.

8 SAFETY REPORTS AND MANAGEMENT OF ADVERSE EVENTS

8.1 Definition

8.1.1 Definition of adverse event

An adverse event (AE) is defined as any adverse unexpected medical event within the period from the signing of the informed consent form, regardless of whether or not considered as related to the study drug. AEs include but are not limited to the following:

- Worsening of pre-existing (before enrollment) medical conditions/diseases (including symptoms, signs, and laboratory test abnormalities);
- Any new adverse medical conditions (including symptoms, signs, and newly diagnosed diseases);
- Clinically significant laboratory test abnormalities.

AEs include SAEs and non-SAEs.

8.1.2 Definition of serious adverse event

SAEs refer to AEs that occur during any study period that meet one or more of the following criteria:

- Leading to death
- Life-threatening (a life-threatening event is defined as an AE when the patient is at immediate risk of death at the time, but does not include the case that may lead to death only when the event worsens)
- Requiring hospitalization or prolonged hospitalization, excluding those as follows:
  - ✓ Hospitalization at a rehabilitation institution
  - ✓ Hospitalization at a sanatorium
  - ✓ General emergency admission
  - ✓ Day surgery (e.g. outpatient/same-day/ambulatory surgery)
  - ✓ Hospitalizations or prolonged hospitalizations due to worsening of an AE are not considered as SAEs. They include hospitalization due to pre-existing disease, without new AEs and exacerbation of pre-existing disease (e.g. hospitalization to test laboratory abnormalities that have been persistent before the study starts); hospitalization for administrative reasons (e.g. annual routine physical examinations); hospitalizations during the study as specified in the protocol (e.g. hospitalization performed in accordance study protocol); elective hospitalization

unrelated to worsening of AEs (e.g. elective surgery); scheduled treatment or surgical procedures, which should be documented in the entire study protocol and/or individual patient's baseline information; and hospitalization merely due to the use of blood products.

- Leading to permanent or significant disability/functional loss
- Congenital anomalies or birth defects
- Important medical events refer to medical events that do not meet any of the above-mentioned criteria, but which may endanger the health of patients, or which may require drugs or surgical interventions for the prevention of serious consequences as listed in the above-mentioned definitions.

8.2 Adverse Event Evaluation

8.2.1 Severity level

The investigator will grade all clinically significant laboratory abnormalities which are recorded as AEs as per CTCAE V5.0.

For AEs which are not defined in CTCAE, the severity level can be described by referring to the following rules.

- Grade 1: Mild; patients are asymptomatic or have mild signs; only the clinical or diagnostic test results are available; no interventional procedure is required.
- Grade 2: Moderate; minimal, local or non-invasive intervention required; limiting age-appropriate instrumental activities of daily living[*].
- Grade 3: Severe or medically significant but not immediately life-threatening; hospitalization or prolonged hospitalization indicated; disabling; limiting self-care activities of daily living[#]. Attention: "Severe" does not necessarily mean "serious".
- Grade 4: Life-threatening consequences; urgent intervention indicated.
- Grade 5: Resulting in death.

[*]: Capability of instrumental activities of daily living refers to the capability of preparing meals, buying groceries or clothes, using phones, managing money, etc.

[#]: Capability of self-care activities of daily living refers to the capability of bathing, putting on and taking off clothes, eating, using the toilet, taking drugs, and being not bedridden.

8.2.2 Causality with study drugs

The investigator should evaluate the causality of the study drug with each case of AE, and answer "yes" or "no" to the question "Do you think that the development of this AE is attributable to the study drug and is there a reasonable possibility?"

When the "reasonable possibility" is determined that an AE is caused by the drug, the following factors should be taken into consideration:

- Time course. Exposure to suspected drugs. Has the patient been exposed to suspicious drugs? Is there a reasonable time relationship between the development of AEs and the suspected drug?
- Consistency with the characteristics of known drugs. Are the AEs consistent with the previously reported events arising from the suspicious drugs (pharmacology and toxicology) or drugs of similar pharmacological types? Can the development of AEs be expected from the pharmacological characteristics?
- Dechallenge. Have the AEs alleviated or improved after the suspicious drugs have been discontinued or reduced?
- There are no alternative factors. The AEs cannot be explained reasonably by other pathology, *e.g.*, underlying diseases, other drugs, and other internal or environmental factors.
- Rechallenge. Do the AEs reappear following the discontinuation of the suspicious drugs?
- Other possible causes. Cannot the AE be explained by other pathogenies, such as underlying disease, other drugs/vaccines, or other hosts/environmental factors?

Reconsideration of the "reasonable possibility" of this AE is required when one or more factors exist.

On the contrary, if the above-mentioned criteria do not apply, or no obvious exposure evidence and reasonable time course are present, or any rechallenge test (if implemented) produces a negative result or another possible cause of AE, no "reasonable possibility" is likely to be present for the causality.

8.3 Adverse Event Documentation

8.3.1 Time limit for collection of adverse event

All AEs, including SAEs, that occur from the signing of the ICF to 30 days after the last dose will be collected, regardless of whether they are observed by the investigator or self-reported by the patients.

From day 30 to day 90 after the last dose, the investigator should report all SAEs and AEs related to study drugs or procedures.

After 90 days from the last dose, the investigator should report any SAEs that are considered related to study drugs or procedures.

Should new anti-tumor treatment is initiated within 90 days after the last dose, only SAEs related to study drugs will be recorded subsequently.

8.3.2 Follow-up of adverse event

The AEs should be followed until the events recover/return to the baseline values, or until the investigator believes that no further follow-up is required for reasonable reasons (if the event cannot be resolved or has already been improved). If the adverse event cannot be resolved, a reasonable explanation is required.

8.3.3 Contents of adverse event documentation

For each case of AEs, the following information should be collected:

- Name of an AE
- Start and end time of an AE
- Whether it is an SAE
- AE severity level (CTCAE grading)
- Causality with the study drug (yes, no)
- Action taken on the study drugs
- AE outcomes

In addition, the following information should be collected for the SAEs:

- Date when the investigator is noticed about the SAE
- Judgment criteria for SAEs
- Date of hospitalization (if applicable)
- Date of discharge from the hospital (if applicable)
- Reasons leading to death (if applicable)
- Date of death (if applicable)
- Autopsy report (if applicable)
- Assessment of causality with the study process

- Assessment of causality with other concomitant medications
- Description of AEs

Descriptions of the AE are as follows:

Diagnosis, signs, and symptoms

Document the definite diagnosis name, if there is one, rather than just listing the independent signs and symptoms (e.g. hepatic failure rather than jaundice, transaminases increased, and asterixis). Signs and symptoms should be reported as separate AEs/SAEs if cannot be attributed to the diagnosis. If it is determined that the signs and symptoms are caused by the diagnosis during the follow-up of AEs/SAEs, then only the diagnosis should be reported, and the symptoms and signs will be included in the diagnosis and will not be documented as AEs; for SAEs, the follow-up update reports should be sent.

AEs secondary to other events

Generally, AEs secondary to other events (such as the result of another event or clinical sequelae) should not be documented, and only the primary events should be documented. In the following exceptional circumstances, secondary events should be recorded separately:

- The secondary events are clinically significant, and the time of occurrence is different from the primary events;
- There is an unclear correlation between the secondary events and the primary events.

Ongoing or recurrent AEs

An ongoing AE refers to an event that does not resolve and is ongoing between two assessment time points.

Recurring AEs refer to AEs that have resolved between the two time points of assessment but subsequently occur again. These events should be independently documented.

Laboratory test abnormalities

All clinically significant laboratory test abnormalities should be reported as AEs. The investigator had responsibilities to review all the laboratory test abnormalities and determine whether the abnormalities should be reported as AEs.

Death

During the entire course of the study, all deaths (barring deaths due to disease progression) that occur within the 90-day safety follow-up period after the last dose should be documented in the Death Report Form in the eCRFs and reported to the sponsor in a timely manner, regardless of the causality with the study drug.

For death with a known cause, recorded the cause of death as an AE and the outcome of the AE as "death" and submitted an SAE report; for a death with an unknown cause, the AE was recorded as "unknown cause of death" in the AE form, and submitted an SAE report. Further investigation was performed for the exact cause of death.

Pre-existing medical conditions

Symptoms/signs presenting during the screening period will be recorded and reported as AEs only if the severity level, frequency, or property becomes aggravated (except for worsening of the studied disease). The relative change should be documented, such as "increased frequency of headaches".

Disease progression

Disease progression is defined as the deterioration of patients' conditions as a result of primary tumors treated by the study drugs. The appearance of new lesions, or the progression of the primary lesion, is considered disease progression. Disease progression will not be reported as an AE. Any deaths, life-threatening events, hospitalization or prolonged hospitalization, permanent or significant disability/incapacity, congenital anomaly/birth defects, or other important medical events caused by disease progression will not be reported as an SAE.

8.4 Reporting of Serious Adverse Event, Pregnancy, and Hepatic Function Abnormal Events

8.4.1 SAE reporting

All SAEs (See Section 8.3.1 for the specific time limit) that occur since the signing of ICF until 90 days (inclusive) after the last dose must be reported. The investigator must fill out the *Innovent SAE Report Form* regardless of whether it is the initial report or a follow-up report. Besides, the investigator must report the SAE to the sponsor (drugsafety@innoventbio.com) within 24 h of noticing the event, as well as to regulatory authorities and EC in accordance with the laws and regulations of China.

For SAEs occurring outside of the above-mentioned period, those considered related to the study drug should also be reported to the sponsor.

The investigator should urgently perform a visit on missing information and provide a complete SAE report for events that result in death or are life-threatening.

8.4.2 Pregnancy reporting

All patients participating in this clinical study who are fertile should take effective contraceptive measures.

If a female patient exposed to the study drug becomes pregnant during the clinical trial, she should terminate the study treatment immediately. The investigator must report to the sponsor within 24 h of noticing the event and submit the "Innovent Clinical Study Pregnancy Report/Follow-Up Form".

Male patients exposed to the study drug can continue the clinical trial if their partners become pregnant during the clinical trial. The investigator must report to the sponsor within 24 h of noticing the event and submit the "Innovent Clinical Study Pregnancy Report/Follow-Up Form".

The investigator must continuously monitor and follow up on the outcomes of the pregnant patients until the 8th week after the patients give birth. The outcomes should be reported to the sponsor.

If the outcome of the pregnancy is stillbirth, spontaneous abortion, fetal malformation (any congenital anomaly/birth defect), or medical abortion, it should be considered as an SAE and the event is required to be reported in accordance with SAE procedures and time limits.

If the patient also experiences a SAE during the pregnancy, the "SAE Report Form" should also be filled out and reported according to SAE's procedures.

8.4.3 Hepatic function abnormal event reporting

Drug-induced liver injury is considered if abnormal AST and/or ALT levels are accompanied with abnormal elevation of TBIL, and the following conditions (Table 14) are met without other possible causes.

Table 14. Liver injuries required to be reported as SAEs

Baseline	Normal (AST/ALT and TBIL)	Abnormal (AST/ALT and TBIL)
Treatment period	ALT or AST ≥ 3 × ULN with TBIL ≥ 2 × ULN and ALP ≤ 2 × ULN and no hemolysis	ALT or AST ≥ 8 × ULN with TBIL increase ≥ 1× ULN or value ≥ 3× ULN

Once obtaining the abnormalities, the patients must return to the study site promptly (ideally within 48 h) and receive an assessment. The assessment must include laboratory tests, detailed medical history, and physical assessment, and the possibility of hepatic tumor (primary or secondary) should be considered.

Other than repeated AST and ALT tests, albumin, creatine kinase, TBIL, direct and indirect bilirubin, γ-GT, PT/INR, and ALP should also be tested. Detailed medical history includes history of alcohol, acetaminophen, soft drugs, various supplements, traditional Chinese medicine, chemical drug exposure, family diseases, occupational exposure, sexual behavior, travel, contact with patients with jaundice, surgery, blood transfusion, hepatic diseases or allergies, cardiac diseases, and immune diseases. Further tests may include the detection for

acute hepatitis A, B, C and E, hepatic imaging (such as the biliary tract), and echocardiography. If a retest showed consistency with the criteria outlined in Table 14 and there are no other possible causes, the possibility of drug-induced liver injury should be considered before all the results of etiological tests are accessible. This potentially drug-induced liver injury should be reported as SAEs.

8.5 Adverse Event of Special Interest

The following AEs are defined as adverse events of special interests (AESIs) in this study:

AESIs arising from IBI308 and IBI305 include but are not limited to:

- Interstitial pneumonia
- Colitis
- Endocrine disorders: type 1 diabetes mellitus, pancreatitis, adrenal insufficiency, hyperthyroidism, and hypophysitis
- Hepatitis, including ALT or AST ≥ 10 × ULN
- Neurological disorders: Guillain-Barré syndrome, myasthenia syndrome or myasthenia gravis, and meningoencephalitis
- Events such as hypersensitivity, infusion-related reactions, cytokine release syndrome, influenza-like illness, systemic inflammatory response syndrome, and systemic immune activation
- Nephritis
- Ocular toxicity (*e.g.*, uveitis, retinitis, and optic neuritis)
- Myositis
- Myopathy, including rhabdomyolysis
- Cardiac disorders ≥ grade 2 (*e.g.*, atrial fibrillation, myocarditis, and pericarditis)
- Vasculitis
- Autoimmune hemolytic anemia
- Severe skin reactions (*e.g.*, stevens-johnson syndrome, dermatitis bullous, and toxic epidermal necrolysis)
- Intestinal obstruction
- Gastrointestinal perforation

- Procedural and wound healing complications
- Hemorrhage
- Fistula
- Hypertension
- Thrombotic event
- Posterior reversible encephalopathy syndrome (PRES)
- Proteinuria
- Ovarian failure
- Cardiac failure congestive

If the criteria for SAEs are met, the SAEs should be reported within the specified time limit (see Section 8.4.1 for details).

8.6 Immune-Related Adverse Event

AEs related to IBI308 exposure may arise from the immunological causes. These irAEs may occur in a short period after the first IBI308 administration or within several months after the last IBI308 administration, and may affect more than one body system at the same time. Therefore, early discovery and initiation of treatment are of vital importance to reduce complications. Based on the available clinical trial data, most irAEs are reversible and can be managed by interruption of IBI308 administration, administration of glucocorticoids and/or other supportive therapies. For suspected irAEs, make sure that proper assessment is conducted to confirm the etiology or rule out other causes. Other procedures or examinations such as bronchoscopy, endoscopy, or skin biopsy may be included as part of this assessment. Based on the severity levels of irAEs, interrupt or terminate IBI308 and treat with glucocorticoid therapy. If the criteria for SAEs are met, the SAEs should be reported within the specified time limit (see Section 8.4.1 for details). For guidelines for the treatment adjustment and toxicity management of potential irAEs, refer to "Manual for the Management of Immune-related Adverse Events" provided by the sponsor.

9 DATA ANALYSIS/STATISTICAL METHODS

9.1 Statistical Analysis Plan

A detailed statistical analysis plan (SAP) will begin to be written after the enrollment of the first patient, and will be finalized before database locking. The SAP will contain details of all analyses.

9.2 Statistical Hypothesis Test

No formal statistical hypothesis test will be performed in the phase II section of this study. In the phase III section, the superiority test will be carried out for IBI308 + IBI305 vs. sorafenib. The test will be performed on the dual efficacy endpoints of OS and PFS assessed by IRRC as per RECIST V1.1.

With respect to the endpoints of OS and PFS: H0: HR ≥ 1 for null hypothesis, and Ha: HR < 1 for alternative hypothesis.

The Bonferroni correction method is used to adjust α to control the overall type I error rate at a two-sided level of significance α = 0.05, i.e., to test OS at the two-sided level of significance α = 0.048, and to test PFS at the two-sided level of significance α = 0.002. If any of the efficacy endpoints of PFS and OS is statistically significant, the superiority will be established statistically.

First, the test of PFS will be performed. If the test result is statistically significant, the two-sided level of significance α = 0.002 allocated to PFS is recovered to the OS hypothesis test (i.e., the overall OS hypothesis will be tested at the significance level α = 0.05). If the test results of PFS are not statistically significant, OS will be tested at the original significance level α = 0.048.

9.3 Analysis Set

Phase II:

The analysis population is the safety set (SS), which is defined as the patients who have received at least one dose of the study drug.

Phase III:

The analysis population includes intent to treat (ITT) analysis set, full analysis set (FAS), safety set (SS), per protocol set (PPS), and PK analysis set:

1. ITT: all randomized patients. The grouping will be performed with randomized treatment groups, regardless of the actual treatment options that patients receive. This dataset will be used as the primary analysis data set for the efficacy assessment of OS and PFS.
2. FAS: Subset of the ITT analysis set, including the patients who had measurable lesions at baseline, and who have received at least one dose of the study drug. This dataset will be used as the primary analysis data set for the efficacy assessment of ORR.

3. PPS: Subset of the FAS, including the patients who have not received the concomitant medication prohibited by the protocol, who have completed the pre-determined minimum study exposure treatment, and in whom no protocol deviation occurs that affects the efficacy assessment. This dataset will be used as the secondary analysis set for efficacy assessment.
4. SS: Patients who have received at least one dose of the study drug.
5. PK concentration set: Including all the patients who have received at least one dose of the study drug and who have at least one effective component concentration datum after the drug administration.

9.4 Distribution of Patients

The number of patients in phases II and those subjected to randomization in phase III (phase III only), those who received treatment, those who terminated the study treatment and terminated the study, and those who experienced major protocol deviations will be counted separately. The main causes for terminating the study drugs and/or terminating the study will be summarized in the eCRF by category. The status (survival, death, withdrawal of ICF, or loss to follow-up) at the data cutoff date and the end of the study will be summarized on the basis of data in the eCRF.

The major protocol deviations will be summarized and listed by category.

9.5 Statistical Analysis Methods

9.5.1 General statistical analysis

Unless otherwise specified, the data in phases II and III will be analyzed separately.

Unless otherwise specified, variable data will be described using the mean ± standard deviation or median (maximum and minimum), and attributes data will be described using frequency and percentage.

All statistical analyses were performed using SAS (V9.2 or higher).

9.5.2 Efficacy analysis

Phase II: Efficacy analysis will be performed on the SS.

Phase III: Efficacy analysis will be performed on the ITT analysis set and FAS. PPS will be used as the secondary analysis set.

9.5.2.1 Primary efficacy endpoint

Phase II:

There will be no primary efficacy endpoints.

Phase III:

Overall survival (OS): Time from randomization to death recorded for any cause. Patients who are still alive at the time of analysis are censored at the last contacted date. The inter-group comparison of OS will be performed using the stratified log-rank test, the median OS and corresponding 95% CI will be estimated using the Kaplan-Meier method, and the survival curves will be plotted. HR and corresponding 95% CI will be estimated by a stratified Cox proportional hazards model with the randomization stratification factor.

For inter-group OS comparison, the auxiliary analysis will include the situation of other anti-tumor treatment after disease progression (presence or absence). The subsequent anti-tumor treatment of special interest refers to other anti-PD-1/PD-L1 monoclonal antibody treatment. The auxiliary analysis will include but not limited to: Censoring patients who received the subsequent anti-PD-1/PD-L1 anti-tumor treatment using the Rank Preserving Structural Failure Time Model, the two-stage model, etc.

Progression-free survival (PFS): According to RECIST V1.1, the time from randomization to the first recorded imaging disease progression or death caused by any reason as assessed by IRRC, whichever occurs first. Subjects who are still alive with no disease progression record at the time of analysis are censored at the last imaging evaluation date. Subjects who are still alive with no imaging evaluation record after baseline are censored at the randomization date. A stratified log-rank test will be used to compare the PFS between treatment arms. The median PFS and corresponding 95% CI will also be estimated using Kaplan-Meier, the survival curve will be plotted, and the 6-month PFS rate will be calculated. The HR and corresponding 95% CI of PFS between the groups will be estimated with a stratified Cox proportional hazards model, with the stratification factor being random stratification factor.

9.5.2.2 Secondary efficacy endpoints

The efficacy indicators in phase II are will be described in a list only.

Phase III:

- Progression-free survival (PFS) assessed by the investigator: According to RECIST V1.1, the time from randomization to the first recorded imaging disease progression or death caused by any reason as assessed by the investigator, whichever occurs first. Patients who are still alive with no disease progression record at the time of analysis are censored at the last imaging evaluation date. Patients who are still alive with no imaging evaluation record after baseline are censored at the randomization date. PFS analysis is the same as IRRC-assessed PFS analysis in the primary efficacy endpoint.

- Objective response rate (ORR): According to RECIST V1.1, the proportion of patients who achieve CR or PR assessed by IRRC and the investigator in the analysis population. For the main analysis of inter-group ORR comparison, the CMH method added with stratification factors will be used for superiority testing; the Miettinen & Nurminen method added with stratification factors will be used to estimate the inter-group rate differences in ORR and their confidence intervals.
- Time to progression (TTP): According to RECIST V1.1, the time from randomization to the first recorded imaging disease progression or death caused by disease progression as assessed by IRRC and the investigator, whichever occurs first. Patients who are still alive with no disease progression record at the time of analysis are censored at the last imaging evaluation date. Patients who are still alive with no imaging evaluation record after baseline are censored at the randomization date. The median TTP and the corresponding 95% CI will be estimated via the Kaplan-Meier method, the survival curve will be plotted, and the 6-month TTP rate will be calculated.
- Duration of response (DOR): According to RECIST V1.1, the time from the first recorded CR or PR to disease progression or death for patients with CR or PR assessed by the investigator and IRRC. Patients who are still alive with no disease progression at the time of analysis are censored at the last imaging evaluation. The median DOR will be estimated via the Kaplan-Meier method and the survival plots will be plotted.
- Disease control rate (DCR): According to RECIST V1.1, the proportion of patients who achieve CR, PR, or SD assessed by the investigator and IRRC in the analysis population. The analysis method is the same as ORR analysis.
- Time to response (TTR): According to RECIST V1.1, the time from randomization to the first recorded CR or PR for patients with CR or PR assessed by the investigator and IRRC. The median TTR will be estimated via the Kaplan-Meier method and the survival plots will be plotted.
- To compare PFS, ORR, TTP, DOR, DCR, and TTR determined by IRRC as per the modified RECIST (mRECIST) in IBI308+ IBI305 group vs. the sorafenib group.

9.5.3 Quality of life analysis

For phase III, the EORTC QLQ-C30 and EORTC QLQ-HCC18 scores at each evaluation time point will be calculated. The overall score and the score for each dimension at each time point, and relative changes from baseline, will be summarized using descriptive statistics.

9.5.4 Safety analysis

The data of phase II and phase III studies will be analyzed separately. The safety analysis will be performed based on SS, with the safety parameters including AEs, laboratory tests, vital signs, ECG, immunogenicity, etc.

9.5.4.1 Drug exposure

The drug exposure and duration of treatment (number of treatment cycles) will be summarized.

9.5.4.2 Adverse event

All AEs will be coded and classified using MedDRA, and graded as per CTCAE V5.0.

The incidences (frequency) of all TEAEs, TEAEs at grade 3 and above, TEAEs related to the study drugs, irAEs, SAEs, SAEs related to study drugs, TEAEs leading to discontinued study medication, and TEAEs leading to study termination will be summarized, and the above-mentioned AEs will be summarized based on SOCs and PTs in MedDRA coding. In addition, the severity levels of TEAEs and relationship with the study drug were also summarized by SOC and PT.

Patients who terminate the treatment due to AEs, development of SAEs, or death will be listed (at least including the start and end date of the AEs, severity levels, their relationship with study drugs, measures taken, and outcomes).

9.5.4.3 Laboratory test

For laboratory tests, the observed values and changes from the baseline will be analyzed using descriptive statistics. The baseline results and the worst results during the trial were presented in a crosstab. Laboratory test abnormalities will be graded and summarized according to CTCATE V5.0.

9.5.4.4 12-lead ECG

Descriptive statistics are used for 12-lead ECG and changes from baseline. A cross-classification table is used to describe ECG and changes from baseline before and after the treatment and data lists will be provided.

9.5.4.5 Vital signs, physical examinations, and other safety-related examinations

Measured values and changes from baseline for vital signs, physical examinations, and other safety-related examination values will be analyzed using descriptive statistics. The baseline results and the worst results during the trial were presented in a crosstab.

ECOG and PS will be analyzed and summarized using descriptive statistics.

9.5.4.6 Immunogenicity indicators

The percentages of patients with positive ADA and NAb from IBI308 and IBI305 will be calculated separately, and summarized using descriptive statistics. The antibody levels in positive patients will be listed.

9.5.5 Compliance analysis

The data of phase II and phase III studies will be analyzed separately. The proportion and frequency of patients violating the expected administration regimen, the proportion of patients in whom the doses of study drugs account for between 80–120% of the those prescribed by the protocol, the proportion of patients completing the study, and the proportion of patients completing different treatment cycles will be summarized.

9.5.6 Patient baseline characteristics

The data of phase II and phase III studies will be analyzed separately. The patients' demographic characteristics (gender, age), tumor diagnosis information (pathological diagnosis, clinical staging, and previous therapy), and other baseline information (height, and weight [body mass index]), vital signs, ECOG PS, laboratory tests, and past/concomitant therapy will be analyzed using descriptive statistics.

9.5.7 Pharmacokinetic (PK) analysis

The data of phase II and phase III studies will be analyzed separately. The PK characteristics in human body will be described, and the main PK parameters will be determined, including but are not limited to area under curve (AUC), C_{max}, CL, V, $t_{1/2}$, and trough concentrations in cycles 1/3/11.

The population pharmacokinetics of IBI308 + IBI305 in the first-line treatment of advanced HCC will be analyzed using descriptive statistics. The PK analysis set is used for the pharmacokinetic analysis.

9.5.8 Exploratory analysis

- The expression level and the distribution of biomarkers in tissues with the potential of predicting therapeutic effects will be analyzed using descriptive statistics, and the potential relationship between the following indicators and the therapeutic effects will be explored using such methods as the applicable efficacy analysis model, with the expression of the following biomarkers as the covariate:

  1) Expression level of PD-L1;

  2) Detection of RNA in tissues;

9.6 Interim Analysis

The interim analysis will not be carried out in the phase II study.

The interim analysis is only carried out in the phase III study. In this study, the number of interim analyses of the OS endpoint will be determined based on the number of OS events observed when 400 PFS events have occurred: If the number of OS events observed is less than 220 when 400 PFS events have occurred, two interim analyses will be performed: The first interim analysis of OS is performed when 400 PFS events are observed, and the second interim analysis is performed when 265 OS events are observed. Otherwise, only one interim analysis is performed when 265 OS events are observed. The α cut-off value of each analysis of OS is allocated to approximate the O'brien-Fleming boundary according to the Lan-Demets method, and adjusted according to the actual number of OS events that have been observed to ensure that the OS analysis is a two-sided test with an overall significance level of α=0.048. For the cut-off values of the two-sided test and the minimum detectable difference of each analysis estimated based on the expected number of OS events when 400 PFS events have occurred, see Table 15.

Table 15. Estimated α cut-off values and minimum detectable value of HR during each analysis of PFS and OS

		First interim analysis			Second interim analysis			Final analysis		
Possible situation	**Endpoint indicator**	**Estimated number of events**	**α cutoff value (two-sided)**	**HR MDD**	**Estimated number of events**	**α cutoff value (two-sided)**	**HR MDD**	**Estimated number of events**	**α cutoff value (two-sided)**	**HR MDD**
1. The occurrence time of 400 PFS events and 265 OS events is similar.	PFS*	400	0.002	0.719	NA					
	OS#	265	0.0102	0.716	NA			408	0.0448	0.811
2. If the number of OS events observed when 400 PFS events have occurred is far less than 265 cases, two interim analyses will be performed. The number of OS events during the first interim analysis is estimated to be 180–220 cases.	PFS*	400	0.002	0.719	NA					
	OS#	180	0.0014	0.604	265	0.0098	0.715	408	0.0447	0.811
		190	0.0019	0.621	265	0.0096	0.715	408	0.0447	0.811
		200	0.0025	0.637	265	0.0094	0.714	408	0.0447	0.811
		210	0.0033	0.652	265	0.0092	0.713	408	0.0045	0.811
		220	0.0042	0.665	265	0.009	0.713	408	0.0446	0.811

MDD = Minimum detectable difference.
*: PFS is only tested at the two-sided level of significance α=0.002 during the first interim analysis.
#: The overall test level of OS is 0.048. If the test level of PFS is recovered to OS, the α boundary value of each analysis of the OS will also be adjusted correspondingly according to the overall test level of OS.

In this study, the occurrence time of the 400 PFS events will be predicted based on the judgment by the investigator. If the actual number of PFS events predicted by IRRC is not equal to 400 cases, the data cut-off date of the 400 PFS events will no longer be adjusted.

9.7 Subgroup Analysis

In phase III, a subgroup efficacy analysis is performed for factors that are clinically considered to affect prognosis, such as gender, age, ECOG PS score, history of viral hepatitis, baseline level of alpha-fetoprotein (AFP), macrovascular invasion, and/or extrahepatic metastasis. The subgroup analysis to be carried out will be defined in detail in the statistical analysis plan.

9.8 Multiple Comparisons and Adjustments

In the phase III study, the test of PFS will be performed first. If the test result is statistically significant, the two-sided level of significance $\alpha = 0.002$ allocated to PFS is recovered to the hypothesis that is not rejected. If the test result of PFS is not statistically significant, the OS will be tested at the original significance level. The implementation of this pre-defined sequence of testing procedure will effectively control the type I error rates of all key hypotheses to the 5% level (two-sided). This study will only conduct an interim analysis on the OS endpoint. The number of interim analyses will be determined based on the number of OS events observed when 400 PFS events have occurred. The cut-off value of the interim analysis is allocated to approximate the O'brien-Fleming boundary according to the Lan-Demets method, and adjusted according to the actual number of OS events that have occurred.

9.9 Eligible Patient Data Lists

The data of patients in phases II and III will be listed separately. In addition to patients' data list, tumor evaluation (date of evaluation, lesion status, evaluation results) and efficacy endpoints of patients who have achieved CR and PR will be listed separately.

The PFS and OS data of all the patients by the end of the study will also be presented separately.

9.10 Sample Size and Power Calculation

Phase II single-arm study, safety run-in period:

A total of 20 patients with advanced HCC are scheduled to be enrolled. The safety and tolerability of combination treatment will be evaluated.

According to the CSCO conference in 2018, the treatment-related adverse events of grade 3 or above as per CTCAE were observed in 17/68 patients receiving atezolizumab + bevacizumab (incidence: 25%; 95% confidence interval: 15–37%). It is predicted in the phase II design of this study that the safety profile of IBI308 in combination with IBI305 will be similar to that of products of the same kind and it is proposed to enroll a total of 20 patients for preliminary safety

assessment. Assuming that observation shows over 8 (40%) of the 20 patients receiving IBI308 + IBI305 have TRAEs at grade 3 or above as per CTCAE, there is about 90% power in inferring that the true incidence of such AEs exceeds 29%, namely, the incidence of TRAEs at grade 3 or above as per CTCAE arising from IBI308 + IBI305 group may be higher than estimation.

Since the actual number of patients receiving screening was more than the planned enrollment number, a total of 24 patients were eventually enrolled in phase II clinical study.

Phase III, randomized, controlled study:

The sample size of this study will be determined according to the calculation of the number of deaths as required to prove efficacy via OS and the number of patients as required to prove efficacy via PFS.

To detect OS improvement at the two-sided significance level of 0.048 using the log-rank test, about 408 deaths can offer approximately 89% power assuming HR = 0.714 (median OS in sorafenib group is 11.5 months vs. 16.1 months in IBI308 + IBI305 group). In the final OS analysis, the minimum detectable difference (MDD) of the OS-based statistical test is corresponding to an HR of 0.810 (median OS improvement of about 2.7 months). The dropout rate is expected to stand at 0.5%/month in the IBI308 + IBI305 group vs. 0.8%/month in the sorafenib group. Assuming that the enrollment period is 18 months, a total of about 546 patients will be randomized to two groups in a 2:1 ratio, namely the IBI308 + IBI305 group (treatment group) or the sorafenib group (control group). A total of 408 deaths will be expected to be observed at 44 months after the enrollment of the first patient. In actual study, if the actual number of enrolled subjects is different from the number of planned enrollment due to unforeseen reasons, such as the enrollment speed is faster than the requirements of the protocol, the end date of screening should be determined in advance to ensure that the number of actually enrolled subjects does not exceed the number of planned enrollment by about 10%, i.e., a maximum of 600 subjects should be enrolled, including 400 subjects in the treatment group and 200 subjects in the control group.

In this study, an interim analysis of OS endpoint will be carried out when 265 (65%) cases of OS events are observed. The cut-off value of the interim analysis approximates the O'brien-Fleming boundary according to the Lan-Demets method. The interim analysis has a two-sided level of significance $\alpha = 0.0102$. Based on the previously assumed enrollment speed and the monthly dropout rate of each group, the interim analysis of OS is expected to start approximately 25 months after the enrollment of the first patient and the primary analysis of PFS endpoint will be performed concurrently. Assuming that the median PFS is 8 months in IBI308 + IBI305 group and 4 months in sorafenib group and HR is 0.5 for PFS, it is estimated that about 400 cases of PFS events will be observed for interim analysis of OS. Therefore, on the two-sided level of significance $\alpha = 0.002$, there will be a power of > 99% based on 400 PFS events. When results of PFS events are within the minimum detectable difference and significant, the HR is about 0.719.

If the number of OS events observed is less than 220 when 400 PFS events have occurred, the primary analysis of PFS and the first interim analysis of OS will be performed simultaneously, while the second interim analysis is performed when 65% (265 cases) of OS events are observed. In addition, the final analysis of OS will be performed when 408 cases of OS events are observed. The α cut-off value of each analysis of OS is allocated to approximate the O'brien-Fleming boundary according to the Lan-Demets method, and adjusted according to the actual number of OS events that have occurred to ensure that the OS analysis is a two-sided test with an overall significance level of $\alpha = 0.048$.

9.11 Methods for Controlling Bias

9.11.1 Randomization and blinding

The patients will be randomized using central randomization in the study, with the stratification factors including macrovascular invasion and/or extrahepatic metastasis (yes vs. no), baseline AFP (< 400 ng/mL vs. ≥ 400 ng/mL), and the score of ECOG performance status (0 vs. 1). The investigator or qualified designee will log in to the system with their respective passwords, and the system will then assign unique random codes to the patients each. Meanwhile, the names of the study drugs which the patients need to take will be obtained through the central randomization system. The independent statistician responsible for randomization should use the stratified blocked randomization to generate a random assignment table (patient random number), in accordance with the ratio of treatment group : control group = 2:1, and the patient random number will be submitted to the randomization system in the form of electronic documents.

9.11.2 Blinding maintenance evaluation

This study is an open-label study and the study drugs are not blinded. In addition, the subjects, investigators, and sponsor are accessible to the treatment information. To ensure the scientificity of the study, the project statistician of the sponsor remains blinded to the information of randomized treatment before the database is locked. The iDMC is set for interim analysis.

9.11.3 Unblinding and emergency unblinding

Not applicable.

10 QUALITY ASSURANCE AND QUALITY CONTROL

In accordance with the Good Clinical Practice (GCP), the sponsor is responsible for the implementation and maintenance of quality assurance and quality control systems as per appropriate standard operating procedures, to ensure the study is implemented and authentic data is collected, documented, and reported in accordance with the requirements of the protocol, GCP, and applicable regulations.

Each study site will implement internal quality management of the study process, data and biological sample collection, documentation, and completion. Particular quality management plans should be developed for each study site.

10.1 Clinical Monitoring

The sponsor or its authorized contract research organization (CRO) will conduct clinical monitoring of this study. The monitor should perform the monitoring in accordance with the standard operating procedures provided by the sponsor or CRO, and has the same rights and responsibilities as the sponsor's medical monitor. The monitor must maintain regular communication with the investigator, authorized research personnel, and sponsor.

Before the start of the study, the associate monitor will assess the qualifications of each study site, and report issues related to facilities, technical equipment, or medical staff to the sponsor. During the study, the monitor is responsible for the monitoring of whether the written ICF from all patients has been obtained and whether the data records are correct and complete. Also, the associate monitor will compare the data entered into the eCRF with the source data, and inform the investigator of any errors or omissions. Besides, the monitor will also control the compliance of the protocol and study procedures for each study site, arrange for the supply of study drugs, and ensure that the drugs are kept under proper conditions.

The monitoring visits should be conducted in accordance with the requirements of relevant laws and regulations. Each site will receive regular monitoring visits from the time the patients are enrolled. After each visit to the investigator, the monitor should submit a written report to the sponsor.

10.2 Quality Assurance Audits

During the study, the sponsor or designee may conduct quality assurance audits on the study sites, database, and related research documents. At the same time, the corresponding regulatory authorities may also inspect the study sites, database, and related research documents at their own discretion. The investigator should notify the sponsor immediately after being notified of inspection from regulatory authorities.

The sponsor's quality assurance department conducted an audit on the clinical study sites, which includes the supply of drugs, required trial documents, records of the informed consent process, and the consistency of medical report forms with the source documents. The content and scope of the audits can also be increased as the circumstance requires. After reasonable notice, the investigator should allow auditors commissioned by the sponsor to conduct audits related to the trials and inspections conducted by the regulatory authorities. The primary purpose of an audit or an inspection was to verify that the rights or health of the patients have been protected, the signing of the informed consent form and the correct implementation of the trial process, and all

data related to the evaluation of the study drugs have been processed, reported and pre-planned. In addition, the protocol, facility, ethical standard operating procedures, GCP and applicable regulatory requirements were consistent. The investigator should have direct access to all trial files, source records, and source data.

11 DATA MANAGEMENT AND RECORD KEEPING

This study will use an electronic data collection (EDC) system, and the research data will be recorded in the eCRFs by the investigator or its authorized personnel. Before the initiation of the study site or data entry, the investigator and authorized personnel should be properly trained and appropriate security measures should be taken for the computer and other equipment.

Data entry into the eCRFs should be completed as soon as possible during or after visiting. The eCRFs should be updated at any time to ensure that they reflect the latest developments of the patients. To avoid differences in outcome evaluations by different evaluators, it is recommended that baseline and all subsequent efficacy and safety evaluations of a given patient should be performed by the same individual. The investigator are required to review the data to ensure the accuracy and correctness of all the data entered into the eCRFs. If no evaluations are conducted during the study or certain obtained information is not evaluable, not applicable, or unknown, the investigator should record the above information in the eCRFs. The investigator needs to sign all verified data electronically.

The CRA will review the eCRFs, and evaluate the completeness and consistency by comparing with the source document to ensure consistency of key data. Data entry, corrections, and modifications should be performed by the investigator or designee. The data in the eCRFs is submitted to the data server and any modifications in the data should be recorded in the audit trail, including reasons, operator names, time, and dates of modification. The roles and permission levels of the personnel responsible for data entry in the study site will be determined in advance. The CRA or data management personnel may raise a query in the EDC in the event of suspicious data. The study site personnel are responsible for dealing with such query. The EDC system will record the audit trail of the query, including the investigator name, time, and date.

Unless otherwise specified, the eCRF should be considered simply as a form for data collection and not a source document. The source documents are used by the investigator or hospital, including all records related to the patients, which are able to demonstrate the presence, inclusion criteria, and participation of patients (laboratory records, ECGs, pharmaceutical records, patient folders, etc.). Permanent copies of study visit records will be considered as source documents, and will be used to record the data of enrolled patients. Data in the eCRFs should be from the source documents and consistent with source data.

The investigator is required to maintain all source documents and to offer the documents to the CRA for review during each visit. In addition, the investigator must submit a complete eCRF for each enrolled patient, regardless of the duration of participation. The protocol number and patient numbers of all supporting documents (such as laboratory records or hospital records) submitted with the eCRFs should be carefully verified. All the personal privacy information (including the subjects' names) should be deleted or made illegible to protect the privacy of the patients. The investigator should verify that the record has been reviewed and that the data are accurate with an electronic signature. The electronic signature will be completed with the investigators' user ID and password. The system will automatically include the date and time of the signature. The investigator could not share the user ID and password with other personnel. If data in the eCRF need to be modified, the procedures defined by the EDC system should be followed. All modifications and reasons for the changes will be recorded in the audit trail.

AEs and concomitant diseases/history will be coded. The dictionary used for coding will be described in the clinical study report.

Records from the clinical trial (such as protocol and protocol revision, completed eCRFs, and signed ICFs) should be kept and managed in accordance with the GCP. The study site should keep these documents for 5 years after the end of the study.

The study documents should be reasonably kept for future reference or data traceability. Security and environmental risks should be considered when storing documents.

No study documents should be destroyed without the written consent from the sponsor and investigators. The investigator/study site may transfer the study documents to other parties that comply with the record-keeping requirements or to another location that meets the record-keeping requirements only after notifying the sponsor and obtaining the written consent.

12 ETHICS

12.1 Ethics Committee

The sponsor or its designee will prepare the relevant documents including the study protocol, informed consent form (ICF), Investigator's Brochure, patient recruitment materials or advertisement, and other documents required by regulations, which are to be submitted to the corresponding EC of the study site for approval. Before the start of the study, written approval from the in-site EC must be obtained and submitted to the sponsor. The written approval from the EC must specify the name, number, version number of the study protocol, version number of other documents (such as ICF), and date of approval. The investigator is required to notify the sponsor of the EC's written comments regarding delay, interruption, and re-approval of the study.

The study site must follow the requirements of the EC in the study site. Modifications of the protocol, ICF, or patient recruitment materials should be submitted to the EC for approval. Local safety reports should be made and updated regularly in accordance with the regulations of the EC, and the final report should be submitted. All the above documents and EC approvals must be provided to the sponsor or its designee.

12.2 Implementation of Ethics

The study process and acquisition of informed consent should be subject to the *Declaration of Helsinki*, relevant GCP requirements, as well as laws and regulations related to the protection of drug and data in China.

The GCP is an international ethical and scientific quality standard for designing, conducting, recording, and reporting clinical studies that involve the participation of human patients. To ensure the rights, safety, and health of patients, this study will be carried out in accordance with GCP and relevant national regulations, as well as ethical principles outlined in the *Declaration of Helsinki*.

The investigator is required to follow the procedures specified in this study protocol and must not change the procedures without the permission from the sponsor. Any protocol deviation must be reported to the EC, sponsor, or regulatory authorities.

12.3 Patient's Informed Consent

Before the start of any study process, the ICF is used to explain the risks and benefits of this study to potential patients, and the expression used in the ICF should be straightforward. The ICF should state that it is voluntarily signed, emphasize the potential risks and benefits of participating in this study, and make it clear that patients may withdraw from the study at any time. The investigator may only enroll a patient after fully explaining the details of the study, answering questions to the patient's satisfaction, giving the patient sufficient time for consideration, and obtaining written consent from the patient or his/her legal representative. All signed ICFs must be kept in the investigator's files or in the patient's folder.

The investigator is responsible for explaining the contents of the ICF and obtaining the ICF signed and dated by a patient or his/her legal representative before starting the study. The investigator should provide the patient with a copy of the signed ICF. The investigator is required to document the process of informed consent in the source study document.

12.4 Protection of Patient Data

Information about data protection and privacy protection will be included in the ICF (or in some cases, in a separate document).

Precautions should be taken to ensure the confidentiality of the documents and prevent the disclosure of information that can determine the identity of the patient. However, under special circumstances, some personnel may be permitted to see the genetic data and personal identification code of a patient. For example, in the event of a medical emergency, the sponsor, its designated physician, or an investigator will have access to the patient identification code and the patient's genetic data. In addition, relevant regulatory authorities may require access to relevant documents.

12.5 Protocol Deviation

Protocol deviation refers to any non-compliance with the study protocol, the "International Conference on Harmonization Good Clinical Practice (ICH GCP) Guidelines", or operating manual. Non-compliance may come from the patients, investigators, or study site personnel. Deviations should be corrected promptly.

13 PUBLISHING OF STUDY DATA

All the data generated in this study are the confidential information owned by the sponsor. The sponsor has the right to publish study results. Information on the publishing policies of the sponsor and investigators will be described in the clinical study agreement.

All the information on this study (not limited to the protocol and Investigator's Brochure) must be kept strictly confidential. The investigator must recognize that the scientific or medical information derived from this study may be of commercial value to the sponsor. The investigator should keep the information and data related to this study confidential. The sponsor must be consulted in advance and written consent must be obtained before publishing any study-related information or conclusions. To protect the rights and interests, the sponsor may request the investigator not to publish information on this study before the investigational product is approved for marketing.

The sponsor has the right to announce or publish information or data related to the study or to report it to the drug administration. The sponsor must obtain the consent of the investigator if the name of the investigator is included in the content of the announcement, publication, or advertising.

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15 PROTOCOL REVISION HISTORY

Version	Date	Description of change	Brief basis
V2.0	May 20, 2019	Revision of version No.	Version revision.
		Update of IBI308 information.	IBI308 has been authorized for marketing; the trade name and package insert have been added, and relevant clinical data have been updated.
		Update of IBI305 information.	IBI305 marketing application has been submitted. IBI305 clinical data have been updated.
		Revision of some of the inclusion/exclusion criteria, and prohibited/permitted combined treatment.	To ensure the safety of subjects and make the description more accurate.
		Revision of primary endpoints to PFS and OS, and of relevant statistical descriptions.	According to the reply from the Center for Drug Evaluation (CDE), which agreed to revise the primary endpoints of ORR/OS to PFS/OS.
		Biomarker revision.	The content of biomarker detection and sample requirements have been revised by referring to the related research and clinical practice.
		Update of the research data of the phase Ib trial (NCT02715531) involving atezolizumab in combination with bevacizumab as the first-line treatment for advanced liver cancer.	The clinical data of related clinical studies have been updated based on the 2018 ESMO.
V3.0	Apr. 20, 2020	Revision of version No.	Version revision.
		Update of the description and the preparation and infusion process of IBI305.	According to the latest Investigator's Brochure V7.0 and the "Instructions on Temperature in the IBI305 Compatibility Process".
		Revision of the statistical analysis.	The description of statistical analysis has been revised after the communication with CDE.
		Revision of biomarker analysis.	Not collected and thus deleted.
		Addition of the description of continued treatment for patients reaching PD or iCPD.	The description of continued treatment after progression has been added.

APPENDIX 1. SCORING CRITERIA FOR PHYSICAL STATUS (ECOG PS)

Activity score	Description
0	Asymptomatic, fully active, and able to carry on all activities without restriction.
1	Symptomatic but completely ambulatory (restricted in physically strenuous activity, and able to carry out work of a light or sedentary nature, for example, light housework, office work).
2	Symptomatic, ambulatory, and capable of all self-care but unable to carry out any work activities; up and about more than 50% of waking hours (< 50% in bed during the day).
3	Symptomatic, capable of only limited self-care, confined to bed or chair; 50% or more of waking hours (> 50% in bed), but not bedbound.
4	Bedbound (completely disabled, cannot carry on any self-care, totally confined to bed or chair).
5	Death.

APPENDIX 2. CREATININE CLEARANCE RATE

Calculation of creatinine clearance rate using the Cockcroft-Gault formula

Calculation formula for serum creatinine with the concentration unit of mg/dL:

$$\text{Creatinine clearance rate in males (mL/min)} = \frac{(140 - \text{age}) \times \text{body weight (kg)}^{a}}{72 \times \text{serum creatinine (mg/dL)}}$$

$$\text{Creatinine clearance rate in females (mL/min)} = \frac{0.85 \times (140 - \text{age}) \times \text{body weight (kg)}^{a}}{72 \times \text{serum creatinine (mg/dL)}}$$

Calculation formula for serum creatinine with the concentration unit of μmol/L:

$$\text{Creatinine clearance rate in males (mL/min)} = \frac{(140 - \text{age}) \times \text{body weight (kg)}^{a}}{0.818 \times \text{serum creatinine (μmol/L)}}$$

$$\text{Creatinine clearance rate in females (mL/min)} = \frac{0.85 \times (140 - \text{age}) \times \text{body weight (kg)}^{a}}{0.818 \times \text{serum creatinine (μmol/L)}}$$

[a]: The units of age and body weight are year and kg, respectively.

APPENDIX 3. RESPONSE EVALUATION CRITERIA IN SOLID TUMORS VERSION 1.1 (RECIST V1.1)

The content as follows is an excerpt from RECIST V1.1.

1. MEASURABILITY OF TUMORS AT BASELINE

1.1 Definitions

At the baseline, tumor lesions/lymph nodes will be categorized as measurable and non-measurable according to the following definitions:

1.1.1 Measurable

Tumor lesions: Must be accurately measured in at least one dimension (longest diameter in the plane of measurement is to be recorded) with a minimum size of:

- 10 mm by CT scan (CT scan slice thickness no greater than 5 mm)
- 10 mm caliper measurement by clinical exam (lesions which cannot be accurately measured with calipers should be recorded as non-measurable)
- 20 mm by chest X-ray
- Malignant lymph nodes: To be considered pathologically enlarged and measurable, a lymph node must be ≥ 15 mm in short axis when assessed by CT scan (CT scan slice thickness recommended to be no greater than 5 mm). At baseline and in follow-up, only the short axis will be measured and followed.

1.1.2 Non-measurable

All other lesions, including small lesions (longest diameter < 10 mm or pathological lymph nodes with ≥ 10 to < 15 mm short axis) as well as truly non-measurable lesions. Lesions considered truly non-measurable include: leptomeningeal disease, ascites, pleural or pericardial effusion, inflammatory breast disease, lymphangitic involvement of skin or lung, abdominal masses/abdominal organomegaly identified by physical exam that is not measurable by reproducible imaging techniques.

1.1.3 Special considerations regarding lesion measurability

Bone lesions, cystic lesions, and lesions previously treated with local therapy require particular comment:

Bone lesions:

- Bone scan, PET scan or plain films are not considered adequate imaging techniques to measure bone lesions. However, these techniques can be used to confirm the presence or disappearance of bone lesions;
- Lytic bone lesions or mixed lytic-blastic lesions, with identifiable soft tissue components, that can be assessed by cross sectional imaging techniques such as CT or MRI can be considered as measurable lesions if the soft tissue component meets the definition of measurability described above;
- Blastic bone lesions are non-measurable.

Cystic lesions:

- Lesions that meet the criteria for radiographically defined simple cysts should not be considered as malignant lesions (neither measurable nor non-measurable) since they are, by definition, simple cysts;
- "Cystic lesions" thought to represent cystic metastases can be considered as measurable lesions, if they meet the definition of measurability described above. However, if non-cystic lesions are present in the same patient, these are preferred for selection as target lesions.

Lesions with prior local treatment:

- Tumor lesions situated in a previously irradiated area, or in an area subjected to other loco-regional therapy, are usually not considered measurable unless there has been demonstrated progression in the lesion. Study protocols should detail the conditions under which such lesions would be considered measurable.

1.2 Specifications by Methods of Measurements

1.2.1 Measurement of lesions

All measurements should be recorded in metric notation, using calipers if clinically assessed. All baseline evaluations should be performed as close as possible to the treatment start and never more than 28 days (4 weeks) before the beginning of the treatment.

1.2.2 Method of assessment

The same method of assessment and the same technique should be used to characterize each identified and reported lesion at baseline and during follow-up. Imaging based evaluation should always be done rather than clinical examination unless the lesion(s) being followed cannot be imaged but are assessable by clinical exam.

Clinical lesions: Clinical lesions will only be considered measurable when they are superficial and ≥ 10 mm in diameter as assessed using calipers (e.g. skin nodules). For the case of skin lesions, documentation by color photography including a ruler to estimate the size of the lesion is suggested. When lesions can be assessed by both clinical examination and imaging, imaging evaluation should be undertaken since it is more objective and may also be reviewed at the end of the study.

Chest X-ray: Chest CT is preferred over chest X-ray, particularly when progression is an important endpoint, since CT is more sensitive than X-ray, particularly in identifying new lesions. However, lesions on chest X-ray may be considered measurable if they are clearly defined and surrounded by aerated lung.

CT, MRI: CT is the best currently available and reproducible method to measure lesions selected for response assessment. This guideline has defined measurability of lesions on CT scan based on the assumption that CT slice thickness is 5 mm or less. When CT scans have slice thickness greater than 5 mm, the minimum size for a measurable lesion should be twice the slice thickness. MRI is also acceptable in certain situations (e.g. for body scans).

Ultrasound: Ultrasound is not useful in assessment of lesion size and should not be used as a method of measurement. Ultrasound examinations cannot be reproduced in their entirety for independent review at a later date and, because they are operator dependent, it cannot be guaranteed that the same technique and measurements will be taken from one assessment to the next. If new lesions are identified by ultrasound in the course of the study, confirmation by CT or MRI is advised. If there is concern about radiation exposure at CT, MRI may be used instead of CT in selected instances.

Endoscopy, laparoscopy: The utilization of these techniques for objective tumor evaluation is not advised. However, they can be useful to confirm complete pathological response when biopsies are obtained or to determine relapse in trials where recurrence following complete response or surgical resection is an endpoint.

Tumor markers: Tumor markers alone cannot be used to assess objective tumor response. If markers are initially above the upper normal limit, however, they must normalize for a patient to be considered in complete response. Because tumor markers are disease specific, instructions for their measurement should be incorporated into protocols on a disease specific basis. Specific

guidelines for both CA-125 response (in recurrent ovarian cancer) and PSA response (in recurrent prostate cancer), have been published. In addition, the Gynecologic Cancer Intergroup has developed CA-125 progression criteria which are to be integrated with objective tumor assessment for use in first-line trials in ovarian cancer.

Cytology, histology: These techniques can be used to differentiate between PR and CR in rare cases if required by protocol (for example, residual lesions in tumor types such as germ cell tumors, where known residual benign tumors can remain). When effusions are known to be a potential adverse effect of treatment (e.g. with certain taxane compounds or angiogenesis inhibitors), the cytological confirmation of the neoplastic origin of any effusion that appears or worsens during treatment can be considered if the measurable tumor has met criteria for response or stable disease in order to differentiate between response (or stable disease) and progressive disease.

2. TUMOR RESPONSOE EVALUATION

2.1 Assessment of Overall Tumor Burden and Measurable Disease

To assess objective response or future progression, it is necessary to estimate the overall tumor burden at baseline and use this as a comparator for subsequent measurements. Only patients with measurable disease at baseline should be included in protocols where objective tumor response is the primary endpoint. Measurable disease is defined by the presence of at least one measurable lesion. In studies where the primary endpoint is tumor progression (either time to progression or proportion with progression at a fixed date), the protocol must specify if entry is restricted to those with measurable disease or whether patients having non-measurable disease only are also eligible.

2.2 Baseline Documentation of "Target" and "Non-Target" Lesions

When more than one measurable lesion is present at baseline, all lesions up to a maximum of five lesions total (and a maximum of two lesions per organ) representative of all involved organs should be identified as target lesions and will be recorded and measured at baseline (this means in instances where patients have only one or two organ sites involved, a maximum of two and four lesions respectively will be recorded).

Target lesions should be selected on the basis of their size (lesions with the longest diameter), be representative of all involved organs, but in addition should be those that lend themselves to reproducible repeated measurements. It may be the case that, on occasion, the largest lesion does not lend itself to reproducible measurement in which circumstance the next largest lesion which can be measured reproducibly should be selected.

Lymph nodes merit special mention since they are normal anatomical structures which may be visible by imaging even if not involved by tumor. Pathological nodes which are defined as measurable and may be identified as target lesions must meet the criterion of a short axis of ≥ 15 mm by CT scan. Only the short axis of these nodes will contribute to the baseline sum. The short axis of the node is the diameter normally used by radiologists to judge if a node is involved by solid tumor. Nodal size is normally reported as two dimensions in the plane in which the image is obtained (for CT scan this is almost always the axial plane; for MRI the plane of acquisition may be axial, saggital or coronal). The smaller of these measures is the short axis. For example, an abdominal node which is reported as being 20 mm × 30 mm has a short axis of 20 mm and qualifies as a malignant, measurable node. In this example, 20 mm should be recorded as the node measurement. All other pathological nodes (those with short axis ≥ 10 mm but < 15 mm) should be considered non-target lesions. Nodes that have a short axis <10 mm are considered non-pathological and should not be recorded or followed.

A sum of the diameters (longest for non-nodal lesions, short axis for nodal lesions) for all target lesions will be calculated and reported as the baseline sum diameters. If lymph nodes are to be included in the sum, then as noted above, only the short axis is added into the sum. The baseline sum diameters will be used as reference to further characterize any objective tumor regression in the measurable dimension of the disease.

All other lesions (or sites of disease) including pathological lymph nodes should be identified as non-target lesions and should also be recorded at baseline. Measurements are not required and these lesions should be followed as "present", "absent", or in rare cases "unequivocal progression". In addition, it is possible to record multiple non-target lesions involving the same organ as a single item on the case record form (e.g. "multiple enlarged pelvic lymph nodes" or "multiple liver metastases").

2.3 Response Criteria

2.3.1 Evaluation of target lesions

Complete Response (CR): Disappearance of all target lesions. Any pathological lymph nodes (whether target or non-target) must have reduction in short axis to < 10 mm.

Partial Response (PR): At least a 30% decrease in the sum of diameters of target lesions, taking as reference the baseline sum diameters.

Progressive Disease (PD): At least a 20% increase in the sum of diameters of target lesions, taking as reference the smallest sum on study (this includes the baseline sum if that is the smallest on study). In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm. (Note: the appearance of one or more new lesions is also considered progression).

Stable Disease (SD): Neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD, taking as reference the smallest sum diameters while on study.

2.3.2 Special notes on the assessment of target lesions

Lymph nodes: Lymph nodes identified as target lesions should always have the actual short axis measurement recorded (measured in the same anatomical plane as the baseline examination), even if the nodes regress to below 10 mm on study. This means that when lymph nodes are included as target lesions, the "sum" of lesions may not be zero even if complete response criteria are met, since a normal lymph node is defined as having a short axis of < 10 mm. Case report forms or other data collection methods may therefore be designed to have target nodal lesions recorded in a separate section where, in order to qualify for CR, each node must achieve a short axis < 10 mm. For PR, SD, and PD, the actual short axis measurement of the nodes is to be included in the sum of target lesions.

Target lesions that become "too small to measure". While on study, all lesions (nodal and non-nodal) recorded at baseline should have their actual measurements recorded at each subsequent evaluation, even when very small (e.g. 2 mm). However, sometimes lesions or lymph nodes which are recorded as target lesions at baseline become so faint on CT scan that the radiologist may not feel comfortable assigning an exact measure and may report them as being "too small to measure". When this occurs it is important that a value be recorded on the case report form. If it is the opinion of the radiologist that the lesion has likely disappeared, the measurement should be recorded as 0 mm. If the lesion is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned. (Note: It is less likely that this rule will be used for lymph nodes since they usually have a definable size when normal and are frequently surrounded by fat such as in the retroperitoneum; however, if a lymph node is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned in this circumstance as well). This default value is derived from the 5 mm CT slice thickness (but should not be changed with varying CT slice thickness). The measurement of these lesions is potentially non-reproducible, therefore providing this default value will prevent false responses or progressions based upon measurement error. To reiterate, however, if the radiologist is able to provide an actual measure, that should be recorded, even if it is below 5 mm.

Lesions that split or coalesce on treatment: When non-nodal lesions "fragment", the longest diameters of the fragmented portions should be added together to calculate the target lesion sum. Similarly, as lesions coalesce, a plane between them may be maintained that would aid in obtaining maximal diameter measurements of each individual lesion. If the lesions have truly coalesced such that they are no longer separable, the vector of the longest diameter in this instance should be the maximal longest diameter for the "coalesced lesion".

2.3.3 Evaluation of non-target lesions

This section provides the definitions of the criteria used to determine the tumor response for the group of non-target lesions. While some non-target lesions may actually be measurable, they need not be measured and instead should be assessed only qualitatively at the time points specified in the protocol.

Complete Response (CR): Disappearance of all non-target lesions and normalization of tumor marker level. All lymph nodes must be non-pathological in size (< 10 mm short axis).

Non-CR/Non-PD: Persistence of one or more non-target lesion(s) and/or maintenance of tumor marker level above the normal limits.

Progressive Disease (PD): Unequivocal progression of existing non-target lesions. (Note: the appearance of one or more new lesions is also considered progression).

2.3.4 Special notes on assessment of progression of non-target lesions

The concept of progression of non-target disease requires additional explanation as follows: When the patient also has measurable disease: In this setting, to achieve "unequivocal progression" on the basis of the non-target disease, there must be an overall level of substantial worsening in non-target disease such that, even in presence of SD or PR in target disease, the overall tumor burden has increased sufficiently to merit termination of therapy. A modest "increase" in the size of one or more non-target lesions is usually not sufficient to quality for unequivocal progression status. The designation of overall progression solely on the basis of change in non-target disease in the face of SD or PR of target disease will therefore be extremely rare.

When the patient has only non-measurable disease: This circumstance arises in some phase III trials when it is not a criterion of study entry to have measurable disease. The same general concepts apply here as noted above, however, in this instance there is no measurable disease assessment to factor into the interpretation of an increase in non-measurable disease burden. Because worsening in non-target disease cannot be easily quantified (by definition: if all lesions are truly non-measurable), a useful test that can be applied when assessing patients for unequivocal progression is to consider if the increase in overall disease burden based on the change in non-measurable disease is comparable in magnitude to the increase that would be required to declare PD for measurable disease: i.e. an increase in tumor burden representing an additional 73% increase in "volume" (which is equivalent to a 20% increase diameter in a measurable lesion). i.e. an increase in peritoneal effusion from "minor" to "large", an increase in lymphatic lesions from "local" to "widespread", or may be described in protocols as "sufficient to a change in therapy". Examples include an increase in a pleural effusion from trace to large, an increase in lymphangitic disease from localized to a distal region, or may be described in

protocols as "require a change in therapy". If "unequivocal progression" is seen, the patient should be considered to have had overall PD at that point. While it would be ideal to have objective criteria to apply to non-measurable disease, the very nature of that disease makes it impossible to do so, therefore the increase must be substantial.

2.3.5 New lesions

The appearance of new malignant lesions denotes disease progression; therefore, some comments on detection of new lesions are important. There are no specific criteria for the identification of new radiographic lesions; however, the finding of a new lesion should be unequivocal: i.e. not attributable to differences in scanning technique, change in imaging modality or findings thought to represent something other than tumor (for example, some "new" bone lesions may be simply healing or flare of pre-existing lesions). This is particularly important when the patient's baseline lesions show partial or complete response. For example, necrosis of a liver lesion may be reported on a CT scan report as a "new" cystic lesion, which it is not.

A lesion identified on a follow-up study in an anatomical location that was not scanned at baseline is considered a new lesion and will indicate disease progression. An example of this is the patient who has visceral disease at baseline and while on study has a CT or MRI brain ordered which reveals metastases. The patient's brain metastases are considered to be evidence of PD even if he/she did not have brain imaging at baseline.

If a new lesion is equivocal, for example because of its small size, continued therapy and follow-up evaluation will clarify if it represents truly new disease. If repeat scans confirm there is definitely a new lesion, then progression should be declared using the date of the initial scan.

While FDG-PET response assessments need additional study, it is sometimes reasonable to incorporate the use of FDG-PET scanning to complement CT scanning in assessment of progression (particularly possible "new" disease). New lesions on the basis of FDG-PET imaging can be identified according to the following algorithm:

Negative FDG-PET at baseline, with a positive FDG-PET at follow-up is a sign of PD based on a new lesion.

No FDG-PET at baseline and a positive FDG-PET at follow-up:

If the positive FDG-PET at follow-up corresponds to a new site of disease confirmed by CT, this is PD.

If the positive FDG-PET at follow-up is not confirmed as a new site of disease on CT, additional follow-up CT scans are needed to determine if there is truly progression occurring at that site (if so, the date of PD will be the date of the initial abnormal FDG-PET scan).

If the positive FDG-PET at follow-up corresponds to a pre-existing site of disease on CT that is not progressing on the basis of the anatomic images, this is not PD.

2.4 Evaluation of Best Overall Response

The best overall response is the best response recorded from the start of the study treatment until the end of treatment taking into account any requirement for confirmation. On occasion a response may not be documented until after the end of therapy so protocols should be clear if post-treatment assessments are to be considered in determination of best overall response. Protocols must specify how any new therapy introduced before progression will affect best response designation. The patient's best response assignment will depend on the findings of both target and non-target disease and will also take into consideration the appearance of new lesions. Furthermore, depending on the nature of the study and the protocol requirements, it may also require confirmatory measurement. Specifically, in non-randomized trials where response is the primary endpoint, confirmation of PR or CR is needed to deem either one the "best overall response".

2.4.1 Time point response

It is assumed that at each protocol specified time point, a response assessment occurs. Table 1 on the next page provides a summary of the overall response status calculation at each time point for patients who have measurable disease at baseline.

When patients have non-measurable (therefore non-target) disease only, Table 2 is to be used.

2.4.2 Missing assessments and inevaluable designation

When no imaging/measurement is done at all at a particular time point, the patient is not evaluable (NE) at that time point. If only a subset of lesion measurements are made at an assessment, usually the case is also considered NE at that time point, unless a convincing argument can be made that the contribution of the individual missing lesion(s) would not change the assigned time point response. This would be most likely to happen in the case of PD. For example, if a patient had a baseline sum of 50 mm with three measured lesions and at follow-up only two lesions were assessed, but those gave a sum of 80 mm, the patient will have achieved PD status, regardless of the contribution of the missing lesion.

2.4.3 Best overall response: all time points

The best overall response is determined once all the data for the patient is known.

Best response determination in trials where confirmation of complete or partial response IS NOT required: Best response in these trials is defined as the best response across all time points (for example, a patient who has SD at first assessment, PR at second assessment, and PD on last assessment has a best overall response of PR). When SD is believed to be best response, it must

also meet the protocol specified minimum time from baseline. If the minimum time is not met when SD is otherwise the best time point response, the patient's best response depends on the subsequent assessments. For example, a patient who has SD at first assessment, PD at second and does not meet minimum duration for SD, will have a best response of PD. The same patient lost to follow-up after the first SD assessment would be considered inevaluable.

Best response determination in trials where confirmation of complete or partial response IS required: Complete or partial responses may be claimed only if the criteria for each are met at a subsequent time point as specified in the protocol (generally 4 weeks later). In this circumstance, the best overall response can be interpreted as in Table 3.

2.4.4 Special notes on response assessment

When nodal disease is included in the sum of target lesions and the nodes decrease to "normal" size (< 10 mm), they may still have a measurement reported on scans. This measurement should be recorded even though the nodes are normal in order not to overstate progression should it be based on increase in size of the nodes. As noted earlier, this means that patients with CR may not have a total sum of "zero" on the case report form (CRF).

In trials where confirmation of response is required, repeated "NE" time point assessments may complicate best response determination. The analysis plan for the trial must address how missing data/assessments will be addressed in determination of response and progression. For example, in most trials it is reasonable to consider a patient with time point responses of PR-NE-PR as a confirmed response.

Patients with a global deterioration of health status requiring discontinuation of treatment without objective evidence of disease progression at that time should be reported as "symptomatic deterioration". Every effort should be made to document objective progression even after termination of treatment. Symptomatic deterioration is not a descriptor of an objective response: it is a reason for stopping study therapy. The objective response status of such patients is to be determined by evaluation of target and non-target disease as shown in Tables 1–3.

Conditions that define "early progression, early death and inevaluability" are study specific and should be clearly described in each protocol (depending on treatment duration, treatment periodicity).

In some circumstances it may be difficult to distinguish residual disease from normal tissue. When the evaluation of complete response depends upon this determination, it is recommended that the residual lesion be investigated (fine needle aspirate/biopsy) before assigning a status of complete response. FDG-PET may be used to upgrade a response to a CR in a manner similar to a biopsy in cases where a residual radiographic abnormality is thought to represent fibrosis or scarring. The use of FDG-PET in this circumstance should be prospectively described in the

protocol and supported by disease specific medical literature for the indication. However, it must be acknowledged that both approaches may lead to false positive CR due to limitations of FDG-PET and biopsy resolution/sensitivity.

Attached Table 1. Time point response: patients with target (+/– non-target) disease

Target lesions	Non-target lesions	New lesions	Overall response
CR	CR	No	CR
CR	Non-CR/Non-PD	No	PR
CR	NE	No	PR
PR	Non-PD or not all evaluated	No	PR
SD	Non-PD or not all evaluated	No	SD
Not all evaluated	Non-PD	No	NE
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD
CR = Complete response	PR = Partial response	SD = Stable disease	PD = Progressive disease and NE = Inevaluable

Attached Table 2. Time point response: patients with non-target disease only

Non-target lesions	New lesions	Overall response
CR	No	CR
Non-CR/Non-PD	No	Non-CR/Non-PD
Not all evaluated	No	NE
Unequivocal PD	Yes or No	PD
Any	Yes	PD

Note: "Non-CR/non-PD" is preferred over "stable disease" for non-target disease. Since SD is increasingly used as endpoint for assessment of efficacy in some trials so to assign this category when no lesions can be measured is not advised.

For equivocal findings of progression (e.g. very small and uncertain new lesions; cystic changes or necrosis in existing lesions), treatment may continue until the next scheduled assessment. If at the next scheduled assessment, progression is confirmed, the date of progression should be the earlier date when progression was suspected.

Attached Table 3. Best overall response when confirmation of CR and PR required

Overall response First time point	Overall response Subsequent time point	Best overall response
CR	CR	CR
CR	PR	SD, PD or PR[a]
CR	SD	SD provided minimum criteria for SD duration met, otherwise, PD
CR	PD	SD provided minimum criteria for SD duration met, otherwise, PD
CR	NE	SD provided minimum criteria for SD duration met, otherwise NE
PR	CR	PR
PR	PR	PR
PR	SD	SD
PR	PD	SD provided minimum criteria for SD duration met, otherwise, PD
PR	NE	SD provided minimum criteria for SD duration met, otherwise NE
NE	NE	NE

Notes: CR = complete response, PR = partial response, SD = stable disease, PD = progressive disease, and NE = inevaluable. Superscript "a": If a CR is truly met at first time point, then any disease seen at a subsequent time point, even disease meeting PR criteria relative to baseline, makes the disease PD at that point (since disease must have reappeared after CR). Best response would depend on whether minimum duration for SD was met. However, sometimes "CR" may be claimed when subsequent scans suggest small lesions were likely still present and in fact the patient had PR, not CR at the first time point. Under these circumstances, the original CR should be changed to PR and the best response is PR.

2.5 Frequency of Tumor Re-Evaluation

Frequency of tumor re-evaluation while on treatment should be protocol specific and adapted to the type and schedule of treatment. However, in the context of phase II studies where the beneficial effect of therapy is not known, follow-up every 6–9 weeks (timed to coincide with the end of a cycle) is reasonable. Smaller or greater time intervals than these could be justified in specific regimens or circumstances. The protocol should specify which organ sites are to be evaluated at baseline (usually those most likely to be involved with metastatic disease for the tumor type under study) and how often evaluations are repeated. Normally, all target and non-target sites are evaluated at each assessment. In selected circumstances certain non-target organs may be evaluated less frequently. For example, bone scans may need to be repeated only when complete response is identified in target disease or when progression in bone is suspected.

After the end of the treatment, the need for repetitive tumor evaluations depends on whether the trial has as a goal the response rate or the time to an event (progression/death). If "time to an event"

(e.g. time to progression, disease-free survival[1], progression-free survival) is the main endpoint of the study, then routine scheduled re-evaluation of protocol specified sites of disease is warranted. In randomized comparative trials in particular, the scheduled assessments should be performed as identified on a calendar schedule (for example: every 6–8 weeks on treatment or every 3–4 months after treatment) and should not be affected by delays in therapy, drug holidays or any other events that might lead to imbalance in a treatment arm in the timing of disease assessment.

2.6 Confirmatory Measurement/Duration of Response

2.6.1 Confirmation

In non-randomized trials where response is the primary endpoint, confirmation of PR and CR is required to ensure responses identified are not the result of measurement error. This will also permit appropriate interpretation of results in the context of historical data where response has traditionally required confirmation in such trials. However, in all other circumstances, i.e. in randomized trials (phase II or III) or studies where stable disease or progression are the primary endpoints, confirmation of response is not required since it will not add value to the interpretation of trial results. However, elimination of the requirement for response confirmation may increase the importance of central review to protect against bias, in particular in studies which are not blinded.

In the case of SD, measurements must have met the SD criteria at least once after study entry at a minimum interval (in general not less than 6–8 weeks) that is defined in the study protocol.

2.6.2 Duration of overall response

The duration of overall response is measured from the time measurement criteria are first met for CR/PR (whichever is first recorded) until the first date that recurrent or progressive disease is objectively documented (taking as reference for progressive disease the smallest measurements recorded on study). The duration of overall complete response is measured from the time measurement criteria are first met for CR until the first date that recurrent disease is objectively documented.

2.6.3 Duration of stable disease

Stable disease is measured from the start of the treatment (in randomized trials, from date of randomization) until the criteria for progression are met, taking as reference the smallest sum on study (if the baseline sum is the smallest, this is the reference for calculation of PD). The clinical relevance of the duration of stable disease varies in different studies and diseases. If the proportion of patients achieving stable disease for a minimum period of time is an endpoint of importance in a particular trial, the protocol should specify the minimal time interval required between two measurements for determination of stable disease.

Note: The duration of response and stable disease as well as the progression-free survival are influenced by the frequency of follow-up after baseline evaluation. It is not in the scope of this guideline to define a standard follow-up frequency. The frequency should take into account many parameters including disease types and stages, treatment periodicity, and standard practice. However, these limitations of the precision of the measured endpoint should be taken into account if comparisons between trials are to be made.

2.7 Progression-Free Survival/Proportion Progression-Free

2.7.1 Phase II trials

This guideline is focused primarily on the use of objective response endpoints for phase II trials. In some circumstances, "response rate" may not be the optimal method to assess the potential anticancer activity of new agents/regimens. In such cases "progression-free survival" (PFS) or the "proportion progression-free" at landmark time points, might be considered appropriate alternatives to provide an initial signal of biologic effect of new agents. It is clear, however, that in an uncontrolled trial, these measures are subject to criticism since an apparently promising observation may be related to biological factors such as patient selection and not the impact of the intervention. Thus, phase II screening trials utilizing these endpoints are best designed with a randomized control. Exceptions may exist where the behavior patterns of certain cancers are so consistent (and usually consistently poor), that a non-randomized trial is justifiable. However, in these cases it will be essential to document with care the basis for estimating the expected PFS or proportion progression-free in the absence of a treatment effect.

APPENDIX 4. RESPONSE EVALUATION AS PER MRECIST (EXCERPTED)

Attached Table 4. Differences between response evaluation as per RECIST V1.1 and response evaluation as per mRECIST:

	RECIST V1.1	**mRECIST**
CR	Disappearance of all target lesions	Arterial phase enhancements of all target lesions disappear
PR	At least a 30% decrease in the sum of diameters of target lesions, taking as reference the baseline sum of the diameters of target lesions	At least a 30% decrease in the sum of diameters of viable (enhancement in the arterial phase) target lesions, taking as reference the baseline sum of the diameters of target lesions
SD	Increase or decrease of the sum of diameters of target lesions does not qualify for partial response or progressive disease	Increase or decrease of the sum of diameters of target lesions (arterial-phase enhancement) does not qualify for partial response or progressive disease
PD	An increase of at least 20% in the sum of the diameters of target lesions, taking as reference the smallest sum of the diameters of target lesions	An increase of at least 20% in the sum of the diameters of viable (enhancing) target lesions, taking as reference the smallest sum of the diameters of viable (enhancing) target lesions

Attached Table 5. Evaluation of overall response as per mRECIST

Target lesion	**Non-target lesion**	**New lesions**	**Overall response**
CR	CR	N/A	CR
CR	IR/SD	N/A	PR
PR	Non-PD	N/A	PPR
SD	Non-PD	N/A	SD
PD	Any	Present or absent	PD
Any	PD	Present or absent	PD
Any	Any	Present	PD

Note: IR means incomplete response

APPENDIX 5. IMMUNE RESPONSE EVALUATION CRITERIA IN SOLID TUMORS (IRECIST)

iRECIST: Guideline on response evaluation in clinical studies for immunotherapy assessment (Appendix) (excerpted)

1. RESPONSE EVALUATION AS PER iRECTIST

Immunotherapy may promote infiltration of immune cells, resulting in transient tumor volume enlargement, or allow undetectable lesions to be detected. This guideline is identical to RECIST V1.1 in many aspects, but evaluation of cases that may not reflect real tumor progression, such as increased tumor burden or emergence of new lesions, is modified.

Here below, critical differences will be described. Each endpoint evaluated as per iRECIST will have a prefix "i". Responses at various time points and BOR as per iRECIST will be documented separately.

1.1 Progression Confirmation

Unlike RECIST V1.1, iRECIST calls for confirmation of PD, while criteria for unconfirmed PD (iUPD) and confirmed PD (iCPD) are introduced. Scan for progression confirmation can be conducted as early as Week 4 after iUPD and no later than Week 8 after iUPD.

PD could be determined as iCPD, if there is still an increase in tumor burden compared with the last measurement, which is evidenced by one or more of the following:

- Tumor burden continues to increase on the basis of PD (compared with the lowest) of target lesions, non-target lesions or new lesions as defined in RECIST V1.1
    a) Target lesion worsens, manifested as an increase in the absolute sum of measurements (SOM) by at least 5 mm
    b) Non-target lesion explicitly progresses, manifested as tumor burden increase
    c) Any previously found new lesion is enlarged (the absolute SOM of the new target lesion increases by at least 5 mm) or any other new lesion is found
- Other lesions (target lesions, non-target lesions or new lesions) without previously found tumor progression are in conformity with the progression criterion in RECIST V1.1, including finding of any other new lesion

If an iUPD is still not confirmed in the next evaluation, then corresponding response evaluation results will be documented (the tumor will be documented as iUPD if it still meets the criteria for iUPD and does not worsen; if, compared with baseline, the tumor meets the criteria for iSD, iPR or iCR, then corresponding results will be documented). As shown in Attached Table 4-2, in the

case that an iUPD fails to be evaluated as iCPD in the next response evaluation, the previous iUPD does not affect documentation of iCR, iPR, or iSD at subsequent time points of evaluation or at the time of BOR evaluation.

1.2 New lesions

New lesions should be measured and evaluated as per RECIST V1.1 (up to 5 lesions in total, no more than 2 lesions per organ, with a longest diameter of at least 10 mm (at least 15 mm for the short axis of a lymph node lesion)), and explicitly recorded as new target lesions or new non-target lesions to be distinguished from the target lesions and non-target lesions at baseline.

New lesions should meet the definitions of new target lesions and new non-target lesions to be recorded as iUPD (or iCPD). Measurements of such target lesions should not be included in the SOM of target lesions defined at baseline. Measurements of these new target lesions should be documented in separate CRF.

PD of a new lesion can be confirmed in the following cases: In imaging evaluation at least 4 weeks (no longer than 8 weeks) after iUPD, it is found that the absolute SOM of the new target lesion increases by at least 5 mm or volume of a new non-target lesion is enlarged (not necessarily unequivocal enlargement) or any other new lesion occurs.

For all the patients, immunity best overall response (iBOR) from the beginning to the end of treatment will be defined according to Attached Table 4-3.

2. DURATION OF RESPONSE AND DURATION OF STABLE DISEASE (RECIST V1.1 VERSUS iRECIST)

DOR starts from the time when the criteria for CR/PR or iCR/iPR are met (whichever is earlier) and ends at relapse or PD, taking as reference the minimal measured value (including baseline) during the study.

Duration of SD starts from the start of treatment and ends at PD, taking as reference the minimal measured value (including baseline) during the study.

Attached Table 6. RECIST V1.1 versus iRECIST

	RECIST V1.1	iRECIST
Definitions of measurable and unmeasurable lesions; quantity and position of target lesion	Lesions having a long axis greater than or equal to 10 mm (15 mm for lymph node lesions) are measurable lesions; up to 5 lesions in total (2 lesions per organ); other lesions are non-target lesions (the short axis of a lymph node lesion should be ≥10 mm)	No change; but new lesions will be separately documented in CRF as per RECIST V1.1 (and will not be included in the SOM of target lesions at baseline)
CR, PR or SD	Cannot have PD before CR, PR or SD	Can have iUPD (one or more instances) but not iCPD before iCR, iPR or iSD

	RECIST V1.1	iRECIST
Confirmation of CR or PR	Required in non-randomized studies only	Identical to RECIST V1.1
Confirmation of SD	Not required	Identical to RECIST V1.1
New lesions	Evaluated as PD without measurement	Evaluated as iUPD, and documented as iCPD if evidenced in the next evaluation by the following: • Occurrence of any other new lesion or • New lesion volume enlargement (the SOM of new target lesions increasing by ≥ 5 mm or enlargement of any new non-target lesion) iCPD can also be evidenced by new lesions not documented previously
Independent blind review and central imaging	Recommended in some cases	Image acquisition is recommended in all studies (but without independent review)
Confirmation of PD	Not required (unless equivocal)	Required
Clinical state of patient	Not considered in evaluation	Consider whether the patient is clinically stable (see definition) to determine whether to continue the medication after iUPD

CR: complete response; PR: partial response; SD: stable disease; PD: progressive disease; iCR: immunity complete response; iPR: immunity partial response; iSD: immunity stable disease; iUPD: immunity - unconfirmed progressive disease; iCPD: immunity - confirmed progressive disease;

Attached Table 7. Responses evaluated as per iRECIST at different time points

Response evaluation				
Target lesion*	**Non-target lesion***	**New lesion***	**Result of response evaluation**	
			Without previous iUPD**	**With previous iUPD**,***
iCR	iCR	No	iCR	iCR
iCR	Non-iCR/Non-iUPD	No	iPR	iPR
iPR	Non-iCR/Non-iUPD	No	iPR	iPR
iSD	Non-iCR/ Non-iUPD	No	iSD	iSD
iUPD unchanged or reduced as compared with the last evaluation	iUPD unchanged or reduced as compared with the last evaluation	Yes	NA	If a new lesion was previously found and increases in volume (SOM of new target lesion increases by ≥ 5 mm or new non-target lesion is enlarged), iCPD is evidenced. If a new lesion is unchanged as compared with the last evaluation (in volume or quantity), then it will still be evaluated as iUPD

Response evaluation				
Target lesion*	**Non-target lesion***	**New lesion***	**Result of response evaluation**	
			Without previous iUPD**	**With previous iUPD**,***
iSD, iPR, iCR	iUPD	No	iUPD	If volume of a non-target lesion increases (not necessarily the unequivocal PD defined in RECIST V1.1), iCPD is evidenced, otherwise the evaluation result remains to be iUPD
iUPD	Non-iCR/Non-iUPD; iCR	No	iUPD	When the following conditions are met, iCPD is evidenced, otherwise the evaluation result remains to be iUPD: • SOM of a target lesion increases by ≥ 5 mm (if not, the evaluation result remains to be iUPD)
iUPD	iUPD	No	iUPD	When the following conditions are met, iCPD is evidenced, otherwise the evaluation result remains to be iUPD: • SOM of any previously found target lesion with iUPD increases by ≥ 5 mm and/or • Any non-target lesion with iUPD is enlarged (previous evaluation—not necessarily unequivocal PD)
iUPD	iUPD	Yes	iUPD	When the following conditions are met, iCPD is evidenced, otherwise the evaluation result remains to be iUPD: • SOM of any previously found target lesion with iUPD increases by ≥ 5 mm and/or • Any non-target lesion with iUPD is enlarged (not necessarily unequivocal PD) and/or • Previously found new lesions increase in volume or quantity
Non-iUPD/PD	Non-iUPD/PD	Yes	iUPD	When the following conditions are met, iCPD is evidenced, otherwise the evaluation result remains to be iUPD: • Previously found new lesions increase in volume or quantity
*: According to RECIST V1.1. If no pseudo-progression occurs, RECIST V1.1 is consistent with iRECIST in definitions of CR, PR, and SD. **: For any type of lesion. ***: Found in the latest evaluation before this evaluation.				

iCR: immunity complete response; iPR: immunity partial response; iSD: immunity stable disease; iUPD: immunity-unconfirmed progressive disease; iCPD: immunity - confirmed progressive disease; SOM: sum of measurements; NA: not applicable; NE: not evaluable

Attached Table 8. BORs pursuant to iRECIST

BOR					
First evaluation	**Second evaluation**	**Third evaluation**	**Fourth evaluation**	**Fifth evaluation**	**iBOR**
iCR	iCR, iPR, iUPD, NE	iCR, iPR, iUPD, NE	iUPD	iCPD	iCR
iUPD	iPR, iSD, NE	iCR	iCR, iUPD, NE	iCR, iPR, iSD, iUPD, iCPD, NE	iCR
iUPD	iPR	iPR, iSD, iUPD, NE	iPR, iSD, iUPD, NE, iCPD	iPR, iSD, iUPD, NE, iCPD	iPR
iUPD	iSD, NE	iPR	iPR, iSD, iUPD, NE	iPR, iSD, iUPD, iCPD, NE	iPR
iUPD	iSD	iSD, iUPD, NE	iSD, iUPD, iCPD, NE	iSD, iUPD, iCPD, NE	iSD
iUPD	iCPD	Any	Any	Any	iCPD
iUPD	iUPD (no iCPD)	iCPD	Any	Any	iCPD
iUPD	NE	NE	NE	NE	iUPD

- For illustrative purpose only—there may be more cases following the same principle
- This table is assumed to be designed for randomized studies and does not require confirmation of CR or PR
- For patients with only non-target lesions at baseline, evaluations at various time points are recorded as iCR or non-CR/non-PD, which are not shown in the table for ease of presentation

iCR: immunity complete response; iPR: immunity partial response; iSD: immunity stable disease; iUPD: immunity-unconfirmed progressive disease; iCPD: immunity - confirmed progressive disease; SOM: sum of measurements; NA: not applicable; NE: not evaluable

APPENDIX 6. CHILD-PUGH CLASSIFICATION SYSTEM FOR EVALUATION

Child-Pugh score is used to assess prognosis of chronic liver diseases, mainly liver cirrhosis. Although initially used to predict operative mortality, it is now used to determine prognosis, required treatment intensity, and the need for liver transplantation.

Scoring

This scoring scale applies to 5 clinical indicators of liver diseases. Each indicator is scored from 1–3, where 3 represents the most serious outcome. See Attached Table 9.

Attached Table 9. Child-Pugh Classification Form for Evaluation

Measurement	**1**	**2**	**3**
TBIL[1] (mg/dL)	<2.0	2.0 ~ 3.0	>3.0
Serum albumin (g/dL)	>3.5	2.8 ~ 3.5	<2.8
INR [2] or Prothrombin time delay (s)	<1.7 <4.0	1.7 ~ 2.3 4.0-6.0	>2.3 >6.0
Ascites	N/A	Mild (easy to be pharmaceutically controlled)	Moderate to severe (poorly controlled)
Hepatic encephalopathy[3]	N/A	Class I–II (mild to moderate)	Class III–IV (severe or coma)

[1] For primary sclerosing cholangitis and primary biliary cirrhosis, the reference range of bilirubin level is modified to reflect high level of conjugated bilirubin present in such diseases. Upper limit of score 1 is 68 μmol/L (4 mg/dL) while upper limit of score 2 is 170 μmol/L (10 mg/dL).

[2] Different measures are taken in different teaching materials and publications. In some old references, PT is used in place of INR.

[3] Hepatic encephalopathy is classified according to West Haven Criteria—semi-quantitative classification of mental state. *Excerpted from: Conn H, Lieberthal M. The hepatic coma syndromes and lactulose. Baltimore: Williams & Wilkins; 1979.*

For primary sclerosing cholangitis and primary biliary cirrhosis, the reference value of bilirubin level is modified to reflect the characteristically high level of direct bilirubin in such diseases. Upper limit of score 1 is 68 μmol/L (4 mg/dL) while upper limit of score 2 is 170 μmol/L (10 mg/dL).

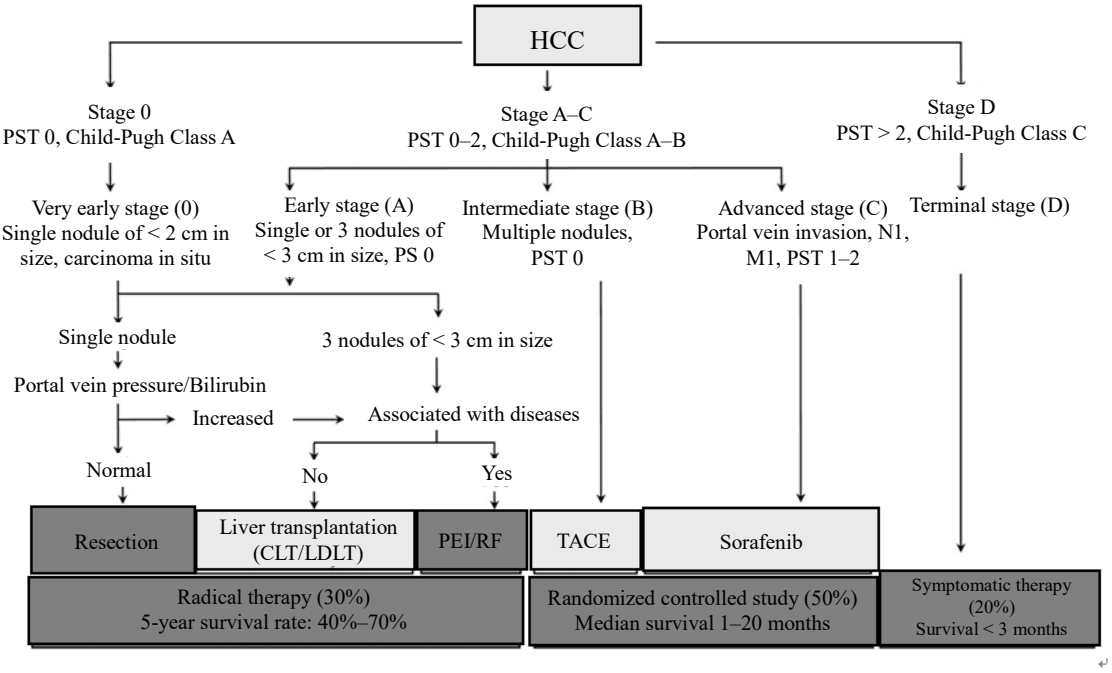
Explanations

Chronic liver diseases are classified into Child-Pugh Class A to C by adding the scores of the indicators described above.

Attached Table 10. Child-Pugh Classification and Prognosis

Score	Grade	One-year survival rate	Two-year survival rate
5~6	A	100%	85%
7~9	B	81%	57%
10~15	C	45%	35%

APPENDIX 7. BARCELONA CLINIC LIVER CANCER (BCLC) STAGING SYSTEM



CLT = Cadaver Liver Transplantation; LDLT = Live Donor Liver Transplantation; PEI = Percutaneous Ethanol Injection; RF = Radio Frequency (Ablation);
TACE = Transarterial Chemoembolization.

APPENDIX 8. CLASSIFICATION, CONVERSION, AND CALCULATION FORMULAS INVOLVED IN CLINICAL STUDIES

Formula	Calculation
Fridericia's formula for QTc	QTcF = QT/(RR^0.33), where RR is standardized heart rate value obtained by dividing 60 by heart rate.
New York Heart Association Classification	Class I: Patients have heart disease but no limitation of physical activity. General physical activity does not result in excessive fatigue, palpitation, asthma or angina. Class II: Patients have heart disease leading to mild limitation on physical activity. The patients are asymptomatic at rest, while general physical activity results in excessive fatigue, palpitation, asthma or angina. Class III: Patients have heart disease leading to significant limitation on physical activity. The patients are asymptomatic at rest, while even minor physical activity can result in excessive fatigue, palpitation, asthma or angina. Class IV: Patients with heart disease who have symptoms of cardiac insufficiency or angina at rest, and any physical activity would increase discomfort.
Endoscopic esophageal varices classification	Mild (G1): Esophageal varices appear linear or slightly tortuous without RC. Moderate (G2): Esophageal varices appear linear or slightly tortuous with RC, or esophageal varices show serpentine tortuous eminences without RC. Severe (G3): Esophageal varices show serpentine tortuous eminences with RC, or esophageal varices appear beaded, nodular or tumor-like (regardless of RC). Note: RC: Red color; red signs, including red wale signs and hemorrhagic spots, are signs indicating that varicose veins have a tendency to bleed.
Conversion of hormone (mg)	Cortisone 25 = Hydrocortisone 20 = Prednisone 5 = Prednisolone 5 = Methylprednisolone 4 = Triamcinolone 4 = Betamethasone 0.6 = Dexamethasone 0.75

APPENDIX 9. CLINICAL DIAGNOSIS CRITERIA FOR HEPATOCELLULAR CARCINOMA (HCC) ESTABLISHED BY AMERICAN ASSOCIATION FOR THE STUDY OF LIVER DISEASES (AASLD) (EXCERPTED TRANSLATION)

1. HCC diagnosis procedure should be initialized if ultrasound suggests nodule diameter > 1 cm; in the case of AFP plus ultrasound, HCC diagnosis procedure should be initialized if AFP > 20 ng/mL.
2. For patients at high risk of HCC, rigorous multi-phase imaging diagnosis criteria should be used to enable non-invasive diagnosis of HCC. Typical image features of multi-phase CT and MRI include nodules ≥ 1 cm in size; arterial-phase enhancement; nodule size in combination with other imaging manifestations, such as venous-phase washout, lesion growth beyond threshold, and marginal enhancement. If high probability of HCC or any other malignancy is considered but not evidenced by the above imaging manifestations, then liver biopsy should be taken into account.
3. Liver biopsy is required to confirm the diagnosis of HCC even with images suggestive of arterial-phase enhancement and venous-phase washout in a patient without liver cirrhosis.
4. If routine histopathological examination fails to diagnose HCC, histological markers GPC3, HSP70, and GS can be assessed to identify severe atypical hyperplasia and HCC.

From biopsy, repeated imaging, and multiple types of imaging, which one should be selected when the nature of hepatic nodules in a patient with liver cirrhosis is uncertain?

Recommendations

(1) For hepatic nodules of uncertain nature, there are many options, including follow-up imaging, use of any other contrast agent or alternative imaging method, and biopsy, though it is impossible to recommend an optimal method. (Quality of evidence: very low; strength of recommendation: conditional)

(2) Routine biopsy of every hepatic nodule of unknown nature is not recommended. (Quality of evidence: very low; strength of recommendation: conditional)

Technical comments

(1) It is probable to get timely diagnosis by biopsy so as to affect treatment decision, but there are risks of hemorrhage and tumor deposit. A negative result may be attributed to failure in harvesting representative tissue.

(2) According to the Liver Imaging Reporting and Data System (LI-RADS) of the American College of Radiology, Organ Procurement and Transplantation Network (OPTN), and the above AASLD guidelines, rigorous imaging criteria for HCC tumors of ≥ 1 cm in diameter have been established. Typical manifestations are significant arterial-phase enhancement, venous-phase enhancement not involving peripheral hepatic tissue, and delayed-phase continuous regression. But the criteria do not apply to lesions of < 1 cm in size.

Translated from excerpt of Julie K. Heimbach, Laura M. Kulik,et al. AASLD Guidelines for the Treatment of Hepatocellular Carcinoma. HEPATOLOGY, VOL. 67, NO. 1, 2018.

Signature Page for Sponsor

Protocol Title: A randomized, open-labeled, multi-center study involving the efficacy and safety of IBI308 in combination with IBI305 vs. sorafenib as the first-line treatment for advanced hepatocellular carcinoma (ORIENT-32)

Project Number: CIBI338B301

Position	Name (Regular Script)	Signature	Date
Senior Director of Medical Science	Wang Yan		
Senior Director of Biostatistics	Sun Xing		

Statement and Signature Page for Investigator

Investigational Drug: Recombinant fully human anti-PD-1 monoclonal antibody (IBI308)

Recombinant Anti-VEGF Humanized Monoclonal Antibody (IBI305)

Study Title: A randomized, open-labeled, multi-center study involving the efficacy and safety of IBI308 in combination with IBI305 vs. sorafenib as the first-line treatment for advanced hepatocellular carcinoma (ORIENT-32)

Approval No.:

Study Protocol No.: CIBI338B301

Investigator's Statement

I have received and read the "Investigator's Brochure" for this study and learned about the properties, action, efficacy, and safety, etc. of recombinant fully human anti-PD-1 monoclonal antibody injection (IBI308) and recombinant human anti-vascular endothelial growth factor monoclonal antibody(IBI305). I have carefully read and learned about this protocol, and agree that this clinical study is ethical. I will get approval from relevant department of my institution.

I will earnestly fulfill my responsibilities as an investigator in accordance with GCP, agree to carry out this clinical study as designed and specified in this protocol, and elaborate study-related information, provisions, and responsibilities to all staff involved in this study. I will guarantee that I have sufficient time to be in charge of and complete this clinical study within protocol-specified duration, and make sure to document all related information and data in the medical record and "Case Report Form" in a truthful, accurate, complete, timely, and lawful manner.

I will strictly abide by *Declaration of Helsinki*, explain to the patients details of this clinical study approved by the ethics committee, and acquire "Informed Consent Form". I will be obliged to make medical decisions related to this clinical study and make sure that the patients get appropriate treatment in the event of any adverse event occurring during the study; should any serious adverse event occur in the study, I will immediately take appropriate therapeutic measures to keep the patients safe and report to the sponsor.

I agree to undergo monitoring and audit by a Clinical Research Associate or auditor designated by the sponsor as well as review by relevant drug administration authority to ensure quality of the clinical study.

I agree to keep confidential all information related to this protocol received or obtained from the study.

Clinical Study Site:

Principal Investigator (Signature): Date: